

Functional universality of critical SK autocorrelations

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Abstract

We prove that the stationary spin autocorrelation of the zero-field Sherrington–Kirkpatrick model at inverse temperature $\beta = 1$ has a common functional scaling limit for Gaussian and Rademacher couplings. With rate-one heat-bath clocks at every site, time is scaled by $n^{2/3}$ and the mean spin autocorrelation is multiplied by $n^{1/3}$. The functions converge in law uniformly on compact positive-time intervals. Their limiting law is not a point mass: it retains sample-to-sample randomness, and its functions decay to zero at large times.

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1 Introduction

The equilibrium spin autocorrelation measures how much a system remembers its initial configuration after averaging over its stationary dynamics. At the critical temperature of the Sherrington–Kirkpatrick model, this memory lives on a time scale that diverges with the number of spins. We determine its limiting random function and prove that Gaussian and Rademacher interactions give the same law on the stated clock.

1.1 Model and result

Let D be either the standard Gaussian distribution or the uniform distribution on $\{-1, 1\}$. For $n \geq 2$, let $(J_{ij})_{i < j}$ be independent with law D , put $J_{ji} = J_{ij}$ and $J_{ii} = 0$, and define

$$\pi_{n,J}(\sigma) = \frac{1}{Z_{n,J}} \exp \left(\frac{1}{\sqrt{n}} \sum_{i < j} J_{ij} \sigma_i \sigma_j \right), \quad \sigma \in \{-1, 1\}^n. \quad (1.1)$$

This is the zero-field model at inverse temperature one. Each site has an independent rate-one clock. When the clock at site i rings, its spin is resampled from its conditional law under $\pi_{n,J}$:

$$\mathbb{P}\{\sigma_i \leftarrow a \mid \sigma_{-i}\} = \frac{\exp(ah_i(\sigma))}{2 \cosh h_i(\sigma)}, \quad h_i(\sigma) = \frac{1}{\sqrt{n}} \sum_{j \neq i} J_{ij} \sigma_j.$$

One unit of this clock contains n site updates on average. Write P_t^J for this reversible semigroup. Its stationary spin autocorrelation and its critical rescaling are

$$C_{n,J}(t) = \frac{1}{n} \sum_{i=1}^n \langle \sigma_i, P_t^J \sigma_i \rangle_{\pi_{n,J}}, \quad (1.2)$$

$$A_{n,J}(s) = n^{1/3} C_{n,J}(sn^{2/3}), \quad s > 0. \quad (1.3)$$

Thermal fluctuations and the randomness of the dynamics have already been averaged in these quantities. Their only remaining randomness is the interaction matrix.

Let $C_{\text{loc}}((0, \infty))$ denote the continuous real functions, with uniform convergence on every compact subinterval of $(0, \infty)$.

Theorem 1.1 (Functional universality). *There is a Borel probability law Q_{crit} on $C_{\text{loc}}((0, \infty))$ such that, for each of the two choices of D ,*

$$\text{Law}_D(A_{n,J}) \implies Q_{\text{crit}} \quad (n \rightarrow \infty).$$

The law Q_{crit} is not a point mass. It assigns probability one to nonzero, nonnegative, nonincreasing functions f satisfying $\lim_{s \rightarrow \infty} f(s) = 0$. The Gaussian and Rademacher models use exactly the normalization in (1.3).

The restriction to positive times is necessary: $A_{n,J}(0) = n^{1/3}$. At every fixed positive time, the limiting value has unbounded support, as proved in Proposition 9.4. The limiting law is described explicitly in Section 2. It is the law of a sum of coordinate autocorrelations for a reversible semigroup in countably many edge coordinates. The equilibrium measure of those coordinates is Gaussian conditioned on a renormalized quadratic constraint; its parameters are the random GOE edge points. A single positive deterministic coefficient accounts for the relaxation of microscopic spin gradients. Section 6 characterizes that coefficient by a determinate moment problem.

1.2 History and relation to earlier work

Sherrington and Kirkpatrick introduced their infinite-range spin-glass model in 1975 [24]. Dynamical mean-field theory subsequently studied relaxation through correlation and response, notably in the soft-spin framework of Sompolinsky and Zippelius [25]. For critical Ising heat-bath dynamics, Billoire and Campbell [5, Equation (7)] tested numerically the finite-size scaling form

$$q_c(n, t) \sim n^{-1/3} F(t/n^{2/3}),$$

with time measured in updates per spin. This is the direct numerical precedent for (1.3). A disorder-averaged scaling curve does not determine whether individual sample curves concentrate; our result identifies their limiting probability law.

Spherical models made the role of the spectral edge especially clear. Kosterlitz, Thouless, and Jones [18] identified the edge mechanism in the spherical spin-glass transition. Cugliandolo and Dean [12] analyzed Langevin relaxation and aging, and Ben Arous, Dembo, and Guionnet [4] proved limiting dynamics and aging results for a soft spherical constraint. At finite size, Fyodorov, Perret, and Schehr [17] related non-self-averaging on the $n^{2/3}$ time scale to edge fluctuations in zero-temperature spherical dynamics. These are spectral and dynamical predecessors, with different temperatures, initial conditions, and spin spaces from stationary critical Ising heat-bath dynamics.

The equilibrium comparison used here has its own history. Comets [11] related Ising and spherical partition functions by averaging over the eigenframe. Baik and Lee [2] studied spherical free-energy fluctuations on the two sides of the transition, and Landon [20] analyzed critical free-energy fluctuations and the edge-scale saddle. Du and Huang [13] obtained a quantitative critical Ising–spherical partition comparison; their quenched overlap theorem [14] describes the random critical overlap law using Gaussian coordinates conditioned on a centered square sum. The conditioned measure used below follows this framework. The additional work here identifies the closed dynamical form, its microscopic coefficient, and the functional limit across two coupling laws.

Rigorous high-temperature mixing developed through spectral functional inequalities and localization. Relevant steps include Bauerschmidt and Bodineau [3], Eldan, Koehler, and Zeitouni [16], and Anari, Koehler, and Vuong [1]. For fixed inverse temperature $\beta < 1/2$, Wang [26] obtains $O_\beta(n \log(n/\epsilon))$ random-site heat-bath mixing, uniformly in external fields, with probability tending to one over the disorder. At zero field, Boban, Li, and Oveis Gharan [6] obtain a spectral gap bounded below by a positive constant for rate-one-per-site dynamics, and polynomial mixing for $\beta < 1/2 + \epsilon_0$, with a universal $\epsilon_0 > 0$, again with high disorder probability. Such fixed-subcritical estimates do not identify dynamics at the endpoint $\beta = 1$.

Two companion manuscripts supply the Gaussian estimates used below. The critical slowing-down manuscript [22, Theorems 1.1 and 1.2] identifies the $n^{2/3}$ scale of covariance and linear relaxation, and gives an obstruction to mixing from most realized equilibrium starts at smaller times. The critical mixing manuscript [21, Theorem 1.1] obtains the matching mixing exponent and functional inequalities. Its comparison of spectral caps with spherical measures provides the starting estimates for our proof. An exponent statement alone does not identify the dynamics at a fixed multiple of $n^{2/3}$; we need quantitative improvements and an identification of the limiting energy. We state the companion inputs with their precise ranges in Section 2 and prove those additional steps here.

Finally, the comparison of coupling laws belongs to the Lindeberg approach to universality [9]. Carmona and Hu [8] proved universality of the SK limiting free energy. The observable here depends on the Gibbs law and the generator simultaneously. Its critical derivative bounds require the quadratic-feature comparison described next.

1.3 The proof and its main ingredients

We first work with Gaussian couplings. Adjoining an independent Gaussian diagonal does not affect the Gibbs law and makes the interaction matrix a GOE matrix $W = U\Lambda U^\top$. The spectral coordinates $x = n^{-1/3}U^\top\sigma$ isolate the fluctuations near the upper edge. Their limiting equilibrium law is a conditioned measure Π , rather than a product of independent Gaussian coordinates. This constraint remains visible in the domain of the limiting gradient form.

There are two separate reasons why equilibrium convergence does not already give the answer. First, a function of the edge coordinates can lower its Dirichlet energy by adding a small microscopic correction. Second, convergence of finitely many edge coordinates gives no bound on the sum over all coordinates in (1.2). The proof treats these issues before comparing the two coupling distributions.

The starting observation is the resolvent trace. If H is minus the heat-bath generator, then for $z > 0$ set

$$B_n(z) = \sum_{a=1}^n \langle x_a, (z + n^{2/3}H)^{-1} x_a \rangle.$$

This quantity is independent of the choice of orthogonal coordinates. It is the Stieltjes transform of the same spectral measure whose Laplace transform is $A_{n,J}$. We establish convergence of these traces and then recover convergence of the functions on every compact positive time interval.

The observations used below have a concrete spectral meaning. At scale $p > 0$, remove from W the matrix

$$B_p = U \operatorname{diag}((\lambda_a - (2 - p^2))_+) U^\top, \quad h_p = B_p \sigma + \mathbf{N}(0, B_p).$$

Conditioning on h_p replaces the interaction by $W - B_p$, whose eigenvalues are capped at $2 - p^2$, and adds the external field h_p . Decreasing p retains fewer observed directions near the spectral edge.

The Gaussian argument has four stages.

1. Spectral observations smooth functions on the cube. We compare their posterior means with a spherical model and control the score of the comparison density. Sections 3 and 4 sharpen the cap estimates to a conditional variance bound of order p^{-2} at cap scale p , without a power loss. The gain comes from averaging the curvature error before compressing it onto a smaller random spectral subspace.
2. In Section 5, slopes of these smoothings converge to weak gradients for Π . We prove that their domain equals the closure of smooth cylinder functions. This identifies the closed energy form on the conditioned equilibrium measure.
3. Section 6 identifies the energy remaining after microscopic correction. Fixed-order contractions specify a deterministic positive measure ω on $[0, \infty)$. Polynomial approximation isolates its atom $c_* = \omega(\{0\})$, producing matching lower bounds and recovery sequences for the limiting form $c_* \int |\nabla F|^2 d\Pi$.
4. Section 7 bounds the tail of the resolvent trace by vector integration by parts. The summable decay of the resulting coefficients makes finite-coordinate convergence sufficient for the whole observable.

For Rademacher couplings we compare basis-independent traces directly. The Lindeberg argument in Section 8 first transfers a trace built from quadratic spin features. Its vanishing controls the fourth-order remainders for the spin trace. This order of comparison is what allows Gaussian and sign couplings, whose fourth moments differ, to have the same limit. Finally, Section 9 proves the functional convergence, decay, and surviving disorder randomness. Appendix A explains why almost-sure decay is compatible with nondecaying functions in the closed topological support.

2 The conditioned edge measure and the Gaussian inputs

The limiting state space is determined by the upper spectral edge of the interaction matrix. The spherical model makes this state space explicit: after diagonalization, its only remaining interaction is a constraint on the sum of the squared coordinates. We construct the resulting probability measure on infinitely many coordinates, following the equilibrium edge-conditioning description of Du–Huang [14, Section 2]. We then explain the change of disorder law that transfers spherical calculations to the cube, and record the cap estimates used in the dynamical argument.

2.1 The heat-bath form and the edge coordinates

Write $\mu = \pi_{n,J}$ and put $T = n^{2/3}$. For a function on the cube, let

$$D_i f(X) = \frac{f(X^{i,+}) - f(X^{i,-})}{2}, \quad t_i(X) = \tanh h_i(X), \quad w_i(X) = 1 - t_i(X)^2,$$

where $X^{i,\pm}$ is obtained by setting the i th spin to ± 1 and $h_i(X) = n^{-1/2} \sum_{j \neq i} J_{ij} X_j$. The nonnegative heat-bath operator and its Dirichlet form are

$$H = \sum_{i=1}^n (X_i - t_i) D_i, \quad \mathcal{E}(f, g) = \mu \left[\sum_{i=1}^n w_i D_i f D_i g \right], \quad \mathcal{E}(f) = \mathcal{E}(f, f). \quad (2.1)$$

Thus $P_t^J = e^{-tH}$. Forms and inner products of vector-valued functions include summation over the vector index. In particular, the clock normalization in (2.1) is rate one at each site.

Until Section 8, the couplings are Gaussian. Let W have off-diagonal entries $J_{ij}/\sqrt{n}$, and add independent diagonal entries with law $N(0, 2/n)$. The extra energy is constant on the cube, so it changes neither μ nor H . The completed matrix is GOE. Diagonalize it as

$$W = U \Lambda U^\top, \quad \lambda_1 \geq \dots \geq \lambda_n, \quad d_a = 2 - \lambda_a, \quad g_{a,n} = n^{2/3} d_a.$$

After independent choices of eigenvector signs, U is Haar orthogonal conditionally on Λ . Its columns are denoted by v_a . Set

$$Y_a = v_a^\top X, \quad x_a = n^{-1/3} Y_a, \quad B_n(z) = \sum_{a=1}^n \langle x_a, (z + TH)^{-1} x_a \rangle_\mu, \quad z > 0.$$

The trace $B_n(z)$ is independent of the chosen orthonormal basis and is the Laplace transform of the rescaled autocorrelation:

$$B_n(z) = \int_0^\infty e^{-zs} A_{n,J}(s) ds.$$

We use x_a also for the coordinate functions of the limiting measure.

Let ρ_{sc} be the semicircle probability measure

$$d\rho_{\text{sc}}(\lambda) = \frac{1}{2\pi} \sqrt{4 - \lambda^2} \mathbf{1}_{[-2,2]}(\lambda) d\lambda.$$

The GOE edge convergence theorem gives the joint convergence of each fixed number of $g_{a,n}$ to a simple, increasing point process $g = (g_1, g_2, \dots)$; see [23]. This definition fixes the sign and scale of the edge process throughout the paper.

Lemma 2.1 (Spectral bounds). *The GOE eigenvalues have the following additional properties.*

1. $\max_a |\lambda_a| \leq 3$ with probability tending to one.
2. There are tight random bounds $K_n \geq 0$ and $C_n > 0$ such that $g_{a,n} \geq C_n^{-1} a^{2/3}$ for every $a > K_n$.
3. For every $\epsilon > 0$,

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left(g_{1,n} + L \leq 0 \text{ or } \left| n^{1/3} - \sum_{a=1}^n \frac{1}{g_{a,n} + L} - \sqrt{L} \right| > \epsilon \right) = 0. \quad (2.2)$$

In the event appearing in the last formula, the sum is evaluated only when $g_{1,n} + L > 0$.

Proof. We use the eigenvalue-only spectral estimate of [22, Proposition 3.1]. For every deterministic sequence $L_n \rightarrow \infty$ with $L_n \leq \log n$, that proposition controls both resolvent traces for $s \in [L_n n^{-2/3}, 5]$, with positive denominators. At $s = L_n n^{-2/3}$, multiplying its first-trace error by $n^{1/3}$ gives

$$O\left(L_n^{-1} + L_n^{-1/8} + L_n^{-1/4} \sqrt{\log(2 + L_n)}\right) = o(1).$$

The semicircle resolvent equals $(2 + s - \sqrt{s(4 + s)})/2$, whose expansion supplies the main term $\sqrt{L_n}$ and an additional error $O(L_n n^{-1/3}) = o(1)$.

The second-trace estimate yields, for the same interval of u ,

$$\#\{a : d_a \leq u\} \leq 4u^2 \sum_a (d_a + u)^{-2} \leq Cnu^{3/2}.$$

Indeed $0 < d_a + u \leq 2u$ for every counted eigenvalue. It follows that $g_{a,n} \geq ca^{2/3}$ once $a > CL_n^{3/2}$. The uniform norm bound is part of the same proposition.

These conclusions hold for every arbitrarily slowly divergent L_n . If (2.2) failed, one could select $L_j \rightarrow \infty$ and $n_j \rightarrow \infty$, with $\log n_j \geq L_j$, along which the failure probability stayed positive, and embed the values $L_{n_j} = L_j$ in a sufficiently slow divergent sequence. This contradicts the preceding estimate. The identical criterion, applied to the least index after which the counting consequence holds, makes that index tight. This proves the second assertion as well. $\square$

Remark 2.2 (Convergence on sets of large disorder probability). We will use constants that are bounded on disorder events of probability arbitrarily close to one. A bound of this kind is called a tight bound. For subsequential arguments, include these bounds, the eigenvalues, and the countably many relevant error quantities in the joint law before taking a Skorokhod representation. On the represented space, finite bounds and coordinate convergence can then be used deterministically. The conditional laws of the remaining finite-dimensional variables lift this representation to the original finite problems. A further diagonal extraction handles quantities that vanish in an iterated limit such as (2.2). The representation itself makes no assertion of convergence for the original sequence: one must identify every subsequential limit. Once this identification is made, the subsequence criterion gives the asserted joint convergence in distribution. In particular, if every subsequence of an intrinsic error quantity has a further represented subsequence tending to zero, that quantity tends to zero in probability on the original disorder space. No uniform integrability on exceptional disorders is required.

A separate uniformity convention is essential for variational arguments. When an estimate is applied to a disorder-dependent test, its quenched inequality is understood to hold on one event for every test in the stated class, with the displayed bounds on its norm, energy, and polynomial growth. One way to obtain such an event, used below, is to bound the error for every test by its stated norms times a common nonnegative random error R_n . Convergence $R_n \rightarrow 0$ then permits arbitrary measurable choices of the test. Convergence on a countable core alone is not used to infer this uniformity.

2.2 Conditioning the infinite Gaussian family

Choose an admissible parameter b with $b + g_1 > 0$, and write $l_a = (g_a + b)^{-1}$. A convenient measurable choice is $b = 1 + (-g_1)_+$; use the corresponding $b_n = 1 + (-g_{1,n})_+$ in finite dimension.

By Lemma 2.1,

$$l_{a,n} = (g_{a,n} + b_n)^{-1} \longrightarrow l_a, \quad \lim_{m \rightarrow \infty} \sup_n \sum_{a > m} l_{a,n}^2 = 0$$

on the represented sets just described, after discarding finitely many initial n . In particular $\sum_a l_a^2 < \infty$.

For independent $Z_a \sim N(0, l_a)$, the centered series $S = \sum_a (Z_a^2 - l_a)$ converges almost surely and in L^2 . Define

$$D(b) = \lim_{L \rightarrow \infty} \left\{ \sqrt{L} - \sum_{a \geq 1} \left(\frac{1}{g_a + b} - \frac{1}{g_a + L} \right) \right\}. \quad (2.3)$$

Here L is restricted to values with $L + g_1 > 0$. The limit is in probability with respect to g ; a deterministic subsequence gives an almost sure version. The next proposition both justifies the limit and gives a pointwise meaning to conditioning on $S = D(b)$.

Proposition 2.3 (The conditioned edge measure). *Formula (2.3) defines a finite random variable, and*

$$D_n(b_n) := n^{1/3} - \sum_{a=1}^n l_{a,n} \implies D(b)$$

jointly with the edge coordinates. Almost surely $\sum_a l_a = \infty$. The law of $(Z_a)_{a \geq 1}$ conditioned on $S = D(b)$ has a version $\Pi = \Pi_g$ characterized by continuous conditional densities. It is independent of the admissible choice of b .

Let μ^{sp} denote normalized surface measure on the sphere of radius $\sqrt{n}$, tilted by $\exp(X^\top W X/2)$. The fixed-coordinate marginals of $n^{-1/3} U^\top X$ under μ^{sp} converge jointly in distribution with the eigenvalues to the random marginals of Π . On the spectral representations of Remark 2.2, this is ordinary weak convergence of measures, including convergence of every fixed polynomial moment. On the sets in Remark 2.2, for every fixed $0 \leq j < \infty$ one has

$$\int |x_a|^j d\Pi \leq C_j l_a^{j/2}, \quad \mu^{\text{sp}} |x_a|^j \leq C_j l_{a,n}^{j/2}, \quad (2.4)$$

uniformly in the retained coordinate a .

For any sufficiently large fixed m , the tail $(x_a)_{a > m}$ under Π has a density relative to $\otimes_{a > m} N(0, l_a)$. Conditional on this tail, the first m coordinates lie on the sphere of squared radius

$$D(b) + \sum_{a \leq m} l_a - \sum_{a > m} (x_a^2 - l_a), \quad (2.5)$$

with the Gaussian angular density on that sphere. In particular, $\sum_a (x_a^2 - l_a) = D(b)$ holds Π -almost surely.

Proof. For fixed admissible L , the identity

$$D_n(b_n) = n^{1/3} - \sum_a \frac{1}{g_{a,n} + L} - \sum_a \left(\frac{1}{g_{a,n} + b_n} - \frac{1}{g_{a,n} + L} \right)$$

separates the divergent trace from an absolutely convergent difference. The tail of the second sum is bounded by a constant times $|L - b_n| \sum_{a > m} a^{-4/3}$ on each set of bounded spectral data, so that sum passes to the limit. Lemma 2.1 then shows that the expressions in braces in (2.3) are Cauchy in probability, and that $D_n(b_n)$ has the asserted joint limit. If $\sum_a l_a$ were finite, those

expressions would tend to $+\infty$ as $L \rightarrow \infty$; this contradicts the finite limit. Changing b changes $D(b)$ by the corresponding absolutely convergent resolvent difference.

We give the density argument explicitly. Let q be the density of S , and let q_m be the density of $\sum_{a>m}(Z_a^2 - l_a)$. After any fixed coordinates are omitted, a sufficiently large fixed group of remaining coordinates supplies an integrable bound for the characteristic function: the modulus of the factor for coordinate a is $(1 + 4t^2 l_a^2)^{-1/4}$. Fourier inversion therefore gives continuous, bounded densities. The same bound applies to the finite sums and proves uniform convergence of their densities when the variances converge and their squared tails vanish.

The centered sum after any finite deletion has support all of $\mathbb{R}$. Its unbounded right support comes from one square. To reach an arbitrarily negative interval, make a sufficiently long finite block of squares small, using $\sum l_a = \infty$, and keep the remaining centered sum in a bounded interval of positive probability. Convolution with a finite block whose square-sum density is positive in the interior of its support now gives $q(s) > 0$ and $q_m(s) > 0$ for every $s \in \mathbb{R}$. Thus the first m coordinates of the required conditional law have density

$$\frac{q_m\left(D(b) - \sum_{a \leq m}(x_a^2 - l_a)\right)}{q(D(b))} \prod_{a \leq m} \frac{e^{-x_a^2/(2l_a)}}{\sqrt{2\pi l_a}}. \quad (2.6)$$

Convolution shows that these densities are consistent. They depend measurably on the spectral data, so their extension is a random probability measure Π_g .

For completeness, they admit a disintegration on the constraint itself. Let q_m^{head} be the density of $\sum_{a \leq m}(Z_a^2 - l_a)$. Give the tail product Gaussian law the weight

$$\frac{q_m^{\text{head}}(D(b) - \sum_{a > m}(x_a^2 - l_a))}{q(D(b))}.$$

Given the tail, use polar disintegration of the first m Gaussian coordinates at the radius (2.5). This produces (2.6), has an absolutely continuous tail law, and enforces the constraint almost surely.

In finite dimension, conditioning independent normals of variances $l_{a,n}$ on $\sum_a x_a^2 = n^{1/3}$ gives exactly the spherical Gibbs measure in the stated coordinates: the term involving b_n in the Gaussian exponent is constant on that sphere. The density convergence above and $q(D(b)) > 0$ imply convergence of each cylinder marginal. Their densities are bounded by a constant times the corresponding product Gaussian density. To retain an arbitrary single coordinate uniformly, choose the integrable Fourier bound from a fixed group of leading coordinates other than that coordinate. This proves (2.4) and uniform integrability of all fixed polynomial moments.

The argument applies to every admissible b on sets where its distance from the pole is bounded below and its magnitude is bounded above; these sets exhaust the admissible choices. The finite conditioned spherical law is independent of b_n , so its limiting cylinder marginals are also independent of b . The conclusion holds simultaneously for all admissible b . First take a common probability-one event for rational parameters. On every compact admissible interval, the resolvent-difference series converges uniformly, so $D(b)$ is continuous. The squared variance tails and the Fourier majorants above are also uniform on that interval. Thus the conditional cylinder densities in (2.6), and their integrals against bounded continuous cylinder tests, depend continuously on b . Equality for rational parameters therefore extends to the whole interval. $\square$

Lemma 2.4 (The bulk spectral weight). *For a spherical Gibbs spin, the random probability measure*

$$\frac{1}{n} \sum_a Y_a^2 \delta_{\lambda_a}$$

converges in probability, conditionally on the spectral data as above, to $(2 - \lambda)^{-1} d\rho_{\text{sc}}(\lambda)$.

Proof. First test against a bounded continuous function supported away from 2. Before conditioning, the Gaussian variance of its integral is $O(n^{-1})$, and its mean converges to the stated integral by the semicircle law. Integrate out a fixed number of leading Gaussian coordinates when imposing the norm constraint. Their square-sum density gives a bounded density ratio, so the concentration persists after conditioning. Each of those leading coordinates has mass $n^{-1/3} x_a^2 = o(1)$ in probability by (2.4). Finally both the finite measure and its proposed limit have mass one, since $\int (2 - \lambda)^{-1} d\rho_{\text{sc}}(\lambda) = 1$. This extends convergence across the edge. The same argument permits bounded tests with cuts at continuity points in the bulk. $\square$

2.3 The limiting autocorrelation and its time constant

For a smooth compactly supported cylinder function F on $\mathbb{R}^{\mathbb{N}}$, consider the quadratic form

$$\mathcal{E}_{\Pi}(F) = \int \sum_a |\partial_a F|^2 d\Pi.$$

Proposition 5.1 will prove that this form is closable and identify its domain. Denote the nonnegative operator associated to its closure by H_{Π} . The law in Theorem 1.1 is specified by

$$\mathcal{A}(s) = \sum_{a \geq 1} \langle x_a, e^{-sc_* H_{\Pi}} x_a \rangle_{\Pi}, \quad s > 0, \quad (2.7)$$

where $c_* > 0$ is deterministic. Finiteness, continuity, and the remaining assertions about this random function are proved in Sections 7 and 9.

The coefficient is $c_* = \omega(\{0\})$, where ω is the deterministic finite positive measure of microscopic heat-bath energies characterized by the moment prescription (6.3) in Section 6. Proposition 6.1 constructs this measure and proves its determinacy; Theorem 6.8 proves that its atom at zero is strictly positive and is the coefficient in (2.7).

2.4 Haar bias and Gaussian cap observations

We now connect the spherical calculations with the cube. All partition functions use probability reference measures: uniform probability on the cube and normalized surface measure on the sphere. Conditional on the spectrum, put

$$L_{\circ}(U) = \frac{Z_{\text{cube}}(W)}{Z_{\text{sp}}(W)}.$$

Haar averaging gives $E_U L_{\circ} = 1$. Define the probability law $\mathbf{P}$, and its expectation $\mathbf{E}$, by drawing U with Haar density $L_{\circ}$, then drawing a cube spin with law μ . Any observation noises introduced below are independent of this spin conditionally on the matrix. The spectrum is held fixed in $\mathbf{P}$ and $\mathbf{E}$.

Lemma 2.5 (Spherical identity under Haar bias). *Under $\mathbf{P}$, $U^\top X$ has exactly the spherical Gibbs law in spectral coordinates. After the row sign change sending X to $\mathbf{1}$, conditional on $y = U^\top X$ the eigenframe is Haar subject to $Uy = \mathbf{1}$. Moreover, on the eigenvalue events used below,*

$$E_U(L_\circ - 1)^2 = o(1). \quad (2.8)$$

Proof. In the joint weighted law, Z_{cube} cancels the Gibbs denominator. For each fixed cube vertex, Haar rotation sends that vertex to uniform surface measure on the sphere of radius $\sqrt{n}$. The remaining energy weight is $\exp(y^\top \Lambda y/2)$, proving the first assertion. The same invariance gives the conditional Haar law on the stated coset. Finally, the uncapped zero-field calculation in the proof of the static cap comparison of [21], Section 15, paragraph “Zero field and overlaps bounded away from zero,” gives $E_U Z_{\text{cube}}^2 / (Z_{\text{sp}})^2 = 1 + o(1)$. Together with $E_U L_\circ = 1$, this is (2.8). $\square$

For example, if $G \geq 0$ is a spin or observation functional, then

$$E_U[\mathbf{1}_{\{L_\circ \geq 1/2\}} Q_\mu G] \leq 2EG.$$

Here Q_μ denotes the ordinary joint law of the Gibbs spin and its observations at fixed disorder. This inequality and Markov’s inequality transfer biased mean estimates to quenched estimates; the discarded disorder event has probability $o(1)$ by (2.8).

The cap at scale $p > 0$ is defined by

$$P_p = \mathbf{1}_{\{d \leq p^2\}}, \quad \mathcal{H}_p = \text{ran } P_p, \quad B_p = (p^2 - d)_+, \quad W_p = W - B_p, \quad k_p = np^3, \quad M_p = \sqrt{np}.$$

Matrix functions in these formulas are conjugated by U when used in physical coordinates. Figure 1 shows how the cap flattens the upper edge and how the removed part becomes observation precision.

Observe $h_p = B_p X + \mathbf{N}(0, B_p)$, with independent Gaussian increments as B_p increases, starting from zero precision if necessary. Bayes’ formula makes the posterior interaction W_p and the posterior field h_p . Write $\mu_{p,h}$ for the cube posterior at a specified field h . Let $\phi_p(h)$ be its log partition function, $m_p(h) = \nabla \phi_p(h)$ its mean, and $K_p(h) = P_p \nabla^2 \phi_p(h) P_p$ its covariance restricted to $\mathcal{H}_p$. Superscript sp denotes the corresponding spherical quantities. At path fields we often suppress the argument h_p .

On the joint space of the eigenframe and the spectral observation path, the density of $\mathbf{P}$ relative to independent Haar measure and the spherical observation law, through scale p , is

$$L_p = \exp\{\phi_p(h_p) - \phi_p^{\text{sp}}(h_p)\}. \quad (2.9)$$

Indeed the likelihood of the observations is a common Gaussian factor multiplied by the posterior partition function divided by the original one; the factor $L_\circ$ cancels the two original partition functions. The spin in spectral coordinates already has the spherical law by Lemma 2.5; its observation noises are independent Gaussian increments with covariance determined by the spectrum. Hence the entire spectral observation path under $\mathbf{P}$ has exactly its spherical law. We normalize its coordinates by

$$u_{p,a} = \frac{(U^\top h_p)_a}{p^2 n^{1/3}}, \quad r = n^{-1/3+\delta}, \quad \delta = 10^{-4}.$$

We will use $r \leq p \leq p_{\text{top}}$, allowing fixed-factor buffers at the endpoints. The values $d_a = p^2$ cause no discontinuity in the observations; they may be omitted in scale integrals, and zero-variance coordinates are omitted from density formulas.

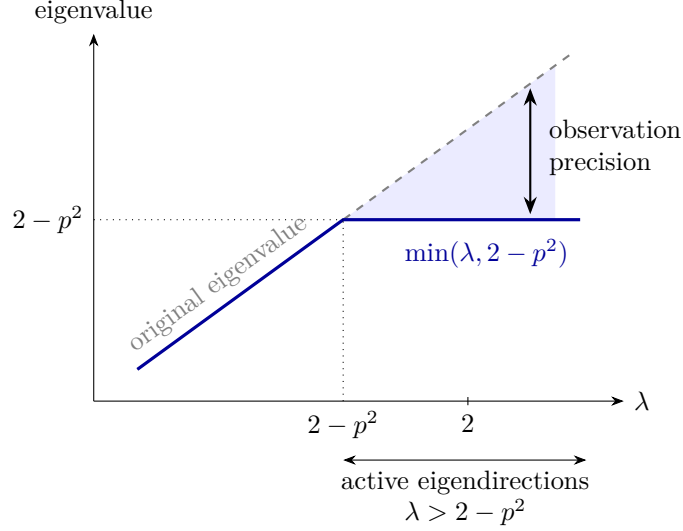


Figure 1: A spectral cap, shown near the upper edge. The posterior interaction W_p retains eigenvalues below $2 - p^2$ and flattens those above it. The removed amount $(\lambda - (2 - p^2))_+$ is the precision of the Gaussian observation in that eigendirection. The drawing is schematic; its geometry illustrates the defining spectral functions, not an eigenvalue density.

2.5 Saddles and cap estimates

For a field $h \in \mathcal{H}_p$, its spherical saddle $\alpha = \alpha_p(h)$ is the unique solution above the pole of

$$R_{p,\alpha} = (\max(d, p^2) + \alpha)^{-1}, \quad n = \text{Tr } R_{p,\alpha} + \|R_{p,\alpha} h\|^2. \quad (2.10)$$

The annulus $|\alpha_p(h)| \leq s_0 p^2$ is separated from zero field. Its radius as a function of α is

$$\ell_p(\alpha) = (p^2 + \alpha) \sqrt{n - \text{Tr } R_{p,\alpha}}.$$

The following statement collects the precise Gaussian inputs needed later. All upper cutoffs and tolerances in it are fixed constants before n tends to infinity.

Proposition 2.6 (Cap inputs). *Fix a finite field radius R and positive fixed error tolerances. There are sufficiently small constants $p_{\text{top}}, s_0 > 0$ such that the following hold with probability tending to one in the range $np^3 \geq c\sqrt{\log n}$ and $p \leq p_{\text{top}}$.*

Spectral and spherical estimates, at every scale. *The spectral dimension satisfies $\dim \mathcal{H}_p \asymp k_p$, $d_1 \geq -o(p^2)$ uniformly in the cap range, and $\text{Tr } R_{p,0}^2 = O(n/p)$. On $|\alpha| \leq s_0 p^2$, $\ell_p(\alpha) \asymp p^2 M_p$ and $\ell'_p(\alpha) \asymp M_p$. For $\|h\| \leq R p^2 M_p$, the saddle precision has gap comparable to p^2 . At $\alpha = \alpha_p(h)$, the density at n of the squared norm of $\mathbf{N}(R_{p,\alpha} h, R_{p,\alpha})$ is comparable to $\sqrt{p/n}$. After multiplication by $e^{v^\top X}$ and renormalization, the Gaussian mean becomes $R_{p,\alpha}(h + v)$ while its covariance remains $R_{p,\alpha}$; its squared-norm density at n is at most a fixed multiple of $\sqrt{p/n}$. For m replicas, where m is fixed, consider rowwise linear or quadratic tilts that preserve Gaussianity. The joint density of their squared norms is at most $C_m(p/n)^{m/2}$ under the following hypotheses. The spectral rows remain independent after the rowwise tilt, the tilted precision is positive definite, and on at least ck_p rows the post-tilt row covariance C_a satisfies $cp^{-2} \text{Id}_m \preceq C_a \preceq Cp^{-2} \text{Id}_m$. All comparison constants here are fixed.*

Curvature estimates, at every scale. *On the annulus, for any prescribed fixed $\epsilon_0 > 0$,*

$$K_p \preceq (1 + \epsilon_0)p^{-2} \text{Id}_{\mathcal{H}_p}, \quad p^2(K_p)_{ee} \leq 1 - \epsilon_1, \quad e = h/\|h\|, \quad \epsilon_1 = 0.1. \quad (2.11)$$

On the entire field ball one also has $K_p \preceq C_R p^{-2} \text{Id}_{\mathcal{H}_p}$, where C_R may be chosen before ϵ_0 .

Static comparisons and cube projection tails, on geometric lists. *For any prescribed fixed geometric list of scales, uniformly over the field ball,*

$$|\phi_p(h) - \phi_p^{\text{sp}}(h)| \leq \epsilon_v k_p, \quad \|P_p m_p(h) - R_{p, \alpha_p(h)} h\| \leq \epsilon_g M_p.$$

Any fixed finite family of such lists is allowed. For each $h \in \mathcal{H}_p$ with $\|h\| \leq R p^2 M_p$, the cube posterior $\mu_{p,h}$ satisfies, uniformly for every $v \geq v_0$ with a sufficiently large fixed threshold v_0 ,

$$\mu_{p,h}\{\|P_p X\| > v M_p\} \leq C \exp(-a k_p v^6).$$

For $v > p^{-1/2}$ the event is empty, since $\|X\| = \sqrt{n}$. Here $a > 0$, and the same bound holds under the uncapped zero-field cube measure μ at the same list scales.

The spherical projection-tail estimate holds at every scale in the stated range, for capped bounded-field and uncapped zero-field measures. Also, at every such cap scale, for independent spherical replicas at a common field in the prescribed ball,

$$(\mu_{p,h}^{\text{sp}})^{\otimes 2}\{|n^{-1} X^\top X'| > \rho\} \leq C \exp(-c n \rho^3), \quad C p \leq \rho \leq \rho_0,$$

for a sufficiently small fixed $\rho_0 > 0$. A fixed finite number of replicas obeys the corresponding bound for any pair.

Source statements and the radial consequence. The spectral, saddle, and density assertions are the conclusions of [21], Sections 15.2–15.3, “Spectral estimates at the bottom scale” and “Spherical saddles, norm densities, and tails.” The static comparison and its projection tails are Proposition 15.1 in that section. Its order of choices is the one stated here: fix the radius, errors, and geometric ratios, then decrease the upper cutoff. The overlap estimate is the spherical estimate preceding that proposition.

The upper covariance bounds are Theorem 18.6, “Uniform curvature of the true cap potentials,” in [21]. Its lower endpoint is $(\sqrt{\log n}/n)^{1/3}$, with fixed-factor changes permitted, so it covers our scale r and its fixed buffers. We also use a consequence retained in that theorem’s proof. Let γ be the fixed geometric ratio chosen in that proof, and set $R_{p,\alpha}^H = (P_p R_{p,\alpha} P_p)|_{\mathcal{H}_p}$. At an intermediate evaluation ratio $x \in [\gamma^2, \gamma]$, Lemma 16.2, “Evaluation with the induction extension,” in [21] bounds the radial entry of $p^2(K_p - R_{p,\alpha}^H)$ above by -0.26 . The theorem’s proof chooses s_0 so that every target saddle lies inside the parent agreement annulus with fixed slack, and applies exactly this deterministic evaluation. Therefore

$$p^2(K_p)_{ee} \leq \frac{1}{1 + \alpha/p^2} - 0.26 \leq \frac{1}{1 - s_0} - 0.26 \leq 0.9$$

after decreasing s_0 . This proves the second part of (2.11) with a fixed tolerance. $\square$

The list restriction in the static comparison is harmless, but its removal requires a short argument because later observation identities integrate over all scales.

Lemma 2.7 (Interpolation between cap scales). *The value, projected-mean, and cube projection-tail conclusions of Proposition 2.6 hold at every scale in the stated range, after prescribing slightly smaller errors on a sufficiently fine fixed geometric list and decreasing the upper cutoff.*

Proof. First use an ordinary fixed-ratio list. Monotonicity of P_p and a fixed-factor change in M_p extend the uncapped projection tail to every p . Removing B_p from the interaction changes its normalizing factor by

$$\mu \exp(-X^\top B_p X/2) \geq \mu\{\|P_p X\| \leq VM_p\} \exp(-\|B_p\|V^2M_p^2/2) \geq e^{-Ck_p},$$

where V is fixed and large. We used $\|B_p\| = O(p^2)$ on the spectral event. Thus the projection tail persists at cap zero field. A field $\|h\| \leq Rp^2M_p$ adds at most Rk_pv on the shell $\|P_p X\| = vM_p$; this is dominated by ak_pv^6 for large v . The zero-field measure is even, so its linear-tilt normalizer is at least one. This gives the same tail on the field ball, with an adjusted starting threshold. The argument also bounds $\|P_q X\|/M_q$ under the p -cap posterior whenever $q \geq p$ and q/p is bounded: use the uncapped P_q tail, the same denominator lower bound, and $P_ph = h \in \mathcal{H}_q$. This is the tail needed to compare two nearby caps. The spherical estimates give the identical conclusion.

Now choose a fixed geometric list with consecutive ratios close to one, and let q be its first scale above p . Then $\mathcal{H}_p \subset \mathcal{H}_q$ and

$$\|W_p - W_q\| \leq C(1 - p/q)q^2, \quad \text{ran}(W_p - W_q) \subset \mathcal{H}_q.$$

For $h \in \mathcal{H}_p$ in the prescribed ball, restrict first to $\|P_q X\| \leq VM_q$. The change in log weight there is at most $C(1 - p/q)V^2k_q$. On its complement, the projection tail controls the exponential of the quadratic change: the exponent $C(1 - p/q)k_qv^2$ is dominated by the negative ak_qv^6 tail. Choose V first and the list ratio second. The resulting log-partition change is as small as desired in units k_q , for both cube and sphere. Their value comparison at q therefore implies the value comparison at p , since k_p/k_q is arbitrarily close to one.

For the mean, use the value comparison on a slightly larger field ball. The frozen Gaussian saddle identity and its norm-density bound give, for any unit $v \in \mathcal{H}_p$ and both signs,

$$\log \mu_{p,h} e^{\pm tv^\top (X - R_{p,\alpha} h)} \leq 2\epsilon_v k_p + \frac{1}{2} t^2 v^\top R_{p,\alpha} v + C.$$

Jensen's inequality, with $t = \theta p^2 M_p$, bounds the projected mean discrepancy divided by M_p by $C\theta + (2\epsilon_v + C/k_p)/\theta$. Choose θ from the required mean tolerance and then ϵ_v sufficiently small. This proves the claim uniformly at every intermediate scale. $\square$

The estimates of this section supply the ordinary Gaussian caps and the spherical description of their observations. The next section derives the quantitative comparison and score estimates needed to retain the exact $n^{2/3}$ time scale.

3 Quantitative estimates for the observations

The coarse cap estimates of Proposition 2.6 locate the relevant fields and bound their covariance by a constant times p^{-2} . For the time normalization in Theorem 1.1, we shall need a relative covariance error that is integrable in logarithmic scale. The purpose of this section is to obtain the averaged estimates from which that improvement will follow. There are two different sources of smallness: a two-replica comparison at the smallest observation scale, and an integrated likelihood estimate at the larger scales.

We retain the notation and conditional-on-spectrum probability $\mathbf{P}$ of Lemma 2.5. Thus $\mathbf{P}$ includes the partition-function bias on the eigenframe and the ordinary Gibbs observation path given that frame; Q_μ denotes the latter path law for fixed disorder. Expectations with respect to

these laws are denoted by $\mathbf{E}$ and $Q_\mu[\cdot]$. Constants in the present section are uniform on each of the tight spectral sets described in Remark 2.2. They may depend on that set, on a fixed moment order, and on fixed field or scale ratios. A bound $C \exp(-n^\xi)$ always has some $\xi > 0$; decreasing ξ absorbs polynomial factors and the $O(\log n)$ scale intervals used below. Recall that

$$r = n^{-1/3+\delta}, \quad \delta = 10^{-4}, \quad k_p = np^3, \quad M_p = (np)^{1/2}.$$

All estimates involving a fixed multiplicative enlargement of the interval of scales have the same meaning, provided its upper endpoint is sufficiently small. In particular, k_p grows as a positive power of n throughout the intervals considered here.

3.1 Location of the path and coordinate moments

The following estimate improves the fixed annulus test to a shrinking annulus. It concerns the ordinary observation path averaged under $\mathbf{P}$; it is not an assertion about every external field.

Lemma 3.1 (The observation annulus). *There are $a, \xi > 0$ and a sufficiently small fixed upper scale $p_{\text{top}} > 0$ such that*

$$|\alpha_p(h_p)| \leq C k_p^{-a} p^2 \quad (r \leq p \leq p_{\text{top}}) \quad (3.1)$$

outside a set of $\mathbf{P}$ -probability at most $C \exp(-n^\xi)$. On disorder events of probability arbitrarily close to one, the same whole-path statement holds under Q_μ , with a possibly smaller ξ .

Proof. Under $\mathbf{P}$, the field in spectral coordinates has exactly its spherical distribution. Generate the spherical spin from independent centered Gaussian coordinates y_i of precision $D'_i = d_i + b_n n^{-2/3}$, conditioned on $\sum_i y_i^2 = n$; the choice of b_n is that of Proposition 2.3. Before this conditioning, write

$$h_{p,i} = B_{p,i} y_i + B_{p,i}^{1/2} z_i, \quad m = p^2 + b_n n^{-2/3},$$

where the z_i are independent standard normals. On an active coordinate $B_{p,i} = m - D'_i$, and direct expansion gives

$$\mathbf{E}_0(\|h_p\|^2 - m^2 \|y\|^2) = -m^2 \text{Tr } R_{p, b_n n^{-2/3}}.$$

Here $\mathbf{E}_0$ denotes the unconditioned Gaussian expectation. The subtraction of $m^2 \|y\|^2$ removes the large contribution of the leading coordinates: its coefficient on the standardized y_i^2 term is $(B_{p,i}^2 - m^2)/D'_i = D'_i - 2m = O(p^2)$.

As a quadratic form in the standardized (y, z) variables, the centered quantity has squared Frobenius norm at most $C p^4 k_p$ and operator norm at most $C p^2 k_p^{1/3}$. Indeed the cross coefficients are $O(p^3/\sqrt{D'_i})$, while the spectral estimates give

$$\min_i D'_i \geq c n^{-2/3}, \quad \sum_{d_i \leq p^2} (D'_i)^{-1} \leq C n p.$$

The inactive coordinates are bounded by the capped second resolvent trace. The Gaussian quadratic-form exponential-moment formula therefore bounds deviations larger than $p^2 k_p^{9/10}$ by $C \exp(-k_p^{a_0})$ for some $a_0 > 0$.

This estimate survives conditioning on $\|y\|^2 = n$. To see this directly, use the same exponential tilt in the Markov bound, with its parameter at most one eighth of the inverse operator norm. The tilted joint Gaussian covariance, and hence its y -marginal covariance, remain comparable to the original ones. A fixed number of leading covariance eigenvalues bounds the density of

$n^{-2/3}\|y\|^2$ from above. Its original density at $n^{1/3}$ is bounded below on the chosen spectral set, by Proposition 2.3. The ratio of norm densities therefore costs only a constant.

After conditioning, the deterministic target for $\|h_p\|^2$ is $\ell_p(b_n n^{-2/3})^2 \asymp p^2 k_p$. The deviation just proved is a relative $O(k_p^{-1/10})$ error, so it first places the inferred saddle inside any fixed surrounding annulus for all sufficiently large n . On that annulus, the derivative of $\ell_p(\alpha)^2$ is comparable to k_p . Consequently the preceding deviation bound implies $|\alpha_p - b_n n^{-2/3}| \leq C p^2 k_p^{-1/10}$. Since $b_n n^{-2/3}/p^2 = O(k_p^{-2/3})$, this proves the fixed-scale assertion.

For the whole path take a sufficiently fine inverse-polynomial mesh in p^2 . The functions B_p , the saddle targets, and their relevant derivatives have polynomial bounds; the positive-part operation defining B_p is continuous at coordinate activations. Conditional on the spin, each coordinate of the noise has independent Gaussian increments. A maximal Gaussian estimate in every mesh interval, followed by a union bound over coordinates and intervals, makes their oscillation smaller than the polynomial tolerance just used. The fixed-scale exponential estimate absorbs this union bound because $k_r = n^{3\delta}$. Finally, Lemma 2.5, Markov's inequality, and $L_\circ \geq 1/2$ outside a vanishing disorder exception transfer the result to Q_μ . $\square$

The signal-plus-noise representation also gives the moment estimates needed to test the field against individual edge coordinates. Every $u_{p,a}$ has moments of every fixed order bounded uniformly in a, p under $\mathbf{P}$. For each fixed finite set of leading indices and every fixed $j < \infty$,

$$\|u_{p,a} - x_a\|_{L^j(\mathbf{P})} \leq \frac{C_j}{pn^{1/3}}, \quad r \leq p \leq p_{\text{top}}. \quad (3.2)$$

Indeed the signal is $(B_{p,a}/p^2)x_a$ and the noise standard deviation is at most $(pn^{1/3})^{-1}$. For a leading coordinate, $|d_a|/p^2 = O((pn^{1/3})^{-2})$. The same estimates hold in $L^j(Q_\mu)$ with tight disorder-dependent constants, simultaneously in p in the sense of moment bounds at each scale: apply the signal-plus-noise formula to the tight quenched moments of x_a furnished by the biased first moment and the concentration of $L_\circ$.

3.2 Two replicas and the smallest scale

In the next lemma the field is deterministic in spectral coordinates. Consequently E_U means an independent Haar average at that fixed field, whereas $\mathbf{E}$ subsequently averages a random field with its spherical observation distribution.

Lemma 3.2 (Replica comparison). *Fix a bounded range for $\|h\|/(p^2 M_p)$ and take the upper scale sufficiently small. Uniformly for fields in this range,*

$$E_U L_p(h)^2 \leq \exp(C + Cnp^4). \quad (3.3)$$

For a sufficiently large fixed C_1 , the contribution to this normalized second moment of cube pairs satisfying $|n^{-1}X \cdot X'| > C_1 p$ is at most $C \exp(-ak_p)$. At $p = r$, the stronger estimates

$$\begin{aligned} E_U (L_r - 1)^2 &\leq Cn^{-1/4}, \\ E_U [L_r^2 \|m_r - m_r^{\text{sp}}\|^2] &\leq CM_r^2 n^{-1/4} \end{aligned} \quad (3.4)$$

hold for the full mean vectors, written in spectral coordinates. The first bound also holds for the uncapped zero-field likelihood $L_\circ$. The pair-tail assertion at scale r also holds in that uncapped case.

Proof. Given the overlap $\rho = n^{-1}X \cdot X'$, the Haar angular integral of a cube pair equals that of a spherical pair. The difference is entirely in their overlap distributions. The replica Gram comparison of [21, Section 15.4, Equation (15.10)] bounds the ratio on overlap cells of width $2/n$ by $\exp(C + Cn\rho^4)$. Above C_1p and below a small fixed cutoff, the spherical pair tail costs $\exp(-an\rho^3)$; choosing the cutoff small pays for the comparison on all dyadic bands. At overlaps bounded away from zero, the normalized zero-field comparison has a strictly negative exponential rate, by [21, Section 15.4, p. 118]. A bounded field changes logarithmic weights by at most $Cn\rho^{5/2}$, which preserves that rate when the upper scale is small. Below C_1p the comparison factor is $\exp(C + Cn\rho^4)$. This proves (3.3) and its pair-tail assertion. The uncapped argument uses the corresponding uncapped spherical tail from [21, Section 15.3].

We give the finer cell estimate, now taking $p = r$ throughout the remaining comparison. On $|\rho| \leq C_1p$, Stirling's formula and the spherical overlap density show that the binomial mass of a cell and its spherical mass differ by a relative error

$$O(np^4 + p + n^{-1}).$$

It remains to compare the angular integrands within one cell. Write a spherical pair on an orthonormal sum-and-difference frame (q, q') . For $A = W_p$ their tilt exponent is

$$\frac{n}{2}((1 + \rho)q^\top Aq + (1 - \rho)q'^\top Aq') + \sqrt{2n(1 + \rho)} h^\top q.$$

Its oscillation on a cell is bounded by $C(|q^\top Aq - q'^\top Aq'| + p^{5/2})$. This is uniformly bounded, and its average against the normalized tilted spherical pair law on these cells is $O(p)$. For the latter assertion, the exact identity

$$q^\top Aq - q'^\top Aq' = \frac{2Y^\top AY' - \rho(Y^\top AY + Y'^\top AY')}{n(1 - \rho^2)}$$

and $\|A\| \leq C$ bound the diagonal terms by $C|\rho|$. It remains to show

$$E_{\text{sp}}|Y^\top AY'| \leq Cnp.$$

At a bounded cap field, apply exponential Markov with parameter cp^2 to either sign of $Y^\top AY'$. The saddle means have squared norm $O(np)$, the covariance trace-square is $O(n/p)$, and a sufficiently small c leaves a fixed precision margin. The Gaussian logarithmic moment generating function is therefore at most Ck_p . Conditioning both norms costs a constant by the joint norm-density bound in [21, Equation (15.6)]. Integrating the resulting tail, beginning at a sufficiently large multiple of np , proves the displayed estimate. For uncapped zero field, freeze a Gaussian parameter equal to a small fixed multiple of p^2 . The same moment generating function bound applies, and the deficit-filling lower density estimate in that subsection costs at most $\exp(Ck_p)$; this gives the same integrated bound.

For completeness, the passage from this averaged oscillation to comparison of angular integrals does not require a pointwise $O(p)$ derivative bound. The oscillation of the exponent on a cell is bounded by a fixed constant, so the mean value theorem bounds the difference of its exponentials by a constant times its integrated absolute derivative. The spherical overlap density varies by a bounded factor on a cell. Summing over cells is thus bounded by the preceding $O(p)$ integral. Slightly enlarging the overlap cutoff pays for boundary cells by the pair tail.

At $p = r$, $np^4 + p + n^{-1} = O(n^{-1/3+4\delta}) = O(n^{-1/4})$. The normalized pair integral consequently equals $1 + O(n^{-1/4})$, which proves the first part of (3.4), since $E_U L_r = 1$. Apply the same

comparison with the test $n\rho$. Its absolute value on the retained cells is at most $Cnp = CM_p^2$, and its change within one cell is bounded by 2. Hence

$$E_U \|L_r m_r\|^2 = \|m_r^{\text{sp}}\|^2 + O(M_r^2 n^{-1/4}).$$

The first-moment identity $E_U[L_r m_r] = m_r^{\text{sp}}$ turns this into an L^2 bound on $L_r m_r - m_r^{\text{sp}}$. Finally,

$$L_r(m_r - m_r^{\text{sp}}) = (L_r m_r - m_r^{\text{sp}}) - (L_r - 1)m_r^{\text{sp}},$$

and $\|m_r^{\text{sp}}\| \leq CM_r$ proves the second part of (3.4). $\square$

Corollary 3.3 (Cube edge marginals). *Jointly with the spectral variables, in the sense of Remark 2.2, the quenched cube integrals of continuous cylinder functions of polynomial growth converge to their integrals under Π . In particular, all fixed moments of any fixed finite set of scaled edge coordinates converge.*

Proof. In the uncapped comparison, insert a bounded Lipschitz cylinder function of the scaled spectral coordinates in each replica numerator. Changing the overlap within a cell moves each scaled coordinate by $O(n^{-5/6})$; the preceding weight comparison therefore remains valid. The variance over Haar frames of the numerator normalized by Z_{sp} tends to zero. Together with $L_o \rightarrow 1$ and Proposition 2.3, this proves convergence of the quenched cube cylinder integrals to Π . Polynomially growing continuous tests follow by the higher moment bounds and truncation. $\square$

Corollary 3.4 (Mean moments and the initial score). *For every fixed $j < \infty$, after reducing the fixed upper scale if needed,*

$$\mathbf{E}(\|m_p\|/M_p)^j + \mathbf{E}(\|m_p^{\text{sp}}\|/M_p)^j \leq C_j. \quad (3.5)$$

At the smallest scale,

$$\mathbf{E} \frac{\|m_r - m_r^{\text{sp}}\|^2}{M_r^2} \leq Cn^{-1/20}, \quad \mathbf{E} \|\nabla_{u_r} \log L_r\|^4 = o(1). \quad (3.6)$$

Proof. Lemma 3.1 confines the random field to a bounded range except with power-exponential probability. At a fixed such field, the Haar-averaged pair-tail numerator divided by the squared spherical partition function is at most Ce^{-ak_p} , by Lemma 3.2. Under the biased frame law $L_p dU$, the set $L_p < e^{-ak_p/4}$ has mass at most $e^{-ak_p/4}$. On its complement, the expectation of the posterior pair-tail probability is at most $e^{ak_p/4}$ times that pair-tail integral. A further Markov bound gives an exponentially small posterior pair tail outside an exponentially small biased exception. Since $\|m_p\|^2 = \mu_{p,h}^{\otimes 2}[X \cdot X']$, this implies $\|m_p\|^2 \leq Cnp$ there. The deterministic bound $\|m_p\| \leq \sqrt{n}$ pays for all exceptions and proves the cube part of (3.5). The spherical part follows by the same argument.

For the first estimate in (3.6), split according to $L_r \geq n^{-0.12}$. On this event (3.4) bounds the biased second moment by $CM_r^2 n^{-0.13}$. The complementary biased probability is at most $n^{-0.12}$; Cauchy–Schwarz and the fourth moment in (3.5) give $CM_r^2 n^{-0.06}$. The field exceptions are negligible. Finally

$$\nabla_{u_r} \log L_r = n^{1/3} r^2 P_r(m_r - m_r^{\text{sp}}), \quad n^{1/3} r^2 M_r = n^{(5/2)\delta}.$$

Interpolation of the decaying second moment with an arbitrarily high fixed moment from (3.5) proves the fourth-moment assertion. $\square$

3.3 Likelihood dissipation and covariance errors

The smallest-scale comparison controls a single observation. At larger scales the useful quantity is the score accumulated along the path. The Gaussian observation construction is a form of stochastic localization [15]; here, conditional on the disorder, the precision follows the deterministic spectral schedule B_p . We derive its likelihood identity below and combine it with the replica estimates just proved. Write $a_p^{\text{err}} = P_p(m_p - m_p^{\text{sp}})$.

Lemma 3.5 (Integrated score). *There is a fixed $\tau > 0$ such that, on each scale interval of a fixed bounded ratio about q ,*

$$\int \|a_p^{\text{err}}\|^2 d(p^2) \leq Ck_q(q^\tau + k_q^{-\tau}) \quad (3.7)$$

outside a set of $\mathbf{P}$ -probability at most $C \exp(-n^\xi)$. The bound holds simultaneously on a geometric list of intervals, with fixed enlargements, and also holds in expectation.

Proof. Use $t = p^2$ as the channel parameter and take the filtration generated by the eigenframe and observations through time t . In particular, every disorder event is measurable at time zero. Between coordinate activations, $\dot{B}_t = P_p$, and the Gaussian likelihood identity gives

$$d \log L_p = a_p^{\text{err}} \cdot d\mathcal{B}_t + \frac{1}{2} \|a_p^{\text{err}}\|^2 dt,$$

where $\mathcal{B}$ is the innovation Brownian motion in the active subspace. The bracket of the martingale is the score integral. There is no singular term at an activation because the added precision starts at zero. These statements also follow by Itô differentiation of the posterior partition function, including the compensating interaction change $-dB_t$.

Put $H_q = Ck_q(q^\tau + k_q^{-\tau})$, with τ small and fixed. At the first endpoint, the likelihood-density identity gives $\mathbf{P}(\log L_p < -H_q) \leq e^{-H_q}$. At the last endpoint, (3.3) gives

$$\mathbf{P}(\log L_p > H_q, h_p \text{ in the bounded range}) \leq \exp(-H_q + C + Ck_q q).$$

Choose C large and use Lemma 3.1 outside the range. An exponential martingale inequality now bounds the event that the bracket exceeds a sufficiently large multiple of H_q while these endpoint inequalities hold. This proves (3.7). The logarithmic number of intervals is absorbed by the power-exponential bound. Both means have norm at most $\sqrt{n}$, so the integral has a polynomial deterministic bound; integrating the exception proves the expectation assertion. $\square$

Let $\mathcal{G}_n$ be the intersection of the disorder events on which the fixed-tolerance annular covariance bounds of Proposition 2.6, the static value and projected-mean comparisons of Lemma 2.7, and the associated projection tails hold on the full scale and fixed field ranges used below. All tolerances and ranges are fixed before taking $n \rightarrow \infty$; their finite intersection has probability tending to one. The next estimate is asserted on $\mathcal{G}_n$; its exceptional probability is measured before any conditioning on that event.

Lemma 3.6 (Integrated covariance error). *Put $D_p^K = p^2 K_p - \text{Id}$ on $\mathcal{H}_p$. After decreasing $\tau > 0$ if necessary,*

$$\int_q^{2q} \frac{\|D_p^K\|_F^2}{k_p} d \log p \leq C(q^\tau + k_q^{-\tau}) \quad (3.8)$$

outside an event whose intersection with $\mathcal{G}_n$ has $\mathbf{P}$ -probability at most $C \exp(-n^\xi)$. Fixed enlargements of the interval are allowed, and the assertions hold simultaneously on a geometric list.

Proof. On the shrinking annulus of Lemma 3.1, the spherical covariance has norm $O(p^{-2})$. By rotation invariance within the flat cap its restriction to $e^\perp$, $e = h_p/\|h_p\|$, is scalar. The frozen Gaussian linear-tilt bound gives an $O(p^{-1})$ error between the spherical radial mean and the saddle mean. Dividing by $\|h_p\| \asymp p^2 M_p$ shows that the transverse covariance differs from $(p^2 + \alpha_p)^{-1} \text{Id}$ by $O(p^{-2} k_p^{-1/2})$. The remaining radial covariance is rank one. Thus its contribution, and the shrinking saddle parameter, are already bounded by the right side of (3.8).

It remains to compare K_p and K_p^{sp} . The spherical posterior, evaluated along the cube observation path, has innovation drift $K_p^{\text{sp}} a_p^{\text{err}} dt$. Subtracting the two posterior mean equations gives

$$da_p^{\text{err}} = (K_p - K_p^{\text{sp}}) d\mathcal{B}_t - K_p^{\text{sp}} a_p^{\text{err}} dt$$

between activations. Multiply $p^2 \|a_p^{\text{err}}\|^2$ by a smooth nonnegative cutoff supported on an enlarged scale interval and equal to one on $[q, 2q]$, with bounded derivative in $\log p$. Itô's formula and integration by parts give

$$\int_q^{2q} \|p^2(K_p - K_p^{\text{sp}})\|_{\text{F}}^2 d\log p \leq C \int_{\text{enlargement}} \|a_p^{\text{err}}\|^2 d(p^2) + C|\mathcal{M}|.$$

Activations add squared new mean coordinates with nonnegative coefficients, so they may be discarded on this side of the inequality. The cutoff is zero at both endpoints; the displayed energy identity is applied over the entire interval, not at a stopping time. Its martingale is

$$\mathcal{M} = 2 \int \chi(t) t \langle a_{\sqrt{t}}^{\text{err}}, (K_{\sqrt{t}} - K_{\sqrt{t}}^{\text{sp}}) d\mathcal{B}_t \rangle.$$

For concentration, stop this stochastic integral alone at the first failure of the annular covariance bounds, and set its integrand to zero on $\mathcal{G}_n^c$, a time-zero event. Its bracket is then bounded by $C \int \|a_p^{\text{err}}\|^2 d(p^2)$. The stopped integral agrees with $\mathcal{M}$ on $\mathcal{G}_n$ and the whole-path annulus event. The martingale inequality, Lemma 3.5, and Lemma 3.1 therefore prove the assertion. Division by $k_q \asymp k_p$ gives the stated normalization. $\square$

We finish with an angular estimate that will identify derivatives of the limiting measure. This estimate uses the remaining Haar randomness without restricting first to $\mathcal{G}_n$.

Lemma 3.7 (Angular score). *For a cap field h , write $\tilde{h} = U^\top h$ and $\tilde{\phi}_p(\tilde{h}) = \phi_p(U\tilde{h})$. Let A be a fixed skew-symmetric matrix supported on finitely many leading spectral coordinates, and set $\mathcal{R} = (A\tilde{h}) \cdot \nabla_{\tilde{h}}$. There is $\tau' > 0$ such that*

$$\int_q^{2q} \frac{\mathbf{E} |\mathcal{R} \tilde{\phi}_p(\tilde{h}_p)|^2}{n^{2/3} p^2} d\log p \leq C_A (q^{\tau'} + k_q^{-\tau'}). \quad (3.9)$$

Proof. The coordinates on which A acts belong to the cap for all sufficiently large n . By (2.9), the joint density of the frame and spectral observation relative to independent Haar and the spherical observation law is L_p . Condition on the physical capped matrix W_p , the physical vector h_p , and the spectral coordinates of that vector. The spherical observation density is now fixed, and L_p depends only on the fixed physical data and spectrum. The remaining rotation of the spectral frame in $e^\perp \cap \mathcal{H}_p$ is therefore exactly Haar. In physical coordinates, the differentiation direction is $UA\tilde{h}_p$, which is orthogonal to h_p and has length $|A\tilde{h}_p|$. The spherical mean in the cap is radial. Haar averaging consequently gives

$$\frac{\mathbf{E} |\mathcal{R} \tilde{\phi}_p(\tilde{h}_p)|^2}{n^{2/3} p^2} \leq C \mathbf{E} \left[|Au_p|^2 \frac{p^2 \|a_p^{\text{err}}\|^2}{k_p} \right].$$

Apply (3.7) in expectation and interpolate with the higher moments in (3.5) and the coordinate moments following Lemma 3.1. Hölder's inequality on probability times logarithmic scale gives (3.9), with a smaller positive exponent. No event depending on the unused spectral refinement was inserted before the Haar average. $\square$

4 Conditional variance at the exact cap scale

The observation removes the part of a test that depends on the leading spectral coordinates. We now show that its remaining variance costs at most Cp^{-2} times the original heat-bath energy. The exponent 2 is essential: integrating a covariance bound $(1 + \varepsilon)p^{-2}$ with a fixed $\varepsilon > 0$ would instead produce a power loss. The integrated estimates of Section 3 allow us to replace this fixed error by one that is integrable in logarithmic scale.

Proposition 4.1 (Conditional variance). *There is a fixed $p_* > 0$ with the following property. For every $\varepsilon > 0$ there are finite constants $C, \xi > 0$ and disorder events $\mathcal{V}_n$ with $\liminf_n \mathbb{P}(\mathcal{V}_n) \geq 1 - \varepsilon$ such that, on $\mathcal{V}_n$, simultaneously for every real function f on the cube and every $r \leq p \leq p_*$,*

$$V_p(f) := Q_\mu[\text{Var}(f \mid h_p)] \leq Cp^{-2}\mathcal{E}(f) + Ce^{-n^\xi}\|f\|_\infty^2. \quad (4.1)$$

The constants and the disorder event are independent of f . The inequality also holds componentwise and may be summed for vector-valued functions.

In particular, the remainder is negligible for any fixed polynomial bound on the supremum norm, including for disorder-dependent tests. We first prove the requisite covariance improvement, then establish a variance bound at a fixed upper scale, and finally integrate the posterior variance identity.

4.1 A sharper upper covariance bound

We work initially on the disorder event $\mathcal{G}_n$ of Section 3. Its fixed tolerances will be chosen below; no tolerance depends on n . Probabilities in the next proposition are measured under $\mathbf{P}$ before restriction to this event.

Proposition 4.2 (Positive covariance error). *There are fixed $a, \xi, C > 0$ and a sufficiently small fixed $p_* > 0$ such that, for each deterministic $p \in [r, p_*]$ away from the activation scales,*

$$p^2 K_p \preceq (1 + C(p^a + k_p^{-a})) \text{Id} \quad (4.2)$$

outside an event $\mathcal{N}_{n,p}$ satisfying $\mathbf{P}(\mathcal{N}_{n,p} \cap \mathcal{G}_n) \leq Ce^{-n^\xi}$. The exceptional event may depend on p ; its probability bound has common constants for the entire scale range. By Fubini's theorem the same bound holds after integration in $d \log p$, with a possibly smaller ξ .

The gain comes from comparing a small cap to a larger one. The radial covariance deficit permits us to put all positive errors in the tangent space of the parent field. Their restriction to the much smaller child space is then controlled by the unused Haar rotation.

Fix a small constant $\gamma > 0$, a child scale c , and a parent scale p such that $x = c/p \in [\gamma, 2\gamma]$. Put $\Delta = B_p - B_c$. Conditional on h_c , the parent field has the transition law

$$d\mathbb{T}_{c,p,h_c}(y) = \frac{e^{\phi_p(y)} dN(h_c, \Delta)(y)}{\int e^{\phi_p(z)} dN(h_c, \Delta)(z)}.$$

The Gaussian is taken on the positive-increment space. The observation precision increment, compressed to the child space, is $\Delta_c = (P_c \Delta P_c)|_{\mathcal{H}_c} = (1 - x^2)p^2 \text{Id}$. Writing $[M]_{\mathcal{H}_c}$ for this compression of any operator M , differentiation of the Gaussian convolution gives

$$K_c = \Delta_c^{-1} [\text{Cov}_{T_{c,p,h_c}}(h_p)]_{\mathcal{H}_c} \Delta_c^{-1} - \Delta_c^{-1}. \quad (4.3)$$

Lemma 4.3 (An annular extension for the transition). *Choose $\eta > 0$ sufficiently small compared with γ^2 , then the coarse curvature tolerance ε_0 sufficiently small compared with η . The annulus width s_0 and the fixed upper scale may be chosen so that the following holds on $\mathcal{G}_n$. For every child field with $|\alpha_c(h_c)| \leq s_0 c^2$, the parent potential has an extension $\bar{\phi}_p$ with upper Hessian bound $(1 + \eta)p^{-2} \text{Id}$ that agrees with ϕ_p near $|\alpha_p| \leq s_0 p^2/2$. Under both the true transition and the transition obtained by using $\bar{\phi}_p$, the complement of this agreement region has probability at most $Ce^{-a_1 k_p}$. The same bound holds with any fixed polynomial moment of $\|h_p\|/(p^2 M_p)$ inserted, after decreasing $a_1 > 0$.*

Proof. We give the construction and its tail comparison, following the cap extension method of [21, Proposition 16.1]. On $|\alpha_p| \leq 3s_0 p^2/4$, take the infimum of the quadratics

$$\phi_p(z) + \nabla \phi_p(z) \cdot (y - z) + \frac{1 + \eta}{2p^2} \|y - z\|^2.$$

Each of these supports lies above ϕ_p on that smaller annulus. Choose s_0 small compared with η and γ^2 . The radial buffer between the smaller and larger annuli then determines a fixed $\beta_{\text{buf}} > 0$ such that every segment starting in the smaller annulus and of length at most $\beta_{\text{buf}} p^2 M_p$ remains in the larger one. For such pairs y, z , the supporting inequality follows by integrating the coarse Hessian bound along their segment. For $d = y - z$ with $\|d\| \geq \beta_{\text{buf}} p^2 M_p$, freeze the spherical saddle at z . The frozen Gaussian bound bounds the spherical logarithmic increment by the quadratic with Hessian $(p^2 + \alpha_p(z))^{-1} \text{Id}$, its saddle gradient, and a bounded density error. The static value and gradient errors from Lemma 2.7 are at most $\epsilon_v k_p$ and $\epsilon_g M_p$. Relative to $\|d\|^2/p^2$, their combined contribution, including the density error, is at most

$$\frac{2\epsilon_v + C/k_p}{\beta_{\text{buf}}^2} + \frac{\epsilon_g}{\beta_{\text{buf}}}.$$

The frozen Hessian is at most $(1 + Cs_0)p^{-2} \text{Id}$. Thus, after choosing s_0 , then β_{buf} , then the static errors, the excess quadratic term dominates this residual for all sufficiently large n . This proves the supporting property, and the infimum is a semiconcave extension agreeing on the smaller annulus.

For the transition tails freeze instead the child saddle $\alpha = \alpha_c(h_c)$. As reference use the Gaussian transition tilted by the parent quadratic $\|y\|^2/(2(p^2 + \alpha))$ with no linear term. Writing $R_u = R_{u,\alpha}$, its law is characterized by

$$R_p h_p \sim N(R_c h_c, R_c - R_p), \quad R_c^{-1} = R_p^{-1} - \Delta.$$

Since the child saddle satisfies its norm constraint, the expected squared norm of h_p is $\ell_p(\alpha)^2$. Its covariance norm is at most $C\gamma^{-2}p^2$. Therefore deviations in radius by $vp^2 M_p$ cost

$$C \exp(-a_0 \gamma^2 k_p v^2), \quad v \geq C(\gamma \sqrt{k_p})^{-1}.$$

The radius $\ell_p(\alpha)$ lies strictly inside the parent agreement annulus, because $|\alpha| \leq s_0 c^2$.

Calibrate the reference quadratic to the spherical potential at this radius. On a bounded range of scaled radii the true potential is at most the reference quadratic plus $\epsilon k_p + O(1)$, with arbitrarily small fixed $\epsilon > 0$, by the frozen norm-density bound and static comparison. Outside a sufficiently large bounded range, the zero-field projection tail gives

$$\phi_p(y) - \phi_p(0) \leq Ck_p \left(1 + \left(\frac{\|y\|}{p^2 M_p} \right)^{6/5} \right).$$

This follows by integrating the tail $\exp(-ak_p v^6)$ against the linear field tilt. Its growth is absorbed by the reference quadratic. For the extension, use the support centered at the point of radius $\ell_p(\alpha)$ in the direction of y . At radial distance $vp^2 M_p$ its excess above the reference is at most

$$\epsilon k_p + \epsilon k_p v + C(\eta + s_0)k_p v^2 + O(1).$$

Choose $\eta + s_0$ small compared with γ^2 , and choose ϵ after the fixed gap to the boundary of agreement. These errors are strictly smaller than the Gaussian deviation cost outside agreement.

The normalizing integrals relative to the reference are bounded below by $\exp(-\epsilon' k_p)$ with any needed fixed $\epsilon' > 0$. Indeed integrate over a sufficiently thin fixed relative shell around $\ell_p(\alpha)$. Both potentials agree there, their values at the central radius are given by static comparison, and their gradients are bounded in units of M_p . Gaussian concentration gives a positive lower bound for the reference mass of this shell for large n . This proves the asserted transition tails; polynomial moments are absorbed by the same Gaussian deviation estimate and the subquadratic bound above. $\square$

The extension permits Brascamp–Lieb’s covariance inequality [7] to be used in the Gaussian convolution. After whitening, the transition has a fixed strict log-concavity margin depending on γ , because $\Delta \preceq (1 - x^2)p^2 \text{Id}$ and $\eta \ll \gamma^2$. If K denotes the extended Hessian, this inequality bounds its covariance by the expectation of

$$\Delta^{1/2}(\text{Id} - \Delta^{1/2} K \Delta^{1/2})^{-1} \Delta^{1/2}.$$

Mollification justifies this formula for the semiconcave extension. Lemma 4.3 then replaces the extended Hessian by the true one inside agreement and replaces the extended transition by the true transition, with exponentially small errors. The covariance comparison includes the means: Cauchy–Schwarz and the polynomial moment tail bound control the difference of first and second moments. In (4.3), only Δ_c is inverted outside this expression, so newly activated directions of arbitrarily small precision cause no loss.

Lemma 4.4 (Removing the radial error). *On the coarse annulus, let $D = p^2 K_p - \text{Id}$, $e = h_p / \|h_p\|$, $z = D_{\perp e}$, and define, on $e^\perp$,*

$$E_p = (D_{\perp\perp} + zz^\top / |D_{ee}|)_+,$$

extended by zero on e . Put $s_p^+ = \|D_+\|$. Then $D \preceq E_p$ and

$$\|E_p\| \leq C s_p^+, \quad \frac{\text{Tr } E_p}{k_p} \leq C \frac{\|D\|_F}{\sqrt{k_p}} + \frac{C}{k_p}. \quad (4.4)$$

Proof. The coarse radial deficit gives $D_{ee} \leq -\varepsilon_1 < 0$. For each transverse vector, maximize the quadratic form of D over its radial coordinate. The resulting form is $D_{\perp\perp} + zz^\top / |D_{ee}|$, proving $D \preceq E_p$. Since $K_p \succeq 0$ and $p^2 K_p \preceq (1 + \varepsilon_0) \text{Id}$, both D and the maximizing radial coordinate

for a unit transverse vector are bounded by constants. Comparing that vector with $D \preceq s_p^+ \text{Id}$ proves the norm estimate. The trace of the rank-one correction is bounded, while the trace of the positive part of $D_{\perp\perp}$ is at most $\sqrt{\dim \mathcal{H}_p} \|D\|_F$. This proves (4.4). $\square$

Define $E_p = 0$ outside the coarse annulus, and also whenever the local covariance inequalities needed in Lemma 4.4 fail. This definition uses only the physical parent data (W_p, h_p) and the spectrum. In particular, it is defined and bounded before imposing $\mathcal{G}_n$. On that event, the local failures never occur on the annulus. By choosing ε_0 small enough, we have $\|E_p\| \ll \gamma^2$.

Insert $K_p \preceq p^{-2}(\text{Id} + E_p)$ in the covariance comparison. The baseline $p^{-2} \text{Id}$ gives $c^{-2} \text{Id}$ in (4.3). For the correction, expand the inverse around $\text{Id} - \Delta/p^2$. Its inverse has norm at most x^{-2} ; because $\|E_p\| \ll x^2$, the whole positive correction is at most twice the first insertion in quadratic-form order. On the child subspace the two outer inverse factors are scalars. Multiplication by c^2 yields

$$s_c^+ \leq Cx^{-2} \mathsf{T}_{c,p,h_c}[\|P_c E_p P_c\|] + C_\gamma e^{-a_1 k_p} \quad (4.5)$$

for every child field in its annulus. The first constant is absolute.

Lemma 4.5 (Compression by the unused frame). *For $x = c/p \in [\gamma, 2\gamma]$,*

$$\|P_c E_p P_c\| \leq C \left(\frac{\text{Tr } E_p}{k_p} + x^3 \|E_p\| \right) \quad (4.6)$$

outside a set of $\mathbf{P}$ -probability at most $C \exp(-a_\gamma k_p)$. This is an unconditional assertion with respect to the unused spectral refinement; its exceptional set may afterwards be intersected with $\mathcal{G}_n$.

Proof. Condition on W_p , the physical field h_p , and its spectral coordinates, as in Lemma 3.7. The remaining frame in $e^\perp \cap \mathcal{H}_p$ is Haar, and E_p is fixed under this conditioning. For a uniformly rotating unit vector v in that space, the Gaussian representation of the uniform sphere gives

$$\mathbb{P} \left(v^\top E_p v > \frac{C(\text{Tr } E_p + t \|E_p\|)}{k_p} \right) \leq C(e^{-t} + e^{-c_0 k_p}).$$

The same bound applies to a fixed vector of norm at most one. Choose a constant-resolution net of the child unit sphere and project it onto $e^\perp$. Its cardinality is at most $\exp(C \dim \mathcal{H}_c) \leq \exp(Cx^3 k_p)$. With $t = C_2 x^3 k_p$, a sufficiently large C_2 pays for the net; taking γ sufficiently small also pays for the second exponential term. The net approximation for a positive quadratic form proves (4.6). $\square$

Proof of Proposition 4.2. Let $\hat{s}_p$ be s_p^+ on the full coarse annulus and zero outside it. We now restrict to $\mathcal{G}_n$. Average (4.5) over $c/(2\gamma) \leq p \leq c/\gamma$ with normalized logarithmic measure, and condition on W and the observation path through c . Combining (4.4) and (4.6), the operator-norm term has coefficient at most $Cx^{-2}x^3 \leq C\gamma$. Choose γ so that this coefficient, denoted by θ , is less than $1/2$. The trace term is controlled by Cauchy–Schwarz and the integrated Frobenius estimate (3.8). We obtain

$$\hat{s}_c \leq \theta \text{avg}_{c/(2\gamma) \leq p \leq c/\gamma} \mathbf{E}[\hat{s}_p \mid W, \mathcal{F}_c] + C_\gamma (c^b + k_c^{-b}) + \text{err}_c, \quad (4.7)$$

where $b > 0$, $\mathcal{F}_c$ is the observation filtration, and $\text{err}_c \geq 0$ satisfies

$$\mathbf{E}[\mathbf{1}_{\mathcal{G}_n} \text{err}_c] \leq C e^{-n^\varepsilon}$$

uniformly at deterministic scales. To verify the exceptional terms in this assertion, first apply (3.8) to the parent scale integral before taking its conditional expectation. Its failures and those of (4.6) have exponential probability and at most polynomial cost. Since W is included in the conditioning, the indicator of $\mathcal{G}_n$ commutes with that conditional expectation. No assertion of Haar independence conditional on $\mathcal{G}_n$ is used. If the child field is outside its annulus, the inequality holds by truncation.

Choose $a \leq b$ sufficiently small that $\theta\gamma^{-a} < 1$. For a sufficiently large C' , the function $b_0(c) = C'(c^a + k_c^{-a})$ dominates its parent average multiplied by θ , plus $C_\gamma(c^b + k_c^{-b})$, with a fixed margin. It also dominates the coarse constant bound at the upper scales where the parent interval no longer fits below p_* . Take positive parts above b_0 in (4.7), use conditional Jensen, and average over $\mathbf{P}$ with $\mathbf{1}_{\mathcal{G}_n}$. At each iteration the scale increases by at least $(2\gamma)^{-1} > 1$, so there are only $O(\log n)$ generations; their errors sum to at most $Ce^{-n^\xi} \sum_{j \geq 0} \theta^j$. Thus

$$\mathbf{E}[\mathbf{1}_{\mathcal{G}_n}(\hat{s}_c - b_0(c))_+] \leq Ce^{-n^\xi}.$$

Enlarging C' and applying Markov's inequality proves the asserted probability bound, since $b_0(c)$ is bounded below by an inverse polynomial on the scale range. Lemma 3.1 removes the truncation. All bounds were uniform at deterministic scales, so Fubini's theorem gives their integrated version. $\square$

4.2 The variance bound at a fixed upper scale

We next obtain the starting variance estimate from the fixed-scale continuation in [21, Section 19.4]. We record precisely why its entropy estimate applies to the perturbations needed here.

Lemma 4.6 (The fixed-scale starting bound). *On disorder events of probability tending to one, for a sufficiently small fixed $p_{\text{top}} > 0$ and every real function f on the cube,*

$$V_{p_{\text{top}}}(f) \leq C\mathcal{E}(f) + Ce^{-an}\|f\|_\infty^2.$$

The event and constants are independent of f .

Proof. The continuation in [21, Section 19.4] increases the observation precision from $B_{p_{\text{top}}}$ through finitely many cap intervals, a bridge to a classical field, and a bounded-duration continuation to a diagonal-interaction endpoint. For an initial change of spin law from μ to ν , let ν_s, μ_s be the two posteriors, let $l_s = D(\nu_s \|\mu_s)$, and put $H(s) = E_{Q_\nu} l_s$. On this continuation, the estimate used is

$$\dot{H}(s) \geq -CH(s) - e^{-an}. \quad (4.8)$$

This is [21, Equation (19.17)]. Its proof uses upper bounds on observation likelihoods and posterior entropy tails, rather than a support restriction on the posterior. We explain its application when the initial density lies between $1/2$ and 2 . Every observation likelihood then has the same bounds, every posterior density ratio lies between $1/4$ and 4 , and $l_s \leq \log 4$.

At the classical part, the ordinary posterior covariance is bounded on a fixed normalized field neighborhood outside an event of probability e^{-an} ; farther along the continuation it is bounded for every field. The linear entropy inequality in that neighborhood gives $\|m_\nu - m_\mu\|^2 \leq Cl_s$ whenever l_s is bounded: a trial field shift of size proportional to $\sqrt{l_s}$ fits the neighborhood. The bounded density ratios pay for the ordinary exceptional paths.

To propagate this mean estimate across a preceding cap or bridge interval, use the approximate gradient g_s furnished by the extended potential there. The transport identity [21, Lemma 18.3]

holds under either initial law. If U is its linear transport from the current time s to the end of the interval, subtraction gives, for a vector v in the current rate subspace,

$$\begin{aligned}\langle v, m_{\nu,s} - m_{\mu,s} \rangle &= E_{Q_{\nu,h}^{\text{cont}}} \langle Uv, m_{\nu,\text{end}} - m_{\mu,\text{end}} \rangle \\ &\quad + (E_{Q_{\nu,h}^{\text{cont}}} - E_{Q_{\mu,h}^{\text{cont}}}) \langle Uv, m_{\mu,\text{end}} - g_{\text{end}} \rangle.\end{aligned}$$

Here the two continuation laws live on the same path space, and the same path function U is used in both expectations. Its operator norm is bounded by a fixed constant on these fixed intervals. At agreement, the terminal approximation error in the tested subspace has bounded norm. Truncate it there. Data processing gives $D(Q_{\nu,h}^{\text{cont}} \| Q_{\mu,h}^{\text{cont}}) \leq l_s$, so Pinsker's inequality bounds the second term by $C\sqrt{l_s}\|v\|$.

The complements have exponentially small averaged squared cost. Indeed, for any nonnegative terminal cost W_1 , the bounded full-path and prefix likelihoods give both

$$E_{Q_\nu} E_{Q_{\nu,h}^{\text{cont}}} W_1 \leq C E_{Q_\mu} W_1, \quad E_{Q_\nu} E_{Q_{\mu,h}^{\text{cont}}} W_1 \leq C E_{Q_\mu} W_1.$$

Apply these inequalities to the polynomially weighted ordinary exceptions in the terminal approximation estimates of [21, Section 19.4]. Conditional Cauchy–Schwarz controls the first transport term by the terminal squared mean difference. Taking the supremum over unit v costs no dimension factor because all the bounds are norm bounds uniform in v . Since terminal expected posterior entropy is at most $H(s)$, backward induction through the finite list yields

$$E_{Q_\nu} \|P_{\text{rate}}(m_{\nu,s} - m_{\mu,s})\|^2 \leq CH(s) + Ce^{-an}.$$

The observation entropy identity [21, Lemma 19.2], with bounded precision rates on these intervals, now proves (4.8). This also proves that its constants apply uniformly to the bounded-density perturbations below.

Normalize $\|f\|_\infty \leq 1$, set $\eta = e^{-n^{1/3}}$, and take

$$\frac{d\nu}{d\mu} = \frac{1 + \eta f}{1 + \eta \mu f}.$$

Every path likelihood and every posterior density ratio is bounded by fixed constants. Taylor's formula for relative entropy gives, uniformly at every posterior,

$$c\eta^2 \text{Var}_{\mu_s}(f) \leq D(\nu_s \| \mu_s) \leq C\eta^2 \text{Var}_{\mu_s}(f).$$

Thus $H(s)$ is comparable to $\eta^2 V_s(f)$. Integrating (4.8) over a bounded duration and dividing by η^2 gives $V_{p_{\text{top}}}(f) \leq C V_{\text{end}}(f) + Ce^{-a'n}$. At the endpoint the posterior is a product measure, so variance tensorization and conditional-variance contraction give

$$V_{\text{end}}(f) \leq E \sum_i \text{Var}(f \mid X_{-i}, \text{endpoint observations}) \leq \sum_i \mu[\text{Var}(f \mid X_{-i})] = \mathcal{E}(f).$$

Rescaling f proves the lemma. □

4.3 Integration of the posterior variance identity

The passage from a posterior covariance bound to a variance bound uses the localization mechanism of [16, Lemma 9, arXiv version 2]. For the deterministic precision path considered here, it takes the following form.

Proof of Proposition 4.1. For fixed W , the posterior mean of f is a martingale in the observation filtration. Its diffusion coefficient, with parameter $t = p^2$, is $P_p \text{Cov}(X, f \mid h_p)$. Consequently

$$-\frac{d}{d(p^2)} V_p(f) = Q_\mu \|P_p \text{Cov}(X, f \mid h_p)\|^2 \leq Q_\mu [\|K_p\| \text{Var}(f \mid h_p)].$$

Integrate the exceptional probabilities in Proposition 4.2 in logarithmic scale, then apply Markov's inequality and Lemma 2.5. On disorder events of probability arbitrarily close to one, intersected with $\mathcal{G}_n$, their total ordinary path mass is at most Ce^{-n^ξ} . The same statement holds for the annulus exceptions used with the coarse bound between p_* and p_{top} . These events are chosen before, and independently of, f .

On the exceptional paths the elementary estimate $\|P_p \text{Cov}(X, f \mid h_p)\|^2 \leq Cn\|f\|_\infty^2$ suffices. On the remaining paths, put $\epsilon_p = C(p^a + k_p^{-a})$ below p_* and use the fixed coarse tolerance above it. Backward integration from p_{top} gives an amplification bounded by

$$\exp \left(\int_{p^2}^{p_{\text{top}}^2} \frac{1 + \epsilon \sqrt{t}}{t} dt \right) \leq Cp^{-2},$$

because

$$\int_r^{p_*} (p^a + k_p^{-a}) d \log p \leq C,$$

and the remaining scales form a fixed interval bounded away from zero. The amplification and the elementary exceptional-path costs are polynomial in n ; they are absorbed by decreasing ξ . Lemma 4.6 therefore gives (4.1). The same integrated exceptional event works for every lower endpoint p , so the assertion is simultaneous in scale as well as in the test. $\square$

5 The gradient on the conditioned edge measure

The variance estimate of Proposition 4.1 says that a function of bounded rescaled energy is almost determined by a sufficiently fine cap observation. We now identify the derivatives of the resulting functions of the edge coordinates. There are two issues. The measure Π is supported on a renormalized quadratic constraint, so its Sobolev domain must first be specified. We then show that the slopes of the observation smoothings converge to derivatives in this domain. The energy bound obtained here has a finite constant; Section 6 will determine the exact coefficient.

Throughout the first part of this section, fix a realization of the edge parameters for which Proposition 2.3 holds. Thus $l_a = (g_a + b)^{-1} > 0$, $\sum_a l_a = \infty$, and $\sum_a l_a^2 < \infty$. All assertions about Π below are deterministic for this realization.

5.1 Angular derivatives and the form domain

A *smooth cylinder function* is a smooth function of finitely many coordinates. We initially take it compactly supported in those coordinates. For a skew-symmetric matrix A with finitely many nonzero entries, set $\mathcal{R}_A = (Ax) \cdot \partial_x$. We say that $(F, G) \in L^2(\Pi) \times L^2(\Pi; \ell^2)$ satisfies the angular weak identities if

$$\int \psi(Ax) \cdot G d\Pi = - \int F \left\{ \mathcal{R}_A \psi - \psi \sum_a (Ax)_a (g_a + b) x_a \right\} d\Pi \quad (5.1)$$

for every such A and every smooth compact cylinder test ψ whose coordinate set contains the indices of A . Finite rotations preserve the quadratic constraint. Their action on the product

Gaussian density gives the second term in braces; the contribution involving b cancels by skew-symmetry. Consequently, smooth cylinders satisfy (5.1) with their Euclidean gradients.

Proposition 5.1 (The conditioned gradient). *The Euclidean gradient on smooth compact cylinder functions is closable from $L^2(\Pi)$ to $L^2(\Pi; \ell^2)$. Its closed graph consists exactly of the pairs satisfying (5.1). In particular, the vector G in that identity is unique; we denote it by ∇F .*

The closed form

$$\mathcal{E}_\Pi(F) = \int |\nabla F|^2 d\Pi$$

has kernel equal to the constants. Each coordinate x_a belongs to its domain and satisfies $\nabla x_a = e_a$, so that $\mathcal{E}_\Pi(x_a) = 1$.

We first verify closability and then prove the density assertion, which is the substantial part of the proposition. By the polar disintegration in Proposition 2.3, the law of the coordinates outside a sufficiently large fixed initial block is absolutely continuous with respect to their product Gaussian law. Under that product law,

$$\sum_{a \leq N} x_a^2 = \sum_{a \leq N} l_a + \sum_{a \leq N} (x_a^2 - l_a) \longrightarrow \infty \quad \text{almost surely.}$$

Thus $x \notin \ell^2$ also holds Π -almost surely. If G and G' both satisfy (5.1) for the same F , the coordinate rotations give

$$x_i(G_j - G'_j) = x_j(G_i - G'_i) \quad \text{for all } i, j,$$

outside one null set. Their difference is therefore proportional to x and must vanish because it belongs to ℓ^2 . The angular graph is closed: every test in (5.1) has bounded coefficients on its finite coordinate support, so the identity passes to strong limits in $L^2(\Pi) \times L^2(\Pi; \ell^2)$. A closed graph containing the cylinder gradient and having a unique vector component proves closability.

Reduction to a product chart. Let (F, G) satisfy (5.1). Conditional on the complement of a finite block, its coordinates lie on a sphere of fixed radius with a smooth positive tilted density. Disintegrating (5.1), including tests in outside coordinates, gives the ordinary tangential weak identities on almost every such sphere. The rotation fields span its tangent spaces. The usual weak chain and product rules therefore apply.

Smooth truncation first reduces to bounded F , with convergence of both F and G . Fix an initial block of size $m > 3$, and write its radius as R . We may next localize to an annulus $\epsilon < R < M$. Near zero, the marginal density bound of Proposition 2.3 gives $\Pi(R \leq \epsilon) = O(\epsilon^m)$. A cutoff with derivative $O(\epsilon^{-1})$ thus has energy $O(\epsilon^{m-2})$ when multiplied by bounded F . At infinity, a cutoff with derivative $O(M^{-1})$ has vanishing cost. The corresponding products with G converge by dominated convergence. It suffices to approximate each localized pair.

Write the first block as $R\vartheta$, with $\vartheta \in S^{m-1}$, and put $y = (x_i)_{i>m}$. The constraint gives

$$R^2 = C - S_y, \quad C = D(b) + \sum_{i \leq m} l_i, \quad S_y = \sum_{i > m} (y_i^2 - l_i).$$

Relative to uniform angle and the product Gaussian law of y , the conditioned measure has density proportional to

$$\mathbf{1}_{\{S_y < C\}} R^{m-2} \exp \left(-\frac{R^2}{2} \sum_{j \leq m} \frac{\vartheta_j^2}{l_j} \right).$$

On the chosen annulus this density is bounded above and below by positive constants. Decompose the first block of G into radial and transverse components G_R, G_ϑ , and let f be the expression of the localized F in this chart. With $a = -G_R/(2R)$, the finite-block identities give

$$\nabla_\vartheta f = RG_\vartheta, \quad \partial_{y_i} f - 2y_i a = G_i \quad (i > m). \quad (5.2)$$

To justify the weak tail identity, condition on the coordinates outside the union of the first block and $\{i\}$. On this $(m+1)$ -coordinate sphere, varying y_i at fixed angle has tangent vector

$$V_i = e_i - \frac{y_i}{R^2} \sum_{j \leq m} x_j e_j.$$

It is a smooth linear combination of the i - j rotation fields on $R > 0$, and its pairing with G is $G_i - (y_i/R)G_R = G_i + 2y_i a$. The conditional density is smooth and positive on each compact chart with $R > 0$, so multiplying tests by its reciprocal converts the tangential weak identity to the ordinary weak coordinate identity. Disintegration then gives (5.2) in the product chart, with Gaussian integration by parts for Gaussian tests. The radial cutoff permits extension by zero without a boundary term. The quantities f , a , $\nabla_\vartheta f$, and the compensated derivative vector $(\partial_{y_i} f - 2y_i a)_{i>m}$ are square integrable for angle times product Gaussian measure.

The domain question has now become an approximation problem for the compensated tail derivatives in (5.2). The following lemma supplies exactly this approximation; the angular variable will be handled afterward by ordinary spherical harmonics.

Lemma 5.2 (Approximation with a compensated derivative). *Let $\gamma = \bigotimes_{i \geq 1} \mathbf{N}(0, l_i)$, where $l_i > 0$, $\sum_i l_i = \infty$ and $\sum_i l_i^2 < \infty$, and let $S_y = \sum_i (y_i^2 - l_i)$. Suppose that $f, a \in \tilde{L}^2(\gamma)$ and $G \in L^2(\gamma; \ell^2)$ satisfy, coordinatewise in the weak sense,*

$$\partial_{y_i} f - 2y_i a = G_i.$$

Then (f, a, G) is a limit in these three spaces of triples

$$\left(P, \partial_s P, (\partial_{y_i} P)_i \right) \Big|_{s=S_y},$$

where $P = P(s, y_1, \dots, y_N)$ is a polynomial. The derivatives in this display treat s and the individual y_i as independent variables.

Proof. Expand f and a in the normalized product Hermite basis. We index the basis by finitely supported multi-indices β and write the coefficients as f_β, a_β . Gaussian integration by parts gives

$$(G_i)_\beta = \frac{\sqrt{\beta_i + 1}}{\sqrt{l_i}} f_{\beta+e_i} - 2\sqrt{l_i} \left(\sqrt{\beta_i} a_{\beta-e_i} + \sqrt{\beta_i + 1} a_{\beta+e_i} \right). \quad (5.3)$$

The formula is valid for weak derivatives: one first uses finite-coordinate Hermite tests with cutoffs, then removes the cutoffs by Cauchy–Schwarz.

Reduction to one chaos grade. Group the unknowns as pairs $(f^{(d)}, a^{(d-2)})$, where the superscript denotes Hermite chaos order and negative orders are omitted. The derivative of $f^{(d)}$ and the creation part of $2y_i a^{(d-2)}$ both have order $d-1$ and must be kept together. The remaining, annihilation part of $2y_i a^{(d)}$ has summed norm at most

$$\left(4 \sum_{i, \beta} l_i (\beta_i + 1) |a_{\beta+e_i}|^2 \right)^{1/2} \leq C\sqrt{d} \|a^{(d)}\|_2.$$

Here $\sup_i l_i < \infty$ follows from $\sum_i l_i^2 < \infty$. It follows from (5.3) that each fixed-grade pair has a square-summable compensated derivative.

These pairs also approximate the full triple in its graph norm. To see this without separating divergent creation terms, multiply the grade- d pair by e^{-sd} , with $s > 0$. Relative to damping the output of (5.3) by e^{-sd} in order $d - 1$, the only discrepancy is the annihilation term, with multiplier $e^{-s(d+2)} - e^{-sd}$. Its squared norm is bounded by

$$Cs^2 \sum_d de^{-2sd} \|a^{(d)}\|_2^2 \leq Cs \|a\|_2^2,$$

which tends to zero. For fixed s , truncating the damped grades then converges in all three norms. It remains to approximate one grade.

The array represented by a fixed grade. Fix d and the finite set O of coordinates with odd multiplicity. The relevant coefficients of f have indices $\mathbf{1}_O + 2j$, where j is a count vector with $|j| = h = (d - |O|)/2$; those of a use $|j| = h - 1$. Different parity sets are orthogonal in the norms of the common derivative-creation terms, so finitely many such sets suffice for approximation. If $h = 0$, only a finite-coordinate monomial is involved. Suppose henceforth that $h \geq 1$.

Set

$$c(j) = \prod_i l_i^{j_i} \frac{\sqrt{(2j_i)!}}{j_i!}, \quad f_{\mathbf{1}_O + 2j} = c(j)u(j), \quad a_{\mathbf{1}_O + 2j} = c(j)v(j).$$

View u and v as symmetric arrays on h and $h - 1$ ordered places. Give each place mass $\mu_i = l_i^2$. The coefficient norms are equivalent to the resulting product L^2 norms: the ordered-place multiplicities and the displayed factorials have ratios bounded above and below for fixed h and O . This comparison includes repeated indices.

For $i \notin O$, the identity

$$\frac{\sqrt{2j_i + 2}}{\sqrt{l_i}} c(j + e_i) = 2\sqrt{l_i} \sqrt{2j_i + 1} c(j)$$

shows that the common term in (5.3) is exactly a multiple of $u(j + e_i) - v(j)$. Its norm is therefore equivalent, again with fixed-grade constants, to the sum over places of

$$\int \sum_i l_i |u(i, i_2, \dots, i_h) - v(i_2, \dots, i_h)|^2 d\mu(i_2) \cdots d\mu(i_h).$$

The finitely many indices in O contribute bounded operators in the coefficient norms, as does the annihilation term at this fixed grade. We have thus reduced the approximation problem to symmetric arrays with a common one-place trace v .

The sum of the one-place forms. For one place, take the base Hilbert space $\mathfrak{H} = \ell^2(\mu)$. We include the scalar trace in the form norm by defining

$$D(q) = \{u \in \mathfrak{H} : \text{there exists } v \in \mathbb{R} \text{ with } u - v \in \ell^2(l_i)\},$$

$$q(u) = |v|^2 + \sum_i l_i |u_i - v|^2.$$

The trace v is unique because $\sum_i l_i = \infty$. This is a densely defined closed form. Indeed, finite-support sequences belong to $D(q)$; and a Cauchy sequence in the form norm has convergent traces

and convergent deviations, whose coordinatewise limits identify the trace and deviation of its limit in $\mathfrak{H}$. Replacing the tail of u by its trace proves that eventually constant sequences are a form core.

On $\mathfrak{H}^{\otimes h}$ take the *sum* of the h coordinate forms. By the slice description of each coordinate form, its norm on symmetric arrays is precisely, up to fixed factors, the norm of u , the norm of v , and the integrated deviations just obtained. In particular, it requires v to belong to the base $(h-1)$ -place Hilbert space; it requires no further form regularity of v .

For completeness, let A_q be the nonnegative selfadjoint operator associated with q . The commuting projections $\mathbf{1}_{[0,M]}(A_q)^{\otimes h}$ converge to the identity in the norm of the sum form. On their ranges the form is bounded by hM times the squared Hilbert norm. Algebraic tensors are dense there, and each factor can in turn be approximated in its one-place form norm by eventually constant sequences. This proves that such tensors are a core for the sum form. Symmetrizing preserves convergence and identifies the slice traces. We may consequently approximate (u, v) by arrays constant in each place outside a finite set. This argument does not require traces on intersections of two or more limiting faces.

Return to polynomials. The correspondence is already visible with one place and even parity. If $u_i = v$ outside a finite set E , the polynomial

$$P(s, y) = vs + \sum_{i \in E} (u_i - v)(y_i^2 - l_i)$$

has $P(S_y, y) = \sum_i u_i(y_i^2 - l_i)$ and $\partial_s P = v$. Its compensated derivatives vanish outside E . Thus the constant tail of a coefficient array is represented by the centered square sum. For several places, its powers give the corresponding tensor coefficients, as the following calculation shows.

Choose a finite set E containing O and all exceptional indices of one of these approximating arrays. A symmetric array constant in each place outside E is determined by its value U_j for every count vector $j = (j_i)_{i \in E}$ with $|j| \leq h$: the remaining $r = h - |j|$ places can have arbitrary indices outside E , including coincident ones. Define

$$S_{E^c}(s, y) = s - \sum_{i \in E} (y_i^2 - l_i), \quad M_j(y) = \prod_{i \in E} y_i^{\mathbf{1}_{O(i)} + 2j_i}.$$

The degree- d Hermite coefficients of $M_j(y)S_{E^c}(S_y, y)^r$ have the prescribed counts j in E . For outside counts k with $|k| = r$, their coefficient is

$$C_E(j) r! \prod_{i \notin E} l_i^{k_i} \frac{\sqrt{(2k_i)!}}{k_i!}, \quad C_E(j) = \prod_{i \in E} l_i^{(\mathbf{1}_{O(i)} + 2j_i)/2} \sqrt{(\mathbf{1}_{O(i)} + 2j_i)!}.$$

This follows directly from the multinomial expansion, and so includes repeated outside indices. Multiplying this polynomial by $U_j c(j)/(r! C_E(j))$, and summing over the finitely many j , realizes exactly the desired top-grade array u . Here $c(j)$ is restricted to E . Differentiation in s replaces $S_{E^c}^r$ by $r S_{E^c}^{r-1}$. Its top coefficients have the same limiting one-place trace, since $r(r-1)! = r!$. They therefore realize v , also by trace uniqueness.

Every such polynomial pair has a square-summable compensated derivative, since differentiation after subtracting $2y_i \partial_s P$ leaves only the finitely many individual-coordinate derivatives. All its moments are finite; the centered sum S_y has exponential moments in a neighborhood of zero. Induction on the grade now removes the lower Hermite grades of these polynomial pairs. More explicitly, their lower grades satisfy the same summability identities and, by induction,

already belong to the polynomial closure. Subtracting their approximants shows that the desired top-grade pair lies in that closure. The fixed-grade approximations and the preceding damping argument prove the lemma. $\square$

Completion of the proof of Proposition 5.1. We can now use Lemma 5.2. Work in the uniform-angle times product Gaussian norm, equivalent to the localized conditioned norm. First project onto finitely many spherical harmonics in ϑ ; the projections converge in the first angular Sobolev norm and commute with the tail identities. Apply the lemma to each resulting coefficient. This yields polynomials in S_y and finitely many y_i , with smooth angular coefficients, converging in f , a , the compensated derivatives, and the angular derivative.

Multiply these approximants by a smooth cutoff of S_y supported in a slightly larger radius annulus and equal to one on the original support. Under multiplication by $\chi(S_y)$ the scalar component becomes $\chi a + \chi' f$, while the compensated derivative becomes χG . This is continuous in all the norms under consideration. Finally substitute $S_y = C - R^2$. The approximants are now genuine cylinder functions in x , smooth because their first-block radii stay away from zero. Their Euclidean gradients have exactly the components in (5.2). Ordinary cutoffs in their finitely many other coordinates make them compactly supported, with convergence in the same norms. Removing the localizations proves equality of the cylinder closure and the angular graph.

It remains to identify the kernel. If $G = 0$, the restriction of F to almost every finite initial-block sphere is constant, since the sphere is connected. Thus F is measurable, modulo null sets, with respect to the complement of every sufficiently large initial block. This completed tail field is trivial. To check this directly under Π , eliminate one fixed initial block. Its tail law has an integrable density with respect to product Gaussian measure. Approximating this density and its product with any bounded observable by cylinder functions shows that events beyond increasing indices are asymptotically independent of that observable, uniformly over the events. An event in every completed tail field is consequently independent of itself and has probability zero or one. Hence F is constant. Conversely constants satisfy the angular identities with $G = 0$ and are in the closed domain by what we have proved.

The same characterization applies to x_a : it is square integrable by Proposition 2.3, and its angular weak gradient is e_a . This proves the last assertion. $\square$

5.2 Observation smoothings and their slopes

We return to the finite SK model and use the quenched subsequential convention of Remark 2.2. Write $T = n^{2/3}$. Consider functions $f = f_n$ such that

$$\mu|f|^2 + T\mathcal{E}(f) \leq C, \quad \|f\|_\infty \leq Cn^D$$

for a fixed exponent D . The bounds below hold on disorder sets of arbitrarily high probability, with finite constants depending on the set but not on f . Terms denoted by $o(1)$ absorb the exponentially small exceptional-path costs multiplied by these polynomial bounds. The fixed exponent D affects these vanishing errors, not the constants multiplying the energy.

For the cap observation h_p , define

$$F_p(h_p) = \mathbb{E}[f \mid h_p], \quad S_p f(X) = \mathbb{E}[F_p(h_p) \mid X], \quad J_p^f(X) = n^{1/3} \mathbb{E}[B_p \nabla_{h_p} F_p(h_p) \mid X].$$

All expectations here are under the ordinary observation law for the fixed disorder. The vector J_p^f is expressed in physical coordinates. Recall that $u_{p,a} = (U^t h_p)_a / (p^2 n^{1/3})$.

Lemma 5.3 (Smoothing estimates). *For $r \leq p \leq p_*$ in the range of Proposition 4.1,*

$$Q_\mu \|\nabla_{u_p} F_p\|^2 \leq CT\mathcal{E}(f) + o(1), \quad (5.4)$$

and

$$\begin{aligned} \|n^{1/3}(f - S_p f)\|_{L^2(\mu)} &\leq Cp^{-1}\sqrt{T\mathcal{E}(f)} + o(1), \\ \|n^{1/3}(D_i S_p f)_i - J_p^f\|_{L^2(\mu; \mathbb{R}^n)} &\leq Cp\sqrt{T\mathcal{E}(f)} + o(1). \end{aligned} \quad (5.5)$$

In particular, J_p^f is bounded in $L^2(\mu; \mathbb{R}^n)$.

Proof. On the active cap, $\nabla_{h_p} F_p = P_p \text{Cov}(X, f \mid h_p)$. Conditional Cauchy–Schwarz therefore gives

$$|\nabla_{h_p} F_p|^2 \leq \|K_p\| \text{Var}(f \mid h_p).$$

Combining the covariance bound (2.11), the annulus estimate (3.1), and the variance estimate (4.1), and multiplying by the squared change-of-variable factor $p^4 n^{2/3}$, proves (5.4). Conditional Jensen and $\|B_p\| \leq Cp^2$ also bound J_p^f . The first line of (5.5) follows from the same Jensen inequality applied to $f - F_p(h_p)$.

For the second line, condition on X and write the channel as $h_p = B_p X + B_p^{1/2} Z$, with Z standard Gaussian on the active space. Flipping spin i translates Z by $-2X_i B_p^{1/2} e_i$. Expand the Gaussian translation in Hermite tensors. The degree-one term in the half-difference is the i th component of J_p^f . For the higher degrees, subtract the constant $f(X)$ from the translated function. The total squared norm of the Hermite coefficients of $n^{1/3}(F_p(h_p) - f(X))$, averaged over X , is at most

$$Cp^{-2}T\mathcal{E}(f) + o(1)$$

by (4.1). The degree- j contraction into the vector indexed by i has operator norm at most

$$\frac{2^j}{\sqrt{j!}} \|B_p\|^{1/2} \left(\max_i (B_p)_{ii} \right)^{(j-1)/2}.$$

Indeed its Gram matrix, apart from the displayed scalar and spin signs, is a Schur power of B_p ; the Schur product bound gives the estimate. Since $\max_i (B_p)_{ii} \leq \|B_p\| \leq Cp^2$, summing degrees $j \geq 2$ costs $O(p^2)$ times the coefficient norm. This proves the second line of (5.5) in the unweighted vector norm. $\square$

5.3 Identification of the limiting derivative

The static convergence of cube cylinder integrals in Corollary 3.3 implies that a bounded $L^2(\mu)$ sequence has a subsequence with a weak cylinder limit $F \in L^2(\Pi)$. Concretely, this means $\mu[f\psi(x)] \rightarrow \Pi[F\psi]$ for a countable dense family of cylinder tests, and then for all such tests by their L^2 bounds. For fixed small $q > 0$, let a bar denote probability averaging in $d \log p$ over $[q, 2q]$.

Proposition 5.4 (Limit of the observation slopes). *Suppose f_n satisfies the preceding bounds and has weak cylinder limit F . Choose a subsequence on which $T\mathcal{E}(f_n)$ tends to its liminf. After further extraction in the quenched convention, there is a sequence $q_j \downarrow 0$ such that the spectral*

components of $\overline{J^f_{q_j}}$ have weak cylinder limits $G^{(j)}$ as $n \rightarrow \infty$. Every weak cluster point of $(G^{(j)})$ in $L^2(\Pi; \ell^2)$ equals ∇F . In particular, F belongs to the form domain of Proposition 5.1 and

$$\|\nabla F\|_{L^2(\Pi; \ell^2)} \leq C \left(\liminf_{n \rightarrow \infty} T\mathcal{E}(f_n) \right)^{1/2}. \quad (5.6)$$

The constant is the same for all sequences with the stated size bounds.

Proof. Pass first to a subsequence on which the energy converges to its liminf. Equation (5.4) bounds the weak spectral vector limits in $L^2(\Pi; \ell^2)$. To identify them, fix A and ψ as in (5.1). We will compare the angular pairing of the slopes at fixed cap scale to an integration by parts at the smallest scale r .

Define

$$T_p^A = Q_\mu[\psi(u_p)\mathcal{R}_A F_p(h_p)],$$

where the rotation acts on spectral coordinates. By (3.2) and (5.4),

$$\mu[\psi(x)(Ax) \cdot U^t J_p^f] - T_p^A = O((pn^{1/3})^{-1}) + o(1).$$

For the finitely many coordinates involved, $B_p/p^2 = I + O((pn^{1/3})^{-2})$. Replacing x by u_p in the test then uses Cauchy–Schwarz and (3.2); compact support of the test, or its fourth-moment version, controls the coordinate factors.

Transport between two scales. Let $r \leq c < p \leq 2c$, with all indices of A active at scale c . The increment covariance $B_p - B_c$ is scalar on those indices. Rotation therefore commutes with the Gaussian convolution expressing the observation bridge. Differentiating its numerator and denominator gives the exact identity

$$\mathcal{R}_A F_c(h_c) = \mathbb{E}[\mathcal{R}_A F_p(h_p) + (F_p(h_p) - F_c(h_c))\mathcal{R}_A \phi_p(h_p) | h_c]. \quad (5.7)$$

In the first term, replacement of $\psi(u_c)$ by $\psi(u_p)$ costs $O((cn^{1/3})^{-1}) + o(1)$, using (3.2) and (5.4). The other term is bounded in absolute value by

$$\|\psi\|_\infty V_c(f)^{1/2} \|\mathcal{R}_A \phi_p\|_{L^2(Q_\mu)}. \quad (5.8)$$

Here $\mathbb{E}[(F_p - F_c)^2] \leq V_c(f)$ follows from the posterior martingale and conditional variance decomposition.

Starting at each $p \in [q, 2q]$, descend by factors of two, using r as the final scale if necessary. The coordinate errors sum to $O((rn^{1/3})^{-1}) = O(n^{-\delta})$. For the score errors, (4.1) bounds (5.8), up to a fixed tight factor, by the root mean square of $\mathcal{R}_A \phi_p$ divided by $n^{1/3}p$. Average over the starting p . In every resulting parent window, (3.9) and Cauchy–Schwarz apply, including a partial final window by enlarging it to its fixed-ratio buffer. Under the biased law the expected contribution of a window of scale Q is at most

$$C_A(Q^{\tau'} + k_Q^{-\tau'})^{1/2}.$$

The comparison through $L_\circ \geq 1/2$ transfers this estimate to ordinary disorder integrals. Its sum is bounded by $C_A(q^{\tau'/2} + k_r^{-\tau'/2})$. Markov's inequality therefore shows that the averaged T_p^A and T_r^A differ by a quantity tending to zero in probability as $n \rightarrow \infty$ and then $q \downarrow 0$. The estimate is independent of the choice of f within the stipulated size and energy bounds.

Integration by parts at the smallest scale. The observation density in the active h_r coordinates is proportional to

$$\exp\left(\phi_r(h_r) - \frac{1}{2}h_r^t B_r^{-1} h_r\right).$$

In u_r coordinates the quadratic coefficient in direction i is $n^{2/3}r^4/B_{r,i}$. Its pairwise differences are

$$\delta_{ij} = n^{2/3}r^4 \frac{d_i - d_j}{B_{r,i}B_{r,j}} \longrightarrow g_i - g_j \quad (5.9)$$

for fixed leading indices. The common divergent part of the coefficient cancels in an angular derivative. Also ϕ_r^{sp} is radial on the cap, so that $\mathcal{R}_A \phi_r = \mathcal{R}_A \log L_r$. Its contribution to the integration-by-parts identity tends to zero by (3.6), the field-coordinate moments, and the same biased-to-quenched comparison. Cauchy–Schwarz applies because F_r is bounded in L^2 and the relevant score has vanishing L^2 norm after multiplication by the compact cylinder test.

For every function $b(h_r)$, $Q_\mu[F_r(h_r)b(h_r)] = Q_\mu[f(X)b(h_r)]$. We may therefore use (3.2) in the remaining terms of the integration-by-parts identity and then the weak cylinder convergence of f . Equations (5.9) and (5.1) have the same limiting quadratic coefficient, since adding the scalar b has no angular effect. We obtain

$$T_r^A \longrightarrow - \int F \left\{ \mathcal{R}_A \psi - \psi \sum_a (Ax)_a (g_a + b) x_a \right\} d\Pi.$$

Together with the preceding scale comparison, this proves that each weak limit G of the averaged slopes satisfies (5.1). Proposition 5.1 identifies it with ∇F , and weak lower semicontinuity in (5.4) proves (5.6).

Finally, these arguments can be realized simultaneously for the countably many scales and tests needed in the sequel. Represent the bounds by tight random constants on the high-probability disorder sets, and include the error variables in the subsequential realization of Remark 2.2. The errors tend to zero along a sufficiently rapidly decreasing outer sequence of scales. On the resulting limit samples, weak Hilbert-space extraction applies to any of the particular sequences f_n under consideration. This establishes the stated quenched form of the proposition. $\square$

6 Microscopic relaxation and the effective coefficient

The weak gradient constructed in Section 5 gives a coarse lower bound on the limiting energy. We now identify its exact coefficient. A correction that vanishes in L^2 can still alter the limiting rescaled Dirichlet energy. We describe these corrections using polynomials in the unscaled heat-bath operator H . Their limiting slope inner products are encoded by a deterministic measure ω on $[0, \infty)$. Its atom at zero will give the effective coefficient.

Throughout this section the couplings are Gaussian. Limits of quenched quantities use the joint spectral convention of Section 2. Assertions about countably many fixed polynomials and cylinder functions may consequently be imposed simultaneously after subsequence extraction. All polynomial degrees are fixed before n tends to infinity. Write

$$\langle A, B \rangle_{w,n} = \mu \left[\sum_i w_i A_i B_i \right], \quad \|A\|_{w,n}^2 = \langle A, A \rangle_{w,n}$$

for vectors of functions on the cube. For a polynomial Q and a fixed leading eigenvector v_a , define $A_a^Q = DQ(H)Y_a$. Its component $(A_a^Q)_i$ is independent of X_i .

6.1 The measure of microscopic energies

Let $\nu_{a,n}$ be the spectral measure of H on Y_a , and put $\omega_{a,n}(d\lambda) = \lambda \nu_{a,n}(d\lambda)$. This is a finite positive measure. Selfadjointness and the formula for the Dirichlet form give

$$\int QQ' d\omega_{a,n} = \langle A_a^Q, A_a^{Q'} \rangle_{w,n}, \quad (6.1)$$

$$\int \lambda^j d\omega_{a,n} = \mu \left[\sum_i w_i D_i Y_a D_i H^j Y_a \right]. \quad (6.2)$$

The factor λ is essential: $\omega_{a,n}$ records energies, rather than the variances recorded by the spectral measure of Y_a .

To specify the limiting measure through an experiment with deterministic spectral data, use only fixed microscopic moments as follows. Take deterministic midpoint quantiles $\lambda_{1,N} \geq \dots \geq \lambda_{N,N}$ of ρ_{sc} , and choose $y \in \mathbb{R}^N$ with $\|y\|^2 = N$ such that

$$\frac{1}{N} \sum_{a=1}^N y_a^2 \delta_{\lambda_{a,N}} \implies \frac{d\rho_{sc}(\lambda)}{2-\lambda}.$$

For example, take y_a^2 proportional to $(2 - \lambda_{a,N})^{-1}$ and normalize their sum to N . Draw U by Haar measure subject to $Uy = \mathbf{1}$, form $W = U\Lambda U^\top$, and define H, w_i, Y_1 as above, always omitting the diagonal of W from the molecular fields. The prescription for its j th moment, for each fixed integer $j \geq 0$, is

$$\int \lambda^j d\omega(\lambda) = \lim_{N \rightarrow \infty} \mathbb{E}_{U|Uy=\mathbf{1}} \left[\sum_{i=1}^N w_i D_i Y_1 D_i H^j Y_1 \Big|_{X=\mathbf{1}} \right], \quad c_* = \omega(\{0\}). \quad (6.3)$$

Proposition 6.1 proves that these limits exist and determine a finite positive measure; Theorem 6.8 proves that its atom at zero is strictly positive and is the coefficient in (2.7).

Proposition 6.1 (Microscopic moment measure). *There is a deterministic finite positive measure ω on $[0, \infty)$ whose moments are given by (6.3). For fixed leading indices a, a' , polynomials Q, Q' , and a bounded continuous cylinder ψ ,*

$$\mu \left[\psi(x) \sum_i w_i (A_a^Q)_i (A_{a'}^{Q'})_i \right] \longrightarrow \mathbf{1}_{\{a=a'\}} \Pi[\psi] \int QQ' d\omega. \quad (6.4)$$

For every $\varepsilon > 0$ one also has

$$\lim_{q \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\mu \left[\left\| P_q(wA_a^Q) - v_a \int Q d\omega \right\|^2 \right] > \varepsilon \right) = 0. \quad (6.5)$$

Here wA denotes entrywise multiplication. For some $c > 0$, $\int e^{c\sqrt{\lambda}} d\omega(\lambda) < \infty$. The measure is uniquely determined by its moments, and polynomials are dense in $L^2((1+\lambda)\omega + \delta_0)$.

The proof has two parts. A fixed-order Haar calculation gives the limiting contractions and their factorization. A separate estimate on iterated differences controls moment growth. This separation avoids applying a diagram expansion at a degree that increases with n .

6.1.1 Conditional Haar contractions

In the biased representation of Lemma 2.5, multiply row i of U by the sampled spin X_i . The spin is then $\mathbf{1}$, and, conditionally on $y = U^t X$, the rotation is Haar subject to $Uy = \mathbf{1}$. The spectral inputs are

$$\frac{1}{n} \sum_a \delta_{\lambda_a} \implies \rho_{\text{sc}}, \quad \frac{1}{n} \sum_a y_a^2 \delta_{\lambda_a} \implies \frac{\rho_{\text{sc}}(d\lambda)}{2 - \lambda},$$

together with bounded $\|\Lambda\|$, $\lambda_a \rightarrow 2$, and $y_a/\sqrt{n} \rightarrow 0$ for each fixed leading index. These inputs hold both for the deterministic data in (6.3) and for a biased Gibbs sample. For the latter, the weighted bulk limit is Lemma 2.4.

The case $Q = Q' = 1$ shows what the calculation must track. Since $D_i Y_a = v_{ia}$, the unrelaxed slope contraction is

$$\sum_i \text{sech}^2(h_i) v_{ia} v_{ia'} = \frac{1}{n} \sum_i \text{sech}^2(h_i) (\sqrt{n} v_{ia}) (\sqrt{n} v_{ia'}).$$

Thus each summand carries two marked eigenvector entries and the common normalization is $1/n$. Its limit can already be read from a finite number of rows. Put $u = y/\sqrt{n}$ and

$$m_n = u^t \Lambda u \longrightarrow 1, \quad s_n^2 = u^t \Lambda^2 u - m_n^2 \longrightarrow 1.$$

These values follow from $\int \lambda \rho_{\text{sc}}(d\lambda)/(2 - \lambda) = 1$ and $\int \lambda^2 \rho_{\text{sc}}(d\lambda)/(2 - \lambda) = 2$. The exact field decomposition is

$$U\Lambda y = m_n \mathbf{1} + \sqrt{n} U(\Lambda u - m_n u).$$

Conditionally on $Uy = \mathbf{1}$, the second term comes from a vector of length s_n rotated uniformly in $\mathbf{1}^\perp$. It therefore tends at each fixed row to a standard normal. The inner products of a marked axis e_a with u and with $\Lambda u - m_n u$ are, respectively, $y_a/\sqrt{n}$ and $(\lambda_a - m_n)y_a/\sqrt{n}$, both tending to zero. Consequently the joint fixed-row limit of the field and the marked entries is $(1 + Z, G_a, G_{a'}, \dots)$, with independent standard normals $Z, G_a, G_{a'}, \dots$ for distinct marked axes. The omitted diagonal W_{ii} tends to zero and does not change this limit.

The same conditional Haar argument at two distinct rows gives independent copies of these limits, with uniformly bounded fixed moments. Row exchangeability then makes the displayed average converge in L^2 to $\mathbf{1}_{\{a=a'\}} m_0$, where

$$m_0 = \mathbb{E}[\text{sech}^2(1 + Z)] = \omega([0, \infty)). \quad (6.6)$$

This is the energy before microscopic correction; the coefficient in the limiting dynamics is instead the atom $c_* = \omega(\{0\})$. For higher moments, a difference of a field function introduces a factor W_{ij} joining two sites. The graph calculation below keeps these matrix factors together with the marked row variables.

A *matrix edge* below means an entry of UMU^t , where M is a bounded spectral function of Λ or a projection onto one of finitely many marked spectral axes. A site vertex may carry a continuous function, of polynomial growth, of the field $(U\Lambda y)_i$ and of the entries $\sqrt{n} v_{ia}$ at the marked axes. In particular it may carry any fixed derivative of $\tanh$.

Here are precise definitions for the contraction statement. Set

$$B = \text{span}\{y/\sqrt{n}, \Lambda y/\sqrt{n}, e_a : a \text{ is marked}\}, \quad R = \text{Id} - P_B.$$

Choose an orthonormal basis $b_1, \dots, b_k$ of B , with $b_1 = y/\sqrt{n}$, and put $\xi_i = (\sqrt{n}(Ub_r)_i)_{1 \leq r \leq k}$. The allowed vertex factors are functions of ξ_i ; the field and marked entries above are linear combinations with convergent coefficients. If a finite multigraph $G = (V_G, E_G)$ has a symmetric matrix M_e at each edge and a vertex function Φ_v , its normalized contraction is

$$S_G = \frac{1}{n} \sum_{\kappa: V_G \rightarrow \{1, \dots, n\}} \prod_{e=\{u,v\} \in E_G} (UM_e U^t)_{\kappa(u), \kappa(v)} \prod_{v \in V_G} \Phi_v(\xi_{\kappa(v)}). \quad (6.7)$$

Loops and multiple edges are allowed. We also use the same expression with κ restricted to be injective. By a normalized spectral trace or a finite spectral path we mean, respectively,

$$\frac{1}{n} \text{Tr}(RM_{e_1} R \cdots M_{e_l} R), \quad b_r^t M_{e_1} R \cdots RM_{e_l} b_s. \quad (6.8)$$

The one-edge path is $b_r^t M_e b_s$. In particular, path endpoints include the field direction $\Lambda y/\sqrt{n}$, not just the spin direction. Only bounded-length words occur in any fixed moment of S_G .

Lemma 6.2 (Connected contractions). *Suppose the edge matrices have uniformly bounded operator norms and that the traces and paths in (6.8) converge for every fixed word. For a fixed connected G and fixed continuous vertex functions of polynomial growth, S_G converges in every fixed L^p under the conditional Haar law to a deterministic limit. The same holds for the sum restricted to injective site labels. Products of finitely many such contractions have the products of their limits.*

The conclusion applies to the contractions defining $\sum_i w_i (A_a^Q)_i (A_{a'}^{Q'})_i$ at the sampled spin, and to their quadratic contractions with an additional P_q edge, for $q > 0$ fixed. In the latter case the limit tends to zero as $q \downarrow 0$ if that edge is replaced by $P_q - P_{\text{mark}}$, where P_{mark} projects onto the marked spectral axes.

Proof. Separate all coincidences among the site labels; the V remaining site vertices have distinct labels. Condition on the image under U of the chosen orthonormal basis of B . The residual map between the orthogonal complements is Haar. The Gram matrix of the first two displayed vectors tends to $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, while the marked axes are asymptotically orthonormal and orthogonal to these two vectors. Thus the basis can be chosen with converging coefficients. For bounds, an arbitrary orthonormal basis suffices.

At any fixed number of distinct rows, its image entries multiplied by $\sqrt{n}$ converge jointly with all moments: the first column equals 1, and the remaining entries tend to independent standard normals. This follows by representing finitely many columns by independent Gaussian vectors projected off $\mathbf{1}$ and then orthonormalizing. At fixed graph and moment orders, products of vertex factors and row polynomials have a common polynomial growth bound. The bounds on higher row moments make these products uniformly integrable, so their expectations converge also when the continuous vertex functions are not polynomials. The vertex functions can therefore be retained in the calculation of residual Haar moments and integrated afterward.

We give the counting argument, using the orthogonal Haar integration formula [10, Section 3.2]. Suppose that splitting the edge endpoints between B and $B^\perp$ leaves e edges and $2b$ residual endpoints. The finite-block endpoints contribute $n^{-(e-b)}$ times a polynomial in the scaled row variables. Odd residual endpoint counts have zero expectation. For $2b$ endpoints the integration formula is a sum over a physical pairing π and a spectral pairing σ , whose coefficient has order

$$O(n^{-b-d(\pi, \sigma)}).$$

Here $d(\pi, \sigma)$ is b minus the number of loops in the paired union. This order also follows directly by inverting the pairing Gram matrix: after removing the diagonal power of n , the off-diagonal entries have orders given by the pairing distances. The leading inverse terms are paths of minimum total distance.

We spell out the spectral wiring represented by a pairing. Split each matrix edge into its two endpoints, declaring each endpoint either finite-block or residual. The $2b$ residual endpoints are paired by σ ; keep the original matrix edges as well. Each connected component of this endpoint diagram is then either a closed alternating cycle, or a path ending at two finite-block endpoints. Summing its spectral indices gives a trace or path of (6.8). Define $t(\sigma)$ to be the number of closed components of this diagram. The notation $t(\pi)$ means the number obtained from exactly this construction with π in place of σ .

Writing $P = \text{Id} - UP_B U^t$, a term of the conditional expectation therefore has the form

$$n^{-(e-b)} W_{g_{n-k}}(\pi, \sigma) \prod_{\{\alpha, \beta\} \in \pi} P_{\kappa(v_\alpha), \kappa(v_\beta)} \prod_{\text{cycles } c} \text{Tr}(M_c) \prod_{\text{paths } p} \langle b_{r_p}, M_p b_{s_p} \rangle,$$

times its vertex factors and a polynomial in the scaled rows. Here v_α is the physical vertex incident to endpoint α , and M_c, M_p abbreviate the compressed products just defined. There are $2e - 2b$ finite-block endpoints, each supplying a row entry $n^{-1/2} \xi_{i,r}$; this explains the factor $n^{-(e-b)}$.

The physical covariance P is the projection off the image of B . Each pairing of unequal site vertices thus costs another factor n^{-1} , with polynomially bounded factors in the scaled rows. Let D be the number of such pairs. Each trace costs at most Cn ; each path costs at most a constant. If the original graph has a components, then

$$t(\pi) \leq e + D - V + a, \quad t(\sigma) \leq t(\pi) + d(\pi, \sigma). \quad (6.9)$$

Indeed, add the D unequal-site pairing connections to the graph, and let $c \leq a$ be the resulting number of components. Its cycle rank is $e + D - V + c$. The residual trace circuits are edge-disjoint circuits in this enlarged graph, so $t(\pi) \leq e + D - V + c \leq e + D - V + a$. Open paths ending in B add no circuits. A pairing switch creates at most one new circuit, which proves the second inequality.

Before summing site labels, the resulting bound is

$$n^{-(e-b)} n^{-b-d(\pi, \sigma)} n^{-D} n^{t(\sigma)} = O(n^{-V+a}).$$

It compensates for the site sum and the a normalizing factors. Every surviving term has a limit: the normalized trace and path factors converge by hypothesis, the row expectations converge by Gaussian orthogonalization, and the rescaled leading Haar coefficients converge by the Gram-matrix expansion.

The same count proves concentration. In a product of normalized sums, identifying vertices in two original components saves a power of n , as does a physical pairing across components. If only the spectral pairing crosses components, consider the first cross-component switch along a shortest pairing path. Before it, each residual circuit or open path lies inside one original component. Joining two circuits loses a circuit; joining paths, or a path and a circuit, cannot create a circuit. This switch therefore gains at most zero circuits, instead of the one allowed by (6.9), and again saves a power of n . Surviving terms remain inside the original components. Their row limits and leading Haar coefficients factor; for the latter, shortest pairing paths cannot join distinct paired

components. Applying this factorization to even powers of centered sums proves convergence in every fixed L^p . The argument also gives bounds uniform over bounded spectral data.

We verify the graph representation for the heat-bath expressions. Repeatedly use the exact rule

$$D_i(FG) = F(X)D_iG + (D_iF)G(X^{(i)}).$$

A difference of an explicit spin identifies an existing site; a difference of a field function adds a factor W_{ij} joining the differentiating site to that field vertex. The induction invariant is that each summand obtained from one copy of Y_a has a connected set of site labels and exactly one marked factor. Initially its summand is $v_{ja}X_j$ on one site. In an application of H , a newly summed site must differentiate an existing factor: it either becomes an existing site or joins one by a field edge. The factor $X_i - t_i$ remains at that same site. Flip evaluations from the product rule introduce only field edges to sites already present. The final difference D_i has the same property, with i retained as the distinguished slope site. A term in which a new difference strikes no factor vanishes, so there is no unattached summation site. The slope inner product identifies the distinguished sites of two such pieces; a projected quadratic contraction joins them by a projection edge instead. Both are connected. Each piece carries one marked factor $v_{ja} = n^{-1/2}(\sqrt{n} v_{ja})$, so the two marks together supply the normalization $1/n$.

The shifts of fields caused by flips may be Taylor expanded to a sufficiently large fixed order. Under conditional Haar, the off-diagonal entries and centered diagonal entries are at most $n^{-0.49}$ outside a probability exponentially small in a positive power of n . Haar concentration on the complement of the prescribed spin direction has bounded Lipschitz constants, and the entry means are $O(n^{-1})$; a union bound gives the assertion. Each fixed expression has only polynomially many summands. A large enough fixed Taylor order therefore makes all remainders vanish in any prescribed fixed moment. Polynomial bounds absorb the exceptional event. For the diagonal in $h_i = (U\Lambda y)_i - W_{ii}$, Taylor-expand about $\text{Tr } \Lambda/n$ and retain the powers of $W_{ii} - \text{Tr } \Lambda/n$ as loop factors in the graph. Their edge matrix is $\Lambda - (\text{Tr } \Lambda/n) \text{Id}$; only the high-order Taylor remainder is discarded by the entrywise bound. This proves the required representation.

For these applications the edge matrices are powers of Λ , fixed spectral cutoffs, marked projections, and the scalar shifts just described. All hypotheses concerning (6.8) follow from the stated spectral inputs. Indeed, finite-rank removal changes a normalized trace by $O(n^{-1})$. Expanding each $R = \text{Id} - \sum_r b_r b_r^t$ in a path reduces it to finitely many products of ordinary mixed spectral inner products between $y/\sqrt{n}$, $\Lambda y/\sqrt{n}$, and the marked axes. Their limits are given by the two empirical spectral laws, the leading eigenvalue limits, and $y_a/\sqrt{n} \rightarrow 0$. At fixed q , the cutoff endpoint $2 - q^2$ is a continuity point of the limiting laws.

Finally set $R_q = P_q - P_{\text{mark}}$ in the extra edge. Normalized traces containing R_q tend to zero as $q \downarrow 0$, because the semicircle mass of $[2 - q^2, 2]$ vanishes. Finite paths also vanish: R_q kills the marked axes and, for every fixed k ,

$$\limsup_{n \rightarrow \infty} \|R_q \Lambda^k y/\sqrt{n}\|^2 \leq \int_{2-q^2}^2 \frac{\lambda^{2k}}{2-\lambda} \rho_{\text{sc}}(d\lambda) \rightarrow 0.$$

Paths containing finite-block compressions reduce to these statements by expanding those projections. There are only finitely many contractions at each fixed order. The limit order is $n \rightarrow \infty$ at fixed q , followed by $q \downarrow 0$. $\square$

The lemma gives deterministic limits for polynomial slope inner products at a sampled spin. A sign change of just one of two distinct marked axes preserves the limiting spin/field Gram

data and reverses the corresponding contraction, so its limit is zero. All fixed leading axes have eigenvalue limit 2 and identical limiting Gram data; hence the diagonal limit is independent of a . By (6.1), it depends on Q, Q' through QQ' .

The moment bounds in Lemma 6.2, together with Lemma 2.5, transfer convergence to constants from the biased representation to quenched Gibbs integrals. Bounded cylinder multipliers may then be inserted using Corollary 3.3. For (6.5), the lemma removes $P_q - P_{\text{mark}}$. The remaining coefficients are $v_{a'} \cdot (wA_a^Q)$, with diagonal limit given by the same moment functional applied to Q and off-diagonal limit zero. It remains to prove that the moment functional determines a measure.

6.2 Growth of the microscopic moments

Lemma 6.2 determines every fixed moment of the microscopic spectral measure. We next control their growth. This will show that those moments determine a measure on $[0, \infty)$ and that polynomials approximate its atom at zero in the weighted norm needed for the correctors. The estimate is deterministic: it uses only a bound on the interaction norm and small individual interaction entries.

For this subsection, let W be a real symmetric $n \times n$ matrix with zero diagonal, and let

$$\mu(X) = Z^{-1} \exp\left(\frac{1}{2} X^t W X\right), \quad X \in \{-1, 1\}^n.$$

Write $h_i(X) = \sum_j W_{ij} X_j$, $t_i = \tanh h_i$, and $w_i = 1 - t_i^2$. The half difference $D_i f$ is independent of X_i . Thus the operator $H = \sum_i (X_i - t_i) D_i$ satisfies

$$\mu[fHg] = \sum_i \mu[w_i D_i f D_i g]. \quad (6.10)$$

In particular, H is selfadjoint and nonnegative on $L^2(\mu)$. For an ordered tuple $S = (s_1, \dots, s_m)$ of distinct sites, set $D_S = D_{s_1} \cdots D_{s_m}$ and $w_S = \prod_{j \in S} w_j$, and define

$$N_m(f)^2 = \sum_{\substack{S \in \{1, \dots, n\}^m \\ \text{entries distinct}}} \mu[w_S |D_S f|^2].$$

The order matters only in this sum: each set of m sites occurs $m!$ times. We put $N_m = 0$ for $m > n$. Subsets of S in the proof below inherit the order of their positions, whereas unions inside a difference operator denote the corresponding collection of distinct sites.

Lemma 6.3 (Weighted difference recursion). *Fix $K < \infty$. Suppose that W is symmetric, has zero diagonal, and obeys*

$$\|W\| \leq K, \quad \epsilon_n := \max_{i,j} |W_{ij}| \leq n^{-49/100}.$$

There is $C = C(K) \geq 1$ such that, for each fixed integer $m \geq 1$, uniformly in these matrices and in real functions f on $\{-1, 1\}^n$,

$$N_m(Hf) \leq \sum_{l=1}^{m+2} (C^{m-l+3} (m+1)^{m-l+2} + o_{n;m}(1)) N_l(f). \quad (6.11)$$

Here $o_{n;m}(1)$ is deterministic, nonnegative, and tends to zero as $n \rightarrow \infty$ with m fixed.

Proof. We first separate the differences that strike a conditional mean from those that remain on f . The exact product rule is

$$D_j(FG) = FD_jG + (D_jF)G(X^{(j)}), \quad G(X^{(j)}) = G(X) - 2X_jD_jG(X).$$

If $i \in S$, applying D_i first to $(X_i - t_i)D_if$ gives D_if ; these terms therefore sum to mD_Sf . For $i \notin S$, the terms in which no difference strikes t_i give HD_Sf . Every remaining term, with its overall minus sign, is

$$(D_At_i) \prod_{j \in B} (-2X_j) D_{\{i\} \cup (S \setminus A) \cup B} f, \quad i \notin S, \quad \emptyset \neq A \subset S, \quad B \subset A. \quad (6.12)$$

Indeed, the iterated product rule first evaluates $D_{\{i\} \cup (S \setminus A)}f$ at X^A , and expanding the flips at the sites of A gives the displayed sum over B .

Fix the positions of A and B , and write $a = |A|$, $b = |B|$ and $l = 1 + m - a + b$. We estimate the sum over i of (6.12) in the norm obtained by summing $\mu[w_S \cdot |^2]$ over S . For positive derivative orders,

$$|\tanh^{(a)} u| \leq C^a a! \operatorname{sech}^2 u.$$

For example, Cauchy's formula on a fixed circle of radius less than $\pi/2$, applied to $\operatorname{sech}^2 z$, proves this inequality: on that circle $|\operatorname{sech}^2 z| \leq C \operatorname{sech}^2 u$ uniformly for real u . The difference integral for D_At_i consequently gives

$$|D_At_i| \leq w_i C^a a! \prod_{j \in A} |W_{ij}|. \quad (6.13)$$

The constant is uniform for each fixed m and all sufficiently large n , since all field shifts in that integral have absolute value at most $2m\epsilon_n$ and $|\log \operatorname{sech}^2 u - \log \operatorname{sech}^2 v| \leq 2|u - v|$.

Suppose first that $b \geq 1$, and select $j_0 \in B$. Apply Cauchy-Schwarz to the sum over i , using $\sum_i W_{ij_0}^2 \leq K^2$ for its first factor. The second factor contains w_i^2 , the squared derivative of f , and $\prod_{j \in A \setminus \{j_0\}} W_{ij}^2$. Put $R = S \setminus A$ and $A_0 = A \setminus B$. The weights $w_R w_B$ occur in both input and output norms and are retained. Bound the additional output weights w_{A_0} by one. Summing the sites of A_0 now costs at most $K^{2(a-b)}$, by row square sums; the remaining $b - 1$ factors at sites of B cost at most $\epsilon_n^{2(b-1)}$. Finally, $w_i^2 \leq w_i$ supplies the input weight at site i . Here only the distinctness restrictions involving the summed-out sites of A_0 are removed; the retained tuple (i, R, B) still consists of distinct sites. After summing that tuple and integrating, the resulting norm is at most

$$C^{a+1} a! \epsilon_n^{b-1} N_l(f), \quad b \geq 1. \quad (6.14)$$

Thus all terms with $b \geq 2$ have coefficients $o_{n;m}(1)$.

When $b = 0$, taking absolute values in the sum over i would lose the operator norm bound on W . Instead expand the difference integral once more:

$$D_At_i = \left(\prod_{j \in A} W_{ij} \right) \left\{ \tanh^{(a)}(h_i) - \tanh^{(a+1)}(h_i) \sum_{j \in A} W_{ij} X_j + R_{i,A} \right\}, \quad (6.15)$$

$$|R_{i,A}| \leq C_m w_i \epsilon_n^2.$$

The first correction has the indicated sign because the field in the integral is $h_i - 2 \sum_{j \in A} s_j W_{ij} X_j$, with each s_j uniform on $[0, 1]$.

Here is the operator estimate for the first two terms. Regard

$$(T_a u)_{j_1, \dots, j_a} = \sum_i \prod_{r=1}^a W_{ij_r} u_i$$

as a map from ℓ^2 of sites to ℓ^2 of ordered site tuples, initially allowing repetitions. Its Gram matrix is $(WW^t)^{\circ a}$, the entrywise a th power, so $\|T_a\| \leq K^a$. To see the norm bound directly, the entrywise product of two matrices is the compression of their tensor product to the span of the vectors $e_i \otimes e_i$; iterate this observation for WW^t . In each first-correction term, one factor W_{ij_r} is replaced by $W_{ij_r}^2$. The matrix $(W_{ij}^2)_{i,j}$ has row and column sums at most K^2 , hence operator norm at most K^2 . The same tensor-product bound therefore controls this corrected map by K^{a+1} . Multiplication by X_{j_r} on its output is an isometry.

Fix $R = S \setminus A$ and apply these operator estimates pointwise in X , before integrating against $w_R \mu$. Restrictions that the sites of A be distinct and avoid R are coordinate projections on the output of these maps. The restriction $i \notin R$ is a coordinate projection on the input, and $i \notin A$ is automatic from $W_{ii} = 0$. The output weights w_A may again be discarded. The common weight w_R remains, and the factors $\tanh^{(a)}(h_i)$ and $\tanh^{(a+1)}(h_i)$ are bounded multiples of w_i . Their multiplication on an input with squared norm $\sum_i w_i |D_{\{i\} \cup R} f|^2$ thus costs no inverse powers of w_i . Summing R proves the required bound $C^{a+1} a! N_l(f)$ for the first two terms of (6.15), after increasing C . For the remainder, signs may be discarded: the matrix with entries $\prod_{r=1}^a |W_{ij_r}|$ has squared Frobenius norm

$$\sum_i \left(\sum_j W_{ij}^2 \right)^a \leq n K^{2a}.$$

Its contribution is at most $C_m \sqrt{n} \epsilon_n^2 N_l(f) = o_{n;m}(1) N_l(f)$. We have proved

$$\text{norm of the fixed-position terms with } b = 0 \leq (C^{a+1} a! + o_{n;m}(1)) N_{1+m-a}(f). \quad (6.16)$$

It remains to control $HD_S f$. This term is measured with $w_S \mu$, so we require an operator estimate in precisely that measure. Fix S and write $\nu = w_S \mu$, without normalizing it. Denote its conditional means and variances by t'_i and w'_i , its heat-bath operator by $H' = \sum_i (X_i - t'_i) D_i$, and its first two difference seminorms by N'_1 and N'_2 , using the measure ν and weights w'_i . We claim that

$$\|HF\|_{L^2(\nu)} \leq C \{N'_2(F) + (1+m)N'_1(F)\}, \quad C^{-1} w_i \leq w'_i \leq C w_i. \quad (6.17)$$

The constants depend only on K , once n is sufficiently large for the fixed m .

The conditional field under ν is exactly

$$h'_i = h_i + \delta_i, \quad \delta_i = D_i \log w_S.$$

Since the first two derivatives of $\log \text{sech}^2 u$ are bounded, the nonnegative deterministic quantities

$$a_i = 2 \sum_{k \in S} |W_{ik}|, \quad L_{ij} = C \sum_{k \in S} |W_{ik} W_{jk}|$$

satisfy $|\delta_i| \leq a_i$ and $|D_j \delta_i| \leq L_{ij}$ for $i \neq j$. They obey

$$\|a\|_2 \leq 2mK, \quad \|a\|_\infty \leq 2m\epsilon_n, \quad \|L\| \leq C m K^2, \quad \left(\sum_j L_{ij}^2 \right)^{1/2} \leq C K a_i.$$

These statements follow respectively from the triangle inequality for columns of W and the expression of L as a sum of nonnegative rank-one matrices. The variance comparison in (6.17) follows from $|\delta_i| \leq a_i = o(1)$ uniformly in i . Taylor's formula for $\tanh$ and sech^2 , now at the tilted fields, also gives

$$\begin{aligned} D_i t'_j &= w'_j(W_{ij} + E_{ij}), & |E_{ij}| &\leq C(L_{ij} + W_{ij}^2), \\ |D_i w'_j| &\leq C w'_j r_{ij}, & r_{ij} &= |W_{ij}| + L_{ij}, \quad i \neq j. \end{aligned} \quad (6.18)$$

In deriving the first bound, the square of L_{ij} is absorbed into L_{ij} since its maximum tends to zero. The absolute error matrix is bounded entrywise by a nonnegative matrix of norm at most $C(1+m)$; indeed (W_{ij}^2) has norm at most K^2 . Also $\sup_i \sum_j r_{ij}^2 \leq C$ and $\max_{i,j} r_{ij} = o(1)$, for n large at fixed m .

For completeness, these coefficient bounds give the operator estimate without an independence assumption on the slopes. Put $F_i = D_i F$ and $F_{ij} = D_i D_j F$. The product rule gives the exact identity

$$D_i H' F = H' F_i + F_i - \sum_{j \neq i} (D_i t'_j)(F_j - 2X_i F_{ij}).$$

Use the version of (6.10) for ν to write

$$\begin{aligned} \|H' F\|_{L^2(\nu)}^2 &= N'_1(F)^2 + \sum_{i \neq j} \nu[w'_j D_j(w'_i F_i) F_{ij}] \\ &\quad - \sum_{i \neq j} \nu[w'_i F_i (D_i t'_j)(F_j - 2X_i F_{ij})]. \end{aligned}$$

In the last sum, the F_j term is bounded by $C(1+m)N'_1(F)^2$: apply the signed norm bound to W with input $(w'_i F_i)_i$, and the absolute matrix bound to (E_{ij}) . The F_{ij} term is bounded by $CN'_1(F)N'_2(F)$, by Cauchy–Schwarz and the row square bounds for r . Finally,

$$D_j(w'_i F_i) = w'_i F_{ij} + (D_j w'_i)(F_i - 2X_j F_{ij}).$$

The first summand contributes $N'_2(F)^2$. The other two summands cost, respectively, $CN'_1(F)N'_2(F)$ and $CN'_2(F)^2$, using (6.18) and the same row square bounds. Consequently

$$\|H' F\|_{L^2(\nu)}^2 \leq C\{N'_2(F)^2 + N'_1(F)N'_2(F) + (1+m)N'_1(F)^2\}.$$

Moreover $|t_i - t'_i| \leq C a_i w_i$, so pointwise Cauchy–Schwarz and $\|a\|_2 \leq C m$ show

$$\|(H - H')F\|_{L^2(\nu)} \leq C m N'_1(F).$$

These two bounds prove (6.17).

Apply (6.17) to $F = D_S f$ and sum squares over the ordered tuples S . A difference at a site already in S vanishes. For all other sites, the variance comparison replaces w'_i by w_i with a fixed factor. Appending one or two ordered sites to S enumerates exactly the ordered tuples in N_{m+1} or N_{m+2} ; there is no further factorial. Minkowski's inequality therefore yields

$$\left(\sum_S \mu[w_S |H D_S f|^2] \right)^{1/2} \leq C\{N_{m+2}(f) + (1+m)N_{m+1}(f)\}.$$

Together with the term $m D_S f$, this controls every contribution except the positional sums already bounded above.

There are $\binom{m}{a}$ choices of the positions of A , and $\binom{a}{b}$ choices for B . The terms with $b = 0$ have $l = m - a + 1$ and coefficients bounded by $C^{a+1}a!\binom{m}{a}$; those with $b = 1$ have $l = m - a + 2$ and coefficients bounded by $C^{a+1}a!\binom{m}{a}$. Use $a!\binom{m}{a} \leq m^a$ and absorb the factor $a \leq 2^a$ by increasing C . Both bounds, as well as the $HD_S f$ and $mD_S f$ contributions, fit the coefficient $C^{m-l+3}(m+1)^{m-l+2}$ in (6.11). The finitely many $b \geq 2$ terms and Taylor remainders combine into $o_{n;m}(1)$. $\square$

We now apply this estimate to the linear functions $Y_a = v_a^t X$, where $\|v_a\|_2 = 1$. Then $N_1(Y_a) \leq 1$ and $N_m(Y_a) = 0$ for $m \geq 2$. For each fixed $j \geq 0$, iterating (6.11) gives

$$\limsup_{n \rightarrow \infty} N_1(H^j Y_a) \leq C^j (2j + 2)^{2j}. \quad (6.19)$$

Here and below a larger constant $C = C(K)$ may be used. To verify the iteration explicitly, a nonzero terminal term follows indices $m_0 = 1, m_1, \dots, m_j = 1$, with $1 \leq m_{r+1} \leq m_r + 2$. Set $k_r = m_r - m_{r+1} + 2 \geq 0$. Then

$$\sum_{r=0}^{j-1} k_r = 2j, \quad \sum_{r=0}^{j-1} (k_r + 1) = 3j, \quad m_r + 1 \leq 2j + 2.$$

The product of the coefficients on this path is at most $C^{3j}(2j+2)^{2j}$. The number of paths is bounded by the number of weak compositions of $2j$ into j parts, $\binom{3j-1}{j-1} \leq 8^j$ for $j \geq 1$. Only finitely many fixed orders occur for a given j , so all $o_{n;m}(1)$ coefficients disappear in this iteration before j is allowed to increase. This proves (6.19); the case $j = 0$ is immediate.

Corollary 6.4 (Determination by moments and polynomial density). *Let the matrices and normalized linear functions above vary with n , and suppose the moments of the finite measures*

$$\omega_{n,a}(B) = \langle Y_a, H \mathbf{1}_B(H) Y_a \rangle_{L^2(\mu)}, \quad B \subset [0, \infty) \text{ Borel},$$

converge at every fixed nonnegative integer order. They converge weakly to a finite measure ω on $[0, \infty)$ with those moments. This measure is uniquely determined among measures on $[0, \infty)$ by its moments, and, for some $c > 0$,

$$\int e^{c\sqrt{\lambda}} d\omega(\lambda) < \infty.$$

Polynomials are dense in both $L^2(\omega)$ and $L^2((1 + \lambda)\omega + \delta_0)$.

Proof. The masses are at most one, and the spectral theorem and (6.10) give

$$\int \lambda^{2j} d\omega_{n,a}(\lambda) = N_1(H^j Y_a)^2.$$

The $j = 1$ bound gives tightness. Bounds at higher even orders give uniform integrability for each fixed power of λ , so every weak subsequential limit has the prescribed moments. In particular, (6.19) implies

$$\int \lambda^{2j} d\omega(\lambda) \leq C^{2j} (2j + 2)^{4j}. \quad (6.20)$$

Define the symmetric square-root lift ρ by

$$\int g(t) d\rho(t) = \frac{1}{2} \int \{g(\sqrt{\lambda}) + g(-\sqrt{\lambda})\} d\omega(\lambda).$$

Its $4j$ th absolute moment is bounded by the right side of (6.20). For an arbitrary integer $k \geq 1$, choose $4j$ between k and $k + 3$ and use Hölder's inequality. Since the mass is at most one, this gives $\int |t|^k d\rho(t) \leq (Dk)^k$ with a constant $D = D(K)$. The exponential series and $k! \geq (k/e)^k$ therefore show that $\int e^{c|t|} d\rho(t) < \infty$ for small enough $c > 0$. This is the claimed square-root exponential moment.

If another measure on $[0, \infty)$ has the same moments, its symmetric square-root lift has the same integer moments as ρ , and the same argument gives an exponential moment. The Fourier transforms of the two lifts extend analytically to a strip about the real axis. Their Taylor series at zero agree, so the identity theorem and uniqueness of Fourier transforms of finite measures identify the lifts, and hence the original measures. Every subsequential limit is therefore ω , proving weak convergence of the full sequence.

For density, let $g \in L^2(\rho)$ be orthogonal to all polynomials. Cauchy–Schwarz implies that the finite signed measure $g\rho$ has an exponential moment with a smaller positive exponent. All derivatives at zero of its Fourier transform vanish. The same analytic argument shows that this transform, and hence $g\rho$, is zero. Thus polynomials are dense in $L^2(\rho)$. Applying the bounded projection $p(t) \mapsto (p(t) + p(-t))/2$ shows that even polynomials are dense in its even subspace. The isometry $f(\lambda) \mapsto f(t^2)$ identifies this subspace with $L^2(\omega)$, proving the first density assertion. Finally, the symmetric square-root lift of $(1 + \lambda)\omega + \delta_0$ is $(1 + t^2)\rho + \delta_0$. It also has an exponential moment after decreasing c , and the same proof establishes the second density assertion. $\square$

Completion of the proof of Proposition 6.1. For the Gaussian matrices in this section, removing the diagonal preserves a bounded operator norm and the entry bound used in Lemma 6.3, on events whose probability tends to one. The fixed-order contractions give moment convergence there. Corollary 6.4 therefore constructs the unique positive measure with those moments and proves the exponential bound and polynomial density. The moments are deterministic and the same for every fixed leading index. Applying the conditional Haar calculation to the deterministic spectral data of (6.3) gives precisely those same moments. Finally the contraction and projection arguments preceding this subsection give (6.4) and (6.5) with this measure. This proves all assertions of the proposition. $\square$

6.3 Correctors and the exact energy

Define $c_* = \omega(\{0\})$. We have proved that this number is deterministic; for now we know only $c_* \geq 0$. Choose real polynomials Q_j with

$$Q_j(0) = 1, \quad Q_j \longrightarrow \mathbf{1}_{\{0\}} \quad \text{in } L^2((1 + \lambda)\omega + \delta_0). \quad (6.21)$$

The density assertion in Proposition 6.1 supplies such polynomials: the extra atom enforces convergence of the values at zero, after which division by those values normalizes them. Consequently

$$\int Q_j^2 d\omega \rightarrow c_*, \quad \int Q_j d\omega \rightarrow c_*, \quad \int \lambda Q_j^2 d\omega \rightarrow 0.$$

The first limit gives the energy of the corrected slopes. The last makes their microscopic divergence negligible.

Lemma 6.5 (Cylinder recovery). *Let F be a smooth compact cylinder, and let Q be a polynomial with $Q(0) = 1$. Define*

$$f_{n,Q}(X) = F(x) + n^{-1/3} \sum_a F_a(x)(Q(H) - 1)Y_a, \quad (6.22)$$

where the sum is over the finitely many coordinates of F . Then $f_{n,Q}$ is bounded by a fixed power of n , converges weakly against cylinders to F , and its squared L^2 norm converges to $\Pi[F^2]$. Moreover,

$$\lim_{n \rightarrow \infty} T\mathcal{E}(f_{n,Q}) = \left(\int Q^2 d\omega \right) \Pi[\|\nabla F\|^2].$$

The exponent in the polynomial bound may depend on Q .

Proof. Set $V_a = (Q(H) - 1)Y_a$. Since $Q - 1$ vanishes at zero,

$$\|V_a\|_2^2 = \int \frac{(Q(\lambda) - 1)^2}{\lambda} d\omega_{a,n}(\lambda)$$

is bounded for fixed Q : the quotient extends to a polynomial at zero. Thus the correction in (6.22) tends to zero in L^2 . Static convergence of cylinders gives the claimed convergence of values and norms. The bounds $\|H\|_{\infty \rightarrow \infty} \leq 2n$ and $\|Y_a\|_{\infty} \leq \sqrt{n}$ give the polynomial supremum bound.

Taylor's formula in the finitely many coordinates of F gives

$$n^{1/3} D_i F(x) = \sum_a F_{,a}(x) v_{ia} + r_i, \quad \sum_i |r_i|^2 = o(1),$$

uniformly in the spin. The discrete product rule then writes the scaled slope of (6.22) as $\sum_a F_{,a}(x) A_a^Q$ plus this remainder and terms $(D_i F_{,a})(x) V_a(X^{(i)})$. Their coefficients satisfy

$$|D_i F_{,a}(x)|^2 \leq C n^{-2/3} \sum_b v_{ib}^2.$$

The microscopic potentials need not be independent of these coefficients. Conditionally on all spins except i , put $p_i(\xi) = (1 + \xi t_i)/2$. The identity

$$\frac{w_i p_i(\xi)}{p_i(-\xi)} = (1 + \xi t_i)^2 \leq 4$$

implies $\mu[w_i |V_a(X^{(i)})|^2] \leq 4\mu[|V_a(X)|^2]$. Summing over i and using the unit norms of the finitely many v_b gives

$$\left\| n^{1/3} D f_{n,Q} - \sum_a F_{,a}(x) A_a^Q \right\|_{w,n} \rightarrow 0.$$

Equation (6.4), with multiplier $F_{,a} F_{,a'}$, now proves the energy limit. $\square$

Lemma 6.6 (Positivity of the effective coefficient). *One has $c_* = \omega(\{0\}) > 0$.*

Proof. Choose a smooth compact cylinder F with $\Pi[\|\nabla F\|^2] > 0$, possible because finite-coordinate marginals have full support. For each fixed j , apply Proposition 5.4 to f_{n,Q_j} . Its limiting gradient is ∇F by Proposition 5.1, and the constant in the coarse bound (5.6) is independent of the fixed polynomial index. Lemma 6.5 therefore yields

$$\Pi[\|\nabla F\|^2] \leq C \left(\int Q_j^2 d\omega \right) \Pi[\|\nabla F\|^2].$$

Let $j \rightarrow \infty$ to obtain $1 \leq C c_*$. Only one fixed-degree sequence is used at a time. There is no need to impose a common polynomial supremum exponent on a sequence whose degree increases with n . $\square$

The remaining task is to test arbitrary finite-model slopes against these corrected ones. Multiplication by a macroscopic cylinder test changes their microscopic divergence by a negligible amount. Here E_i denotes conditional expectation over spin i , and D_w^* is the adjoint of D for the weighted slope pairing.

Lemma 6.7 (Multiplied divergence). *For a fixed polynomial Q , fixed leading index a , and smooth compact cylinder ψ ,*

$$D_w^*(\psi(x)A_a^Q) = \psi(x)HQ(H)Y_a + o_{L^2(\mu)}(1), \quad \|HQ(H)Y_a\|_2^2 \longrightarrow \int \lambda Q^2 d\omega.$$

Proof. Conditional single-site integration gives the exact formula

$$D_w^*(\psi A_a^Q) = \sum_i (X_i - t_i)(E_i \psi)(A_a^Q)_i.$$

Replacing $E_i \psi$ by ψ gives $\psi HQ(H)Y_a$. The cylinder Lipschitz estimate is $\sum_i |E_i \psi - \psi|^2 \leq Cn^{-2/3}$ pointwise. Since $(A_a^Q)_i$ is independent of X_i , Cauchy–Schwarz gives

$$\begin{aligned} & \left\| \sum_i (X_i - t_i)(E_i \psi - \psi)(A_a^Q)_i \right\|_2^2 \\ & \leq Cn^{-2/3} \mu \left[\sum_i (X_i - t_i)^2 |(A_a^Q)_i|^2 \right] = Cn^{-2/3} \|A_a^Q\|_{w,n}^2 = o(1). \end{aligned}$$

Finally, $\|HQ(H)Y_a\|_2^2 = \int \lambda Q^2 d\omega_{a,n}$. □

We can now state the form convergence used for the resolvent limit. Weak cylinder convergence means convergence of the integrals against every smooth compact cylinder test, in the quenched represented sense of Section 2. Strong convergence additionally requires convergence of the squared L^2 norms.

Theorem 6.8 (Relaxed energy). *The constant $c_* = \omega(\{0\})$ is deterministic and strictly positive. On any quenched represented subsequence, suppose that $\|f_n\|_\infty \leq Cn^D$ for fixed C, D , that the L^2 norms and $T\mathcal{E}(f_n)$ are bounded, and that f_n converges weakly against cylinders to F . Then F is in the closed form domain on $L^2(\Pi)$ and*

$$\liminf_{n \rightarrow \infty} T\mathcal{E}(f_n) \geq c_* \Pi[\|\nabla F\|_{\ell^2}^2]. \quad (6.23)$$

Conversely, for each smooth compact cylinder F , the functions f_{n,Q_j} in (6.22) converge strongly to F for every fixed j , are polynomially bounded at that fixed j , and satisfy

$$\lim_{j \rightarrow \infty} \lim_{n \rightarrow \infty} T\mathcal{E}(f_{n,Q_j}) = c_* \Pi[\|\nabla F\|^2].$$

All assertions for countably many chosen sources and tests hold on a common represented event. In particular the lower bound and recovery use the same deterministic coefficient.

Proof. Only the lower bound remains. Pass to a subsequence attaining the lower limit of the scaled energies. Proposition 5.4 gives the weak gradient $G = \nabla F$ and its coarse bound. For finitely many smooth compact cylinders ψ_a , put $Z_{n,j} = \sum_a \psi_a(x)A_a^{Q_j}$. Equation (6.4) gives

$$\lim_{j \rightarrow \infty} \lim_{n \rightarrow \infty} \|Z_{n,j}\|_{w,n}^2 = c_* \sum_a \Pi[\psi_a^2].$$

We identify its pairing with $n^{1/3}Df_n$.

Fix $q > 0$ small and use the logarithmic averages over $[q, 2q]$ in Lemma 5.3. By (5.5), $r_{n,q} = n^{1/3}(f_n - \overline{Sf_{n,q}})$ has L^2 norm at most C/q . Lemma 6.7 bounds the limiting absolute value of its slope pairing against $\psi_a A_a^{Q_j}$ by

$$\frac{C\|\psi_a\|_\infty}{q} \left(\int \lambda Q_j^2 d\omega \right)^{1/2}.$$

This vanishes when $j \rightarrow \infty$ at fixed q , after $n \rightarrow \infty$.

For the smoothed part, the second estimate of (5.5) replaces $n^{1/3}D\overline{Sf_{n,q}}$ by $\overline{Jf_{n,q}}$, with pairing error at most $Cq \limsup_n \|Z_{n,j}\|_{w,n}$. This vector belongs to $\mathcal{H}_{2q}$ and has bounded unweighted L^2 norm. For any fixed polynomial index J , (6.5) and weak convergence of the averaged slopes therefore give

$$\mu[\psi_a(x)\overline{Jf_{n,q}} \cdot wA_a^{Q_J}] \longrightarrow \left(\int Q_J d\omega \right) \Pi[\psi_a G_a]$$

as $n \rightarrow \infty$ and then $q \downarrow 0$ along the gradient extraction.

The last step fixes the polynomial before taking $q \downarrow 0$, whereas the residual estimate takes $j \rightarrow \infty$ first. The following bound reconciles these orders without uniform graph estimates in the degree:

$$\limsup_{n \rightarrow \infty} \|w(A_a^{Q_j} - A_a^{Q_J})\|_{L^2(\mu; \mathbb{R}^n)} \leq \left(\int (Q_j - Q_J)^2 d\omega \right)^{1/2}.$$

It follows from $0 \leq w_i \leq 1$ and (6.4). For fixed J , its right side tends as $j \rightarrow \infty$ to $\|\mathbf{1}_{\{0\}} - Q_J\|_{L^2(\omega)}$, and then tends to zero as $J \rightarrow \infty$. Boundedness of $\overline{Jf_{n,q}}$ makes this estimate uniform in the cap scale. All these bounds may be expressed with upper and lower limits, so no prior convergence of the raw slope pairings is required. We first take the finite-size limit, then send $j \rightarrow \infty$ at fixed q to remove the residual, approximate the remaining test by a large fixed J , send $q \downarrow 0$, and finally send $J \rightarrow \infty$. The limiting cross pairing is $c_* \sum_a \Pi[\psi_a G_a]$.

Apply the elementary inequality

$$\|n^{1/3}Df_n\|_{w,n}^2 \geq 2\langle n^{1/3}Df_n, Z_{n,j} \rangle_{w,n} - \|Z_{n,j}\|_{w,n}^2$$

and these iterated estimates to obtain

$$\liminf_n T\mathcal{E}(f_n) \geq 2c_* \sum_a \Pi[\psi_a G_a] - c_* \sum_a \Pi[\psi_a^2].$$

Finite vectors of smooth compact cylinders are dense in $L^2(\Pi; \ell^2)$. Optimizing proves (6.23). The remaining claims follow from Lemmas 6.5 and 6.6. $\square$

The coefficient in this theorem is the atom $\omega(\{0\})$, whereas $\omega([0, \infty))$ is the unrelaxed energy of a leading linear coordinate. Although $\omega_{a,n}(\{0\}) = 0$ for every finite system, its weak limit can acquire mass from eigenvalues tending to zero. The correctors isolate precisely this mass. Theorem 6.8 and cylinder density now supply the energy estimates needed to pass to Gaussian resolvents.

7 Convergence of the Gaussian resolvent trace

The form convergence in Section 6 identifies each fixed edge coordinate. The autocorrelation, however, sums over every coordinate, and the static variances are not summable. We now prove the additional trace estimate that permits this sum to pass to the limit. Its mechanism is angular integration by parts: an edge gap of order $a^{2/3}$ gives a factor $a^{-2/3}$ in the dual norm of the a th source. The squares of these factors are summable. The variational passage from forms to individual resolvents is the standard one in the theory of varying Hilbert spaces [19, Theorem 2.4]; the summation over sources requires the additional estimates below.

Throughout this section the disorder is Gaussian. Write

$$B_g(z) := \sum_{a \geq 1} \langle x_a, (z + c_* H_\Pi)^{-1} x_a \rangle_\Pi, \quad z > 0. \quad (7.1)$$

The sum is initially allowed to be infinite. Recall that $B_n(z) = \sum_{a=1}^n \langle x_a, (z + TH)^{-1} x_a \rangle_\mu$ and $T = n^{2/3}$.

Proposition 7.1 (Gaussian trace convergence). *For almost every edge configuration g , the series defining $B_g(z)$ is finite for every $z > 0$. For every finite collection $z_1, \dots, z_k > 0$,*

$$(B_n(z_1), \dots, B_n(z_k)) \implies (B_g(z_1), \dots, B_g(z_k)).$$

Moreover, for each $z, \epsilon > 0$,

$$\lim_{\ell \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P} \left\{ \sum_{a > \ell} \langle x_a, (z + TH)^{-1} x_a \rangle_\mu > \epsilon \right\} = 0.$$

We separate fixed-coordinate convergence from the estimate uniform in the number of coordinates. For a vector-valued function f , let

$$\|f\|_{z,n}^2 = z\mu|f|^2 + T\mathcal{E}(f),$$

where the energy includes the sum over vector components. The notation $\|f\|_\infty$ means $\sup_X |f(X)|$ in the Euclidean norm of the target space, or its Hilbert–Schmidt norm for matrix targets. We allow a polynomial number of components: summing an exponentially small scalar remainder over that many components still gives a negligible error for each fixed polynomial supremum bound. The resolvent identity in variational form is

$$\langle S, (z + TH)^{-1} S \rangle_\mu = \sup_f \{ 2\mu[S \cdot f] - \|f\|_{z,n}^2 \} = \left(\sup_{\|f\|_{z,n} \leq 1} \mu[S \cdot f] \right)^2.$$

This formula applies to both scalar and vector sources.

Lemma 7.2 (Fixed coordinates). *For every fixed ℓ and every finite collection of positive z , the corresponding sums over $1 \leq a \leq \ell$ converge jointly to the same restrictions of (7.1).*

Proof. Use the quenched subsequential realization from Section 2. For a fixed coordinate the source has bounded L^2 norm on the represented sequence, by the static moment convergence. Its resolvent optimizer therefore has bounded L^2 norm and bounded scaled energy. Markov contraction also gives a polynomial supremum bound: $\|x_a\|_\infty \leq n^{1/6}$ and

$$\|(z + TH)^{-1} x_a\|_\infty \leq z^{-1} n^{1/6}.$$

The lower form bound (6.23), together with weak lower semicontinuity of the mass term, bounds the upper limit of the variational expression. Source pairings pass to the limit by first cutting off x_a , using static cylinder convergence, and then removing the cutoff with the fixed-coordinate moment bounds.

Conversely, the correctors (6.22) give the variational lower bound for every smooth compact cylinder test: first take $n \rightarrow \infty$ at a fixed recovery polynomial Q_j , and then take $j \rightarrow \infty$. The polynomial supremum exponent may depend on j , which is harmless in this order of limits. The cylinder core in Proposition 5.1 extends it to the full form domain. The two variational bounds identify the limiting resolvent. The same argument uses one subsequence for the finitely many sources and parameters under consideration, proving the joint assertion. $\square$

7.1 Moving a vector pairing to the smallest cap

For a vector test $f = (f_a)_{a>\ell}$, put $F_p = \mathbb{E}[f \mid h_p]$, componentwise. We use the ordinary observation law Q_μ , and all its expectations below are at fixed disorder. The scale is $r = n^{-1/3+\delta}$, with the fixed δ chosen in Section 2.

Lemma 7.3 (Observation replacement). *Uniformly over vector tests satisfying $\|f\|_{z,n} \leq 1$ and a fixed polynomial supremum bound,*

$$\sum_{a>\ell} \mu[f_a x_a] = n^{-1/3} \sum_{a>\ell} Q_\mu[(F_r)_a (U^t m_r)_a] + o_{\mathbb{P}}(1). \quad (7.2)$$

The error is uniform in ℓ and in the number of vector components.

Proof. The posterior means of f and X are martingales as the observation precision increases. If $c < p \leq 2c$, orthogonality of martingale increments bounds the change in their pairing by

$$n^{-1/3} (Q_\mu |F_p - F_c|^2)^{1/2} (Q_\mu |m_p - m_c|^2)^{1/2} \leq C n^{-2/3} c^{-1} (Q_\mu |m_p|^2)^{1/2} + o(1).$$

Here (4.1) bounds the first increment by $C n^{-1/3} c^{-1}$; its exponentially small remainder can be summed over at most n components with polynomially bounded supremum. The moment estimate (3.5), with $M_p = (np)^{1/2}$, makes the last bound of order $n^{-1/6} p^{-1/2}$, apart from tight constants. On a geometric list beginning at r , these deterministic weights sum to $O(n^{-\delta/2})$.

For completeness, no simultaneous bound on the maximum of all these random constants is required. Restrict first to the common disorder events on which (4.1) holds with a fixed constant. Average the weighted sum over the biased rotation law, use (3.5), and then use $L_o \geq 1/2$ and Markov's inequality to return to ordinary disorder probability. The probability of the discarded disorder events can be made arbitrarily small.

At the fixed upper endpoint p_* , the difference from the original pairing is bounded directly using (4.1) and $|X| = \sqrt{n}$. It is $O(n^{-1/6})$. Combining this endpoint estimate with the geometric sum proves the lemma. All inequalities use vector norms, so their constants do not grow with the vector dimension. $\square$

7.2 Angular integration by parts and a summable tail

We next bound the right side of (7.2). By (3.6), replacing m_r there by m_r^{sp} costs $o_{\mathbb{P}}(1)$, uniformly over the tests in the lemma. Indeed its annealed squared error is bounded by a constant times $n^{-2/3} M_r^2 n^{-1/20} = n^{\delta-1/20}$. On the cap annulus, radial symmetry of the spherical posterior gives

$$n^{-1/3} (U^t m_r^{\text{sp}})_a = t(\|h_r\|) u_a, \quad u = u_r,$$

on the active coordinate space, and zero on its orthogonal complement. The saddle bounds make t bounded. Insert a smooth radial cutoff supported in the annulus and equal to one on a smaller annulus containing the event in (3.1). The discarded contribution is negligible by its power-exponential probability and the polynomial bounds. Write $\theta(u)$ for the resulting bounded radial multiplier.

Fix $m_0 = 50$ and define $R_0 = \sum_{j=1}^{m_0} u_j^2$. On the tight spectral sets and for sufficiently large n , these leading coordinates are all active. For $a > \ell$ active and $j \leq m_0$, let δ_{aj} be the coefficient in (5.9). Taking ℓ beyond a tight index cutoff gives $\delta_{aj} \geq ca^{2/3}$. Define

$$\begin{aligned} \mathcal{V}_a &= \sum_{j=1}^{m_0} \frac{u_j}{R_0 \delta_{aj}} (u_j \partial_a - u_a \partial_j), \\ |\mathcal{V}_a| &\leq Ca^{-2/3} \left(1 + \frac{|u_a|}{\sqrt{R_0}}\right), \quad |\operatorname{div} \mathcal{V}_a| \leq Ca^{-2/3} \frac{|u_a|}{R_0}. \end{aligned} \tag{7.3}$$

The vector field is tangent to Euclidean spheres. It therefore annihilates θ and every radial term in the spherical log density. The normalization of δ_{aj} gives the exact identity

$$\mathcal{V}_a \left(-\frac{1}{2} h_r^t B_r^{-1} h_r \right) = -u_a.$$

We record the integrability needed to use these fields. In the biased law $\mathbf{P}$, the first m_0 signal coordinates have a uniformly bounded joint density on each tight spectral set, by the conditioning construction of Π and its finite-dimensional version. Their signal multipliers $B_{r,j}/r^2$ are bounded away from zero. Adding independent observation noise preserves a uniform density bound. Hence, for every $q < m_0/2$,

$$\mathbf{E} R_0^{-q} \leq C_q.$$

For example, integrate the bounded density over the unit ball in $\mathbb{R}^{m_0}$ and use $R_0^{-q} \leq 1$ outside that ball. All individual u_a also have uniformly bounded moments of every fixed order. Thus, with $s_\ell = \sum_{a>\ell} a^{-4/3}$ and with sums restricted to active indices,

$$\mathbf{E} \sum_{a>\ell} |\operatorname{div} \mathcal{V}_a|^2 \leq C s_\ell, \quad \left\| \sum_{a>\ell} |\mathcal{V}_a|^2 \right\|_{L^2(\mathbf{P})} \leq C s_\ell.$$

The second estimate follows from Minkowski's inequality and

$$\left(\mathbf{E} \frac{u_a^4}{R_0^2} \right)^{1/2} \leq (\mathbf{E} u_a^8)^{1/4} (\mathbf{E} R_0^{-4})^{1/4}.$$

In particular, the dimensions of the active cap never enter these bounds.

The ordinary observation density in u is proportional to

$$L_r(u) \exp \left\{ \phi_r^{\text{sp}}(h_r) - \frac{1}{2} h_r^t B_r^{-1} h_r \right\}.$$

Integration by parts therefore gives

$$\begin{aligned} \sum_{a>\ell} Q_\mu[\theta(F_r)_a u_a] &= \sum_{a>\ell} Q_\mu[\theta \mathcal{V}_a(F_r)_a] \\ &\quad + \sum_{a>\ell} Q_\mu[\theta(F_r)_a \operatorname{div} \mathcal{V}_a] \\ &\quad + \sum_{a>\ell} Q_\mu[\theta(F_r)_a \mathcal{V}_a \log L_r]. \end{aligned}$$

These three terms correspond respectively to differentiating the test, the vector field, and the comparison density. The first is bounded by Cauchy–Schwarz using $Q_\mu \sum_a |\mathcal{V}_a|^2$ and $Q_\mu \sum_a |\nabla_u (F_r)_a|^2$; the latter is bounded by the vector version of (5.4). The second uses $Q_\mu |F_r|^2 \leq z^{-1}$ and the divergence estimate above. For the third, put $W_\ell = \sum_{a>\ell} |\mathcal{V}_a|^2$. Then

$$\mathbf{E} \sum_{a>\ell} |\mathcal{V}_a \log L_r|^2 \leq \|W_\ell\|_{L^2(\mathbf{P})} (\mathbf{E} |\nabla_u \log L_r|^4)^{1/2} = o(1) s_\ell$$

by (3.6). Transferring these nonnegative bounds to ordinary disorder with $L_o \geq 1/2$ and Markov’s inequality shows that the entire pairing is $O_\mathbb{P}(s_\ell^{1/2}) + o_\mathbb{P}(1)$, uniformly over the tests. More precisely, for each $\eta > 0$ one first chooses a deterministic index L_η and bounds on the spectral, variance, and gradient constants, outside disorder events of probability at most η . The displayed moment estimates then hold with a constant C_η for every $\ell \geq L_\eta$, uniformly in n . This is the uniformity in the tail index that will be used below. The integration by parts is justified first with a cutoff away from $R_0 = 0$ and then without it. If $\chi(R_0/\varepsilon)$ is such a cutoff, then

$$|\mathcal{V}_a \chi(R_0/\varepsilon)| \leq C a^{-2/3} \frac{|u_a|}{R_0} \mathbf{1}_{\{R_0 \leq 2\varepsilon\}}.$$

The squared norms of these cutoff terms sum to a quantity tending to zero by the coordinate and inverse moments above. They are therefore harmless against the vector F_r by Cauchy–Schwarz. At fixed n the observation density is smooth on its active space.

Proof of Proposition 7.1. Let $B_{n,>\ell}(z)$ denote the tail trace in the proposition. If it is at least ϵ , its normalized optimizer is

$$f = \frac{(z + TH)^{-1}(x_a)_{a>\ell}}{\sqrt{B_{n,>\ell}(z)}}.$$

It has $\|f\|_{z,n} = 1$ and $\|f\|_\infty \leq n^{1/6}/(z\sqrt{\epsilon})$, since $|(x_a)_{a>\ell}| \leq n^{1/6}$ pointwise. It is therefore an admissible test for the uniform estimates above. Lemma 7.3 and the angular calculation imply that its pairing with the source vanishes in probability in the double limit $n \rightarrow \infty$, then $\ell \rightarrow \infty$. That pairing equals $\sqrt{B_{n,>\ell}(z)}$. In the localized form just described, Markov’s inequality gives, for each fixed $\epsilon > 0$ and every $\ell \geq L_\eta$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}\{B_{n,>\ell}(z) > \epsilon\} \leq \eta + C_\eta \epsilon^{-1} s_\ell.$$

Letting $\ell \rightarrow \infty$ and then $\eta \downarrow 0$ proves the asserted tail estimate.

Combine it with Lemma 7.2. The tail estimate and tightness of one sufficiently long fixed-coordinate sum first give tightness of the full $B_n(z)$. Each limiting finite partial sum is therefore bounded by the same tightness bounds, uniformly in its length: apply convergence of that partial sum and its pointwise bound by $B_n(z)$. Monotonicity now shows that the limiting series is finite almost surely. The tail estimate then proves full trace convergence, jointly at finitely many parameters. Finiteness at all positive z follows first at positive rationals and then everywhere by resolvent monotonicity. $\square$

7.3 A quadratic feature needed for disorder comparison

One more Gaussian estimate will allow the trace to be compared across entry distributions. Regard XX^t/n as a vector in the space of matrices with the Hilbert–Schmidt inner product, and

set

$$B_n^{[2]}(z) = \sum_{i,j=1}^n \left\langle \frac{X_i X_j}{n}, (z + TH)^{-1} \frac{X_i X_j}{n} \right\rangle_\mu.$$

This feature has pointwise squared norm one, so $0 \leq B_n^{[2]}(z) \leq z^{-1}$.

Proposition 7.4 (Vanishing quadratic-feature trace). *For Gaussian disorder and every $z > 0$,*

$$B_n^{[2]}(z) \longrightarrow 0 \quad \text{in probability.} \quad (7.4)$$

Proof. Test the feature against a vector f with $\|f\|_{z,n} \leq 1$ and a polynomial supremum bound. By (4.1), replacing f by F_r in this pairing costs $O(n^{-\delta}) + o(1)$: the feature has norm one and $Q_\mu |f - F_r|^2 \leq Cr^{-2}n^{-2/3} + o(1)$. The remaining pairing uses the conditional mean $M_r^{[2]} = \mathbb{E}[XX^t/n \mid h_r]$. If X, X' are independent posterior replicas, then

$$|M_r^{[2]}|_{\text{HS}}^2 = \mathbb{E} \left[\left(\frac{X \cdot X'}{n} \right)^2 \middle| h_r \right].$$

The pair-tail estimate following (3.3), the annulus estimate (3.1), and the biased-to-quenched transfer used for (3.5) give

$$\mathbb{E}|M_r^{[2]}|_{\text{HS}}^2 \leq Cr^2 + C \exp(-n^\xi), \quad Q_\mu |M_r^{[2]}|_{\text{HS}}^2 = o_{\mathbb{P}}(1).$$

For the first inequality, split posterior overlaps at $C'r$, and on the complementary event use the likelihood-threshold argument from (3.5). The second follows by $L_\circ$ comparison and Markov's inequality. The n^2 matrix components cause no loss in (4.1), because their exponentially small remainders remain negligible after summation. The posterior Cauchy–Schwarz bound now makes the pairing vanish uniformly. As in the preceding proof, whenever the trace exceeds a fixed ϵ , the normalized resolvent optimizer has a fixed polynomial supremum bound; in this case even a constant bound suffices. The variational formula proves (7.4). $\square$

8 Comparison with sign disorder

We now transfer the Gaussian resolvent limits to sign couplings. The comparison uses the physical spin coordinates, so it requires no information about eigenvectors for the sign model. Its two inputs from Section 7 are the joint Gaussian trace limit (7.1) and the vanishing quadratic-feature trace (7.4). The latter will control fourth derivatives of the former. We use the Taylor replacement principle of [9]; its application here rests on averaged derivative estimates for these dynamical traces.

Let

$$\nu_t = (1-t)\mathbf{N}(0,1) + t \frac{1}{2}(\delta_{-1} + \delta_1), \quad 0 \leq t \leq 1,$$

and write $\mathbb{P}_t, \mathbb{E}_t$ for independent disorder entries with law ν_t . Thermal expectations continue to be denoted by μ . For a vector-valued feature S define its resolvent trace by

$$B_S(z) = \langle S, (z + TH)^{-1} S \rangle_\mu, \quad T = n^{2/3}, \quad z > 0,$$

where the inner product sums the feature coordinates. The two features used below are

$$S^{[1]}(X) = n^{-1/3}X, \quad S^{[2]}(X) = \frac{XX^t}{n}.$$

The second takes values in matrices with the Hilbert–Schmidt norm. Set $B_n^{[k]}(z) = B_{S^{[k]}}(z)$; thus $B_n^{[1]}(z) = B_n(z)$ by orthogonal invariance of the sum defining the trace. Positivity of H gives the useful deterministic bounds

$$0 \leq B_n^{[1]}(z) \leq \frac{n^{1/3}}{z}, \quad 0 \leq B_n^{[2]}(z) \leq \frac{1}{z}. \quad (8.1)$$

Theorem 8.1 (Trace universality). *For every $z > 0$,*

$$\sup_{0 \leq t \leq 1} \mathbb{E}_t B_n^{[2]}(z) \longrightarrow 0.$$

For each finite collection $z_1, \dots, z_m > 0$, the vector $(B_n(z_1), \dots, B_n(z_m))$ under sign disorder converges in law to the Gaussian limiting vector in (7.1). In particular the coefficient c_ in that limit is unchanged.*

The proof first compares the bounded second-feature traces. Their vanishing then supplies the additional estimate needed to compare the first-feature traces. We begin with a static estimate that applies throughout the mixture.

8.1 Uniform static overlap control

For two independent spins under μ , write $\rho(X, X') = n^{-1} X \cdot X'$. The following estimate also controls correlations after multiplying μ by a density that depends on the disorder.

Lemma 8.2. *For every $0 < \epsilon \leq 1$ there is $a_\epsilon > 0$ such that*

$$\sup_{0 \leq t \leq 1} \mathbb{P}_t \left\{ \mu^{\otimes 2}(|\rho| \geq \epsilon) > e^{-a_\epsilon n} \right\} \longrightarrow 0. \quad (8.2)$$

Consequently, for fixed $C, K < \infty$, uniformly over nonnegative, possibly disorder-dependent functions p satisfying $\mu p \leq 1$ and $\|p\|_\infty \leq Cn^K$, the quantities

$$\frac{1}{\binom{n}{2}} \sum_{i < j} |\mu[pX_i X_j]|$$

converge to zero in probability, uniformly in t . They also converge to zero in expectation, uniformly in t .

Proof. Let Z denote the partition function with uniform cube probability as its reference measure. The moment generating function $m_t(v) = \mathbb{E}_t e^{vJ_{12}}$ satisfies

$$m_t(v) \leq e^{v^2/2}, \quad \log m_t(n^{-1/2}) \geq \frac{1}{2n} - \frac{C}{n^2},$$

with constants independent of t . If $Z_A^{(2)}$ is the two-replica partition function restricted to an event A depending on ρ , independence of the entries therefore gives

$$\frac{\mathbb{E}_t Z_A^{(2)}}{(\mathbb{E}_t Z)^2} \leq C \mathbb{E}_{\text{cube}} \left[e^{n\rho^2/2} \mathbf{1}_A \right]. \quad (8.3)$$

Here the expectation on the right is over two independent uniform cube spins. Indeed the pair exponent uses $(X_i X_j + X'_i X'_j)^2 = 2 + 2X_i X'_i X_j X'_j$; summing over $i < j$ produces $n\rho^2/2$ after division by the squared first moment, with only a bounded factor remaining.

Put

$$I(r) = \frac{(1+r)\log(1+r) + (1-r)\log(1-r)}{2}, \quad -1 \leq r \leq 1,$$

with $0 \log 0 = 0$. The binomial bound $\mathbb{P}_{\text{cube}}(\rho = r) \leq e^{-nI(r)}$ and the fact that there are $n+1$ possible overlaps show that the right side of (8.3) is at most $C(n+1)$. On $A = \{|\rho| \geq \epsilon\}$ it is at most

$$C(n+1)e^{-n\kappa_\epsilon}, \quad \kappa_\epsilon = I(\epsilon) - \epsilon^2/2 > 0. \quad (8.4)$$

For the strict inequality, $I'(r) - r = \text{arctanh}(r) - r > 0$ for $0 < r < 1$, and continuity handles $r = 1$.

We next obtain a lower bound for the quenched partition function. The unrestricted second-moment estimate and Paley–Zygmund imply

$$\mathbb{P}_t\{Z \geq \tfrac{1}{2}\mathbb{E}_t Z\} \geq \frac{c}{n+1}.$$

Clip every entry to $[-\log n, \log n]$, and let Z_L be the resulting partition function. The probability that any entry changes is at most $Cn^2e^{-(\log n)^2/2}$, uniformly in t . Changing one clipped entry changes $\log Z_L$ by at most $2\log n/\sqrt{n}$. Bounded differences hence gives, for each $\eta > 0$,

$$\mathbb{P}_t\{|\log Z_L - \mathbb{E}_t \log Z_L| > \eta n\} \leq 2\exp\left(-\frac{c\eta^2 n}{(\log n)^2}\right).$$

This upper bound is smaller than $c/(2(n+1))$ for large n . The preceding Paley–Zygmund event thus forces $\mathbb{E}_t \log Z_L \geq \log \mathbb{E}_t Z - \eta n$ for all sufficiently large n , after adjusting η by a fixed factor. Concentration once more shows that, for every fixed $\eta > 0$,

$$\sup_t \mathbb{P}_t\{Z < e^{-\eta n} \mathbb{E}_t Z\} \longrightarrow 0. \quad (8.5)$$

Use (8.4) and Markov's inequality to bound $Z_A^{(2)} \leq e^{-3\kappa_\epsilon n/4}(\mathbb{E}_t Z)^2$ with probability tending to one. On the event in (8.5) with $\eta = \kappa_\epsilon/8$, division by Z^2 proves (8.2) with $a_\epsilon = \kappa_\epsilon/2$.

For the last assertion, an exact identity is

$$\frac{1}{\binom{n}{2}} \sum_{i < j} \mu[pX_i X_j]^2 = \frac{n\mu^{\otimes 2}[p(X)p(X')\rho^2] - (\mu p)^2}{n-1}. \quad (8.6)$$

On the event supplied by (8.2), the replica integral on the right is at most $\epsilon^2 + C^2 n^{2K} e^{-a_\epsilon n}$. First let $n \rightarrow \infty$ and then $\epsilon \downarrow 0$. Cauchy–Schwarz gives the asserted average of absolute correlations. This average is bounded by $\mu p \leq 1$, which also gives the uniform convergence in expectation. $\square$

8.2 Perturbing one interaction

Fix an arbitrary disorder matrix, one of the two features S , and $z > 0$. All estimates in this subsection are deterministic unless an expectation over disorder is displayed. Let $e = \{i, j\}$ be an unordered pair, set $q = X_i X_j$, and add θ to its interaction coefficient $J_{ij}/\sqrt{n}$. Writing $u = \tanh \theta$, the perturbed Gibbs probability has density $(1 + uq)/(1 + u\mu q)$ relative to μ .

It is convenient to work first with the unnormalized measure $(1 + uq)\mu$. Let $M(u)$ be the mass-plus-energy form for its heat-bath operator:

$$M(u)(f, g) = z\mu[(1 + uq)f \cdot g] + T\mathcal{E}_u^{\text{un}}(f, g), \quad b(u)(g) = \mu[(1 + uq)S \cdot g].$$

Here $\mathcal{E}_u^{\text{un}}$ is the heat-bath Dirichlet form integrated against $(1 + uq)\mu$. Put $M = M(0)$, write $\|g\|_M^2 = M(g, g)$ and $\|\cdot\|_*$ for its dual norm, and define

$$Y = M^{-1}b(0) = (z + TH)^{-1}S, \quad B = B_S(z) = M(Y, Y), \quad d = (1 + B)^{1/2}.$$

The feature coordinates are contracted in every form and dual pairing. Expand $M(u) = \sum_{a \geq 0} u^a M_a$ and $b(u) = b_0 + ub_1$; the indices a here denote Taylor coefficients, not sites. The quantities that measure the effect of the pair are

$$\Gamma_e = \frac{T}{d^2} \sum_{k \in e} \mu[w_k |D_k Y|^2], \quad R_e = \frac{1}{d} \|b_1 - M_1 Y\|_*. \quad (8.7)$$

We use avg_e for the average over the $\binom{n}{2}$ pairs. Counting each site in $n - 1$ pairs gives

$$0 \leq \Gamma_e \leq 1, \quad \text{avg}_e \Gamma_e = \frac{2}{n} \frac{T\mathcal{E}(Y)}{1 + B} \leq \frac{2}{n}. \quad (8.8)$$

Gaussian and sign entries have the same first three moments, so the comparison uses a fourth-order Taylor remainder. The derivative estimates below will require fourth-power averages of R_e .

Lemma 8.3 (Residual estimates). *Uniformly over the disorder and the pair e ,*

$$R_e^2 \leq \frac{2T}{z} \Gamma_e. \quad (8.9)$$

For $S = S^{[2]}$ one also has $R_e^2 \leq C_z$, whereas for $S = S^{[1]}$,

$$R_e^2 \leq C_z (1 + n^{1/3} \sqrt{B_n^{[2]}(z)}). \quad (8.10)$$

Consequently $\text{avg}_e R_e^2 \leq C_z n^{-1/3}$. Its fourth-power analogue is $\text{avg}_e R_e^4 \leq C_z n^{-1/3}$ for $S^{[2]}$, and

$$\text{avg}_e R_e^4 \leq C_z (n^{-1/3} + \sqrt{B_n^{[2]}(z)}) \quad \text{for } S^{[1]}. \quad (8.11)$$

Proof. For a site $k \notin e$, the energy multiplier in $M(u)$ is $1 + uq$. At an endpoint $k \in e$, let s be the other spin in the pair. Its conditional mean changes from t_k to $(t_k + us)/(1 + ust_k)$, while the marginal measure on X_{-k} acquires the factor $1 + ust_k$. Thus its energy term has the multiplier

$$\frac{1 - u^2}{1 + ust_k} \quad (8.12)$$

relative to $\mu[w_k D_k f \cdot D_k g]$. Since $|t_k| \leq 1$, these formulas show that $\|M_a\|_{M \rightarrow M^*} \leq C^a$ and, for $a \geq 2$, that M_a uses only the two endpoint energies. In particular,

$$\begin{aligned} \|M_a Y\|_* &\leq C^a d \sqrt{\Gamma_e}, \quad |M_a(Y, Y)| \leq C^a d^2 \Gamma_e \quad (a \geq 2), \\ \|M_1 Y\|_* &\leq C \sqrt{B}. \end{aligned} \quad (8.13)$$

Write $E_k g = \mu[g \mid X_{-k}]$. Testing the equation $MY = b_0$ against qg gives the exact identity

$$(b_1 - M_1 Y)(g) = T \sum_{k \in e} \mu[w_k D_k Y \cdot s E_k g]. \quad (8.14)$$

To verify it, the mass terms and all sites outside e cancel. At an endpoint, $D_k(qg) = s(g(X_k = 1) + g(X_k = -1))/2$, whereas the first coefficient in (8.12) is $-st_k$. The remaining sum is $sE_k g$. Cauchy–Schwarz and conditional Jensen now give

$$|(b_1 - M_1 Y)(g)|^2 \leq d^2 \Gamma_e T \sum_{k \in e} \mu[w_k |E_k g|^2] \leq \frac{2T}{z} d^2 \Gamma_e \|g\|_M^2,$$

which proves (8.9).

By (8.13), $R_e^2 \leq C_z(1 + \|b_1\|_*^2)$. For the second feature, $\|qS^{[2]}\|_{L^2(\mu)} = 1$, so $\|b_1\|_*^2 \leq 1/z$. For the first feature, use the stationary heat-bath process at the fixed disorder to write

$$\begin{aligned} \|b_1\|_*^2 &= n^{1/3} \int_0^\infty e^{-zs} \mathbb{E}_\mu[q(X_0)q(X_{Ts})\rho(X_0, X_{Ts})] ds \\ &\leq n^{1/3} \left(\frac{1}{z} \int_0^\infty e^{-zs} \mathbb{E}_\mu[\rho(X_0, X_{Ts})^2] ds \right)^{1/2} = n^{1/3} \sqrt{B_n^{[2]}(z)/z}. \end{aligned}$$

The last equality is the Hilbert–Schmidt feature identity $\langle XX^\dagger/n, X'X'^\dagger/n \rangle = \rho(X, X')^2$. This proves (8.10). Finally, (8.9) and (8.8) give the second-power average. Multiplying it by the respective pointwise bound on R_e^2 proves both fourth-power estimates. $\square$

The residual is small on average because a single pair touches only two site energies. The other contribution to a trace derivative is a correlation with q . The static overlap estimate controls precisely this contribution. Define

$$h_e = |\mu q| + d^{-2} |M_1(Y, Y)| + R_e + \sqrt{\Gamma_e}. \quad (8.15)$$

Lemma 8.4 (Averaged sensitivities). *For either feature, $\text{avg}_e(h_e + h_e^4) \leq C_z$ deterministically. There is a deterministic sequence $\epsilon_n \rightarrow 0$ such that, uniformly $t \in [0, 1]$,*

$$\mathbb{E}_t \text{avg}_e(h_e + h_e^4) \leq \epsilon_n \quad \text{for } S^{[2]}, \quad (8.16)$$

$$\mathbb{E}_t \text{avg}_e(h_e + h_e^4) \leq \epsilon_n + C_z \mathbb{E}_t \sqrt{B_n^{[2]}(z)} \quad \text{for } S^{[1]}. \quad (8.17)$$

Proof. The first two terms in (8.15) are bounded by constants. For the second of them introduce the nonnegative function

$$p_Y = \frac{z|Y|^2 + T \sum_k w_k |D_k Y|^2}{d^2}.$$

Its mass is $B/(1+B) \leq 1$. The Markov resolvent representation yields $\|Y\|_\infty \leq \|S\|_\infty/z$, with vector norm intended, and hence $\|p_Y\|_\infty \leq C_z n^2$ for the first feature and $\|p_Y\|_\infty \leq C_z n^{5/3}$ for the second. The endpoint formula gives

$$\left| d^{-2} M_1(Y, Y) - \mu[q p_Y] \right| \leq 2\Gamma_e.$$

Lemma 8.2, first with $p = 1$ and then with $p = p_Y$, therefore shows that these first two terms have vanishing pair averages in expectation, uniformly in t . Their fourth powers have the same property because the terms are bounded. The remaining terms are controlled by (8.8), Lemma 8.3, and Cauchy–Schwarz:

$$\text{avg}_e R_e \leq C_z n^{-1/6}, \quad \text{avg}_e \sqrt{\Gamma_e} \leq \sqrt{2/n}.$$

Combining these estimates proves (8.16) and (8.17). The deterministic bound follows from the same estimates and $B_n^{[2]}(z) \leq 1/z$. $\square$

8.3 Fourth derivatives of bounded trace observables

The first trace need not be bounded uniformly in n . We compare it through the bounded coordinate $(1 + B)^{-1}$. The next lemma turns the preceding sensitivities into the derivative estimate needed for replacement.

Lemma 8.5. *Fix finitely many traces $B_a = B_n^{[k_a]}(z_a)$, with $k_a \in \{1, 2\}$ and $z_a > 0$, and a smooth function F on a neighborhood of $[0, 1]^m$. For an arbitrary base disorder and pair e , let $B_a(\theta)$ denote the traces after adding θ to the interaction coefficient of e , and let $\mathfrak{h}_e = \sum_{a=1}^m h_{e,a}$, with each sensitivity computed at the base disorder. Then for all sufficiently large n ,*

$$\sup_{|\theta| \leq 2 \log n / \sqrt{n}} \left| \frac{d^4}{d\theta^4} F \left(\frac{1}{1 + B_1(\theta)}, \dots, \frac{1}{1 + B_m(\theta)} \right) \right| \leq C_F (\mathfrak{h}_e + \mathfrak{h}_e^4). \quad (8.18)$$

At every real base disorder, derivatives of orders at most four also have a deterministic bound polynomial in n .

Proof. We first consider one trace. Complexify the finite-dimensional form space, extending the forms and dual pairings complex bilinearly. The coefficient bounds for $M(u)$ imply that $M^{-1/2}M(u)M^{-1/2}$ differs from the identity in norm by $O(|u|)$ on a fixed disk. Its inverse is therefore analytic and uniformly bounded on a smaller fixed disk. Put $r(u) = b(u) - M(u)Y$ and $A(u) = b(u)M(u)^{-1}b(u)$. Completing the quadratic expression around Y gives the algebraic identity

$$A(u) = 2b(u)Y - M(u)(Y, Y) + r(u)M(u)^{-1}r(u).$$

The coefficient of u in $(A(u) - B)/d^2$ is

$$\frac{M_1(Y, Y) + 2(b_1 - M_1 Y)(Y)}{d^2},$$

whose absolute value is at most Ch_e . For higher coefficients, (8.13) gives

$$\frac{\|r_1\|_*}{d} = R_e, \quad \frac{\|r_a\|_*}{d} \leq C^a \sqrt{\Gamma_e} \quad (a \geq 2).$$

The direct term $-M_a(Y, Y)/d^2$ is at most $C^a \Gamma_e$. Coefficients of the residual quadratic term are bounded, using the analytic inverse, by $C^a(R_e^2 + R_e \sqrt{\Gamma_e} + \Gamma_e) \leq C^a h_e^2$. Thus every coefficient of order $a \geq 2$ in $(A(u) - B)/d^2$ is bounded by $C^a h_e^2$.

The actual trace divides $A(u)$ by the total mass $1 + u\mu q$. Thus, with

$$\delta(u) = \frac{B(u) - B}{d^2},$$

its linear coefficient is $O(h_e)$ and its coefficients of order $a \geq 2$ are bounded by $C^a(h_e + h_e^2)$. From (8.10) and (8.1), $h_e \leq C_z(1 + n^{1/6})$ for the first feature; for the second it is bounded. Consequently $|u|h_e = o(1)$ uniformly for $|u| \leq 2 \log n / \sqrt{n}$. Summing the coefficient bounds on this interval gives

$$\sup |\delta(u)| = o(1), \quad |\delta'(u)| \leq Ch_e, \quad |\delta^{(a)}(u)| \leq C(h_e + h_e^2), \quad 2 \leq a \leq 4.$$

Since

$$\frac{1}{1 + B(u)} = \frac{1}{d^2} \frac{1}{1 + \delta(u)},$$

the chain rule bounds its fourth derivative, and that of any fixed smooth test of it, by $C(h_e + h_e^4)$. The derivatives of $u = \tanh \theta$ through order four are bounded on the interval in question, so the same estimate holds for interaction displacement θ .

For several traces apply the multivariable chain rule and bound every mixed product using $\mathfrak{h}_e + \mathfrak{h}_e^4$. Finally, at any real base point the coefficient argument applies at $\theta = 0$, where the denominator of the compactification is at least one. The bound $h_e \leq C(1 + n^{1/6})$ supplies the claimed polynomial bounds, uniformly in that base point. $\square$

8.4 Replacement and the two-feature argument

We have obtained a small average fourth derivative under a complete iid mixture disorder. Taylor replacement removes an entry before expanding, so its expansion point does not itself have that law. The following independent completion lets us use the averaged estimates without changing the Taylor coefficients.

Let

$$\Phi(J) = F\left((1 + B_n^{[k_1]}(z_1))^{-1}, \dots, (1 + B_n^{[k_m]}(z_m))^{-1}\right).$$

Differentiating the finite product mixture measure gives exactly

$$\frac{d}{dt} \mathbb{E}_t \Phi = \sum_e \mathbb{E}_{t,-e} [\mathbb{E}_\xi \Phi(J_{-e}, \xi) - \mathbb{E}_g \Phi(J_{-e}, g)], \quad (8.19)$$

where ξ is a symmetric sign, g is standard normal, and $\mathbb{E}_{t,-e}$ averages all entries other than e . For fixed J_{-e} , Taylor-expand $x \mapsto \Phi(J_{-e}, x)$ at zero through degree three. These polynomial terms cancel in (8.19), since ξ and g have the same first three moments.

Independently of both proposed values, draw J_e^0 with law ν_t , and write $J^0 = (J_{-e}, J_e^0)$. If $|x|, |J_e^0| \leq \log n$, every point of the Taylor segment between 0 and x differs from J_e^0 by at most $2 \log n$. Lemma 8.5, with J^0 as its base, therefore bounds the Taylor remainder by

$$C_F \frac{|x|^4}{n^2} (\mathfrak{h}_e(J^0) + \mathfrak{h}_e(J^0)^4). \quad (8.20)$$

The factor n^{-2} changes four derivatives in the interaction coefficient $J_e/\sqrt{n}$ into four derivatives in J_e . The factor $|x|^4$ is measured from the Taylor center 0. The independently completed entry is used only to bound the derivatives along the segment, so it does not enter this factor.

The events $|g| > \log n$ and $|J_e^0| > \log n$ have probability smaller than every fixed inverse power of n , uniformly in t . The boundedness of Φ , together with the polynomial derivative bounds in Lemma 8.5, bounds the Taylor polynomial and its remainder by a polynomial in n times $1 + |x|^4$. Gaussian tail moments thus make the omitted contributions $o(n^{-2})$, uniformly in J_{-e} and t . On the remaining event, enlarge the right side of (8.20) by removing its indicator. Its $|x|^4$ factor is independent of the completed base J^0 and has bounded expectation. Moreover, if e is chosen uniformly, J^0 has the full iid mixture law independently of e . Summing the remainders in (8.19) therefore yields

$$\left| \frac{d}{dt} \mathbb{E}_t \Phi \right| \leq C_F \mathbb{E}_t \text{avg}_e (\mathfrak{h}_e + \mathfrak{h}_e^4) + o(1), \quad (8.21)$$

uniformly for $0 \leq t \leq 1$.

Proof of Theorem 8.1. First use only second-feature traces in (8.21). For finitely many traces, $\mathfrak{h}_e + \mathfrak{h}_e^4 \leq C_m \sum_a (h_{e,a} + h_{e,a}^4)$. Equation (8.16) therefore makes the mixture derivative tend to zero

uniformly in t . In particular, for one z , choose the test $F(v) = 1 - v$. The Gaussian convergence (7.4) and the bound $B_n^{[2]}(z) \leq 1/z$ imply

$$\sup_t \mathbb{E}_t \frac{B_n^{[2]}(z)}{1 + B_n^{[2]}(z)} \longrightarrow 0.$$

Since $b \leq (1 + 1/z)b/(1 + b)$ for $0 \leq b \leq 1/z$, this proves the first assertion of the theorem and, by Jensen, $\sup_t \mathbb{E}_t \sqrt{B_n^{[2]}(z)} \rightarrow 0$.

Now use first-feature traces. Equation (8.17) makes their right side in (8.21) tend to zero uniformly in t . Integrating from $t = 0$ to $t = 1$ shows equality of the limiting expectations of every smooth test of finitely many compactified traces. Such tests determine probability laws on $[0, 1]^m$. By Proposition 7.1, the Gaussian trace vector converges to the finite vector in (7.1); its compactification therefore has all coordinates strictly positive almost surely. The continuous mapping theorem for the inverse map $v \mapsto v^{-1} - 1$ recovers the same joint limit for the sign traces. $\square$

9 The functional limit and its randomness

Proposition 7.1 and Theorem 8.1 identify the same finite collections of resolvent traces for Gaussian and sign disorder. We first turn these statements into convergence of autocorrelation functions. We then prove decay and show that the edge dependence produces a random, rather than deterministic, limit.

9.1 From resolvents to continuous functions

Let E_n be the spectral resolution of the nonnegative operator TH , and define the finite measure

$$\nu_n(I) = \sum_{i=1}^n \langle n^{-1/3} X_i, E_n(I) n^{-1/3} X_i \rangle_\mu$$

for Borel sets $I \subset [0, \infty)$. Basis invariance gives

$$B_n(z) = \int \frac{d\nu_n(\lambda)}{z + \lambda}, \quad A_{n,J}(s) = \int e^{-s\lambda} d\nu_n(\lambda).$$

Although $\nu_n([0, \infty)) = n^{1/3}$ diverges, its division by $1 + \lambda$ has tight mass. This is the reason to use resolvents rather than the unweighted spectral measures.

Lemma 9.1 (A continuity principle for spectral measures). *Let ν_n and ν be random nonnegative measures on $[0, \infty)$ such that their integrals against $(1 + \lambda)^{-1}$ are finite almost surely, for every n and for ν . Suppose the corresponding integrals against $(z + \lambda)^{-1}$ converge jointly in law for every finite collection of positive z , including $z = 1$. Then*

$$\left(s \mapsto \int e^{-s\lambda} d\nu_n(\lambda) \right) \implies \left(s \mapsto \int e^{-s\lambda} d\nu(\lambda) \right)$$

in $C_{\text{loc}}((0, \infty))$.

Proof. Set $d\eta_n = (1 + \lambda)^{-1} d\nu_n$ and regard η_n as a finite measure on the compact space $K = [0, \infty]$, giving no mass to infinity. Its total mass is the Stieltjes transform at $z = 1$, and is therefore tight. Finite measures on K with bounded total mass form a compact set, so the laws of η_n are tight.

For $z > 0$ the function

$$k_z(\lambda) = \frac{1 + \lambda}{z + \lambda}, \quad k_z(\infty) = 1,$$

is continuous on K . A countable dense family of these functions, including $k_1 = 1$, determines a finite measure even when that measure has mass at infinity. Indeed their closed linear span contains 1 and $r_z(\lambda) = (z + \lambda)^{-1}$, with $r_z(\infty) = 0$. The partial-fraction identity $r_z r_w = (r_z - r_w)/(w - z)$, followed by uniform limits when $w = z$, shows that this span is an algebra. It separates the points of K , so it is dense in $C(K)$. Consequently every subsequential law is the law of $d\eta = (1 + \lambda)^{-1} d\nu$, with no mass at infinity. Here joint convergence on the countable determining family identifies the law of the random measure, rather than only its mean.

For $s > 0$, define $h_s(\lambda) = (1 + \lambda)e^{-s\lambda}$ and $h_s(\infty) = 0$. On each compact interval $a \leq s \leq b$, $a > 0$, the map $s \mapsto h_s$ is continuous in the uniform norm on K . Thus weak convergence of finite measures on K gives convergence of their h_s integrals uniformly on $[a, b]$: a finite uniform-norm net of the functions h_s reduces the assertion to finitely many continuous tests, while weak convergence bounds the total masses. The resulting map from measures to $C_{\text{loc}}((0, \infty))$ is continuous. The continuous mapping theorem proves the claim. $\square$

Proposition 9.2 (Identification of the functional limit). *The series (2.7) has a Borel version $g \mapsto \mathcal{A}_g$ with values in $C_{\text{loc}}((0, \infty))$. For both Gaussian and symmetric-sign entries,*

$$\text{Law}(A_{n,J}) \Longrightarrow Q_{\text{crit}} = \text{Law}(\mathcal{A}_g),$$

where the same edge law and the same deterministic coefficient c_* are used in (2.7).

Proof. For a fixed edge configuration g , let E_g be the spectral resolution of $c_* H_{\Pi}$ and put

$$\nu_g(I) = \sum_{a \geq 1} \langle x_a, E_g(I) x_a \rangle_{\Pi}.$$

This countable sum defines a nonnegative measure. By Proposition 7.1, its division by $1 + \lambda$ has finite mass $B_g(1)$ almost surely. It follows in particular that the series in (2.7) defines a continuous function $\mathcal{A}_g$ on $(0, \infty)$.

These constructions are measurable in g . The density prescription for Π makes each cylinder integral measurable. Resolvent pairings can then be computed by the variational supremum over a countable family of smooth compact cylinders. The family can be chosen independently of g by ordinary approximation of a function and its first derivatives on finite-dimensional compact sets; it is a form core by Proposition 5.1. Thus the resolvent pairings, the divided spectral measure determined by them, and $\mathcal{A}_g$ are measurable.

Apply Lemma 9.1 to the finite-dimensional trace convergence in Proposition 7.1 and Theorem 8.1. For either entry law,

$$\text{Law}(A_{n,J}) \Longrightarrow Q_{\text{crit}} := \text{Law}(\mathcal{A}_g) \quad \text{in } C_{\text{loc}}((0, \infty)).$$

The common coefficient is the deterministic c_* of Theorem 6.8; no further change of time is involved. $\square$

9.2 Decay and positive values

The spectral representation makes $\mathcal{A}_g$ nonnegative and nonincreasing. Proposition 5.1 identifies the kernel of H_Π as the constants. Each x_a is centered by sign symmetry of Π , so ν_g has no atom at zero. For any fixed $s_0 > 0$, $e^{-s_0\lambda}$ is integrable against ν_g . Dominated convergence for $s \geq s_0$ therefore gives

$$\lim_{s \rightarrow \infty} \mathcal{A}_g(s) = 0.$$

There is also a useful positive lower bound. Put $v(g) = \Pi[x_1^2]$, which is finite by the coordinate moment bounds and strictly positive by full support of the first-coordinate marginal. The coordinate is in the form domain with energy one, by Proposition 5.1. Jensen's inequality for its normalized spectral measure gives

$$\mathcal{A}_g(s) \geq \langle x_1, e^{-sc_* H_\Pi} x_1 \rangle_\Pi \geq v(g) \exp\{-sc_*/v(g)\} > 0. \quad (9.1)$$

It remains to show that these functions do not have a deterministic law. For this we will vary a leading edge point while retaining absolute continuity with respect to the original edge law.

9.3 Finite shifts of the edge process

Let $\mathbf{P}$ denote the law of the increasing edge sequence g . For an integer m and a vector $h \in \mathbb{R}^m$, define

$$T_h g = (g_1 + h_1, \dots, g_m + h_m, g_{m+1}, \dots),$$

and let $\mathcal{O}_h$ be the event that this sequence is still strictly increasing. We use the restriction of $\mathbf{P}$ to this event, without normalizing its mass.

Lemma 9.3 (Absolute continuity under finite shifts). *For every fixed m and $h \in \mathbb{R}^m$,*

$$(T_h)_*(\mathbf{P}|_{\mathcal{O}_h}) \ll \mathbf{P}.$$

In particular, lowering g_1 by any fixed positive amount gives a law absolutely continuous with respect to $\mathbf{P}$.

Proof. We prove the claim by finite-dimensional GOE density ratios. Let $\mathbf{P}_n$ denote the ordered GOE edge law in dimension n . When a sequence-valued argument is needed, append the values $g_{n,n} + j$, $j \geq 1$, after its final coordinate. This convention does not affect any fixed coordinate for large n , so $\mathbf{P}_n \implies \mathbf{P}$ in the product topology; it also commutes with T_h whenever $n > m$. In the variables $g_{a,n} = n^{2/3}(2 - \lambda_a)$, the ordered density is proportional to

$$\exp \left\{ -\frac{n}{4} \sum_{a=1}^n (2 - n^{-2/3} g_{a,n})^2 \right\} \prod_{i < j} (g_{j,n} - g_{i,n}).$$

Translation of the first m variables has Jacobian one. Its Gaussian log-density ratio is $n^{1/3} \sum_{k \leq m} h_k + O(1)$ on sets where the first m coordinates are bounded. For a fixed $k \leq m$, the spectral estimates of Section 2 give, tightly,

$$\sum_{m < i \leq n} \frac{1}{g_{i,n} - g_{k,n}} = n^{1/3} + O(1), \quad \sum_{m < i \leq n} \frac{1}{(g_{i,n} - g_{k,n})^2} = O(1).$$

To see the first assertion, compare the reciprocals with $(g_{i,n} + b_n)^{-1}$, where $g_{1,n} + b_n \geq 1$. Their difference is summable by the square-tail bound, while $n^{1/3} - \sum_i (g_{i,n} + b_n)^{-1}$ is tight. Tight separation of the first finitely many eigenvalues controls the remaining denominators.

The logarithm of the Vandermonde ratio involving indices $k \leq m < i$ contains

$$\sum_{m < i \leq n} \log \left(1 - \frac{h_k}{g_{i,n} - g_{k,n}} \right).$$

Beyond a sufficiently large fixed index, the fractions have absolute value at most $1/2$. Taylor's inequality and the preceding two bounds show that this sum is $-h_k n^{1/3} + O(1)$. The finitely many remaining factors, including pairs within the first m coordinates, are bounded away from zero after imposing positive margins on the old and shifted gaps. The linear terms cancel those of the Gaussian factor. Consequently the new-to-old density ratio has a positive lower bound on sets of probability arbitrarily close to one, after any fixed localization inside the strict-order event.

Here is the measure-theoretic direction of the resulting implication. Let χ be a continuous function of finitely many leading coordinates, with $0 \leq \chi \leq 1$, supported where the shifted order has a positive margin and the leading coordinates are bounded. For every $\epsilon > 0$, the density-ratio bound supplies $c_\epsilon > 0$ such that, for every bounded nonnegative continuous function f of the edge sequence,

$$P_n(f) \geq c_\epsilon (P_n[\chi f \circ T_h] - \epsilon \|f\|_\infty).$$

The common spectral exceptional sets have probability at most ϵ for all sufficiently large n . Pass to the edge limit. For a compact subset of a P -null set, take continuous tests between zero and one, equal to one on that compact subset and supported in open sets of arbitrarily small P -mass. The inequality bounds its $(T_h)_*(\chi P)$ -mass by ϵ . Inner regularity gives the same bound for every P -null set. Letting $\epsilon \downarrow 0$ proves absolute continuity for this localized measure. Finally exhaust $\mathcal{O}_h$ by such continuous localizations. This proves the lemma without requiring a uniform bound on the density ratio over the entire edge law. $\square$

Proposition 9.4 (Randomness of the limit). *For every fixed $s > 0$, the finite random variable $\mathcal{A}_g(s)$ is essentially unbounded. In particular Q_{crit} is not a point mass.*

Proof. Lower g_1 by $t > 0$, leaving the other points fixed, and denote the resulting sequence by $g^{(t)}$. Choose the auxiliary parameter b large enough for both Gaussian prescriptions. Only l_1 changes, and (2.3) shows

$$D_{g^{(t)}}(b) - D_g(b) = -(l_1^{(t)} - l_1).$$

At a common coordinate sequence x , the finite centered-square sums in the two prescriptions differ by $-(l_1^{(t)} - l_1)$. Thus, wherever these sums converge, $S_{g^{(t)}}(x) = S_g(x) - (l_1^{(t)} - l_1)$. Consequently

$$S_{g^{(t)}}(x) - D_{g^{(t)}}(b) = S_g(x) - D_g(b).$$

The two renormalized constraints therefore coincide. Comparing the first Gaussian coordinate in their conditioning densities gives

$$\frac{d\Pi_{g^{(t)}}}{d\Pi_g}(x) = \frac{\exp(tx_1^2/2)}{\Pi_g[\exp(tx_1^2/2)]}.$$

The denominator is finite: increase b until $g_1 + b > t$ and use the bounded conditional first-coordinate density relative to the centered Gaussian of variance $(g_1 + b)^{-1}$. The independence of the conditioned law from b then gives the displayed identity for Π_g .

For a nonnegative random variable with unbounded support, exponential tilts with parameter tending to infinity have means tending to infinity. Indeed, for any M , the mass below M after tilting is at most $e^{-t/2} \Pi_g[x_1^2 \leq M] / \Pi_g[x_1^2 \geq M+1]$. It follows that $v(g^{(t)}) \rightarrow \infty$ as $t \rightarrow \infty$.

Suppose that $v(g)$ had an essential finite upper bound. By Lemma 9.3, this same bound would hold for $v(g^{(t)})$ almost surely for every positive integer t . Intersecting these countably many full-measure events contradicts the preceding divergence. Therefore $v(g)$ is essentially unbounded. The lower bound (9.1) proves the same assertion for $\mathcal{A}_g(s)$ at every fixed positive s . $\square$

Completion of the proof of Theorem 1.1. Proposition 9.2 proves convergence of the laws in $C_{\text{loc}}((0, \infty))$ for both entry distributions and identifies their common limit by (2.7). The same deterministic $c_* > 0$ occurs in both models. The spectral representation and the trivial form kernel prove nonnegativity, monotonicity, and almost-sure decay. Equation (9.1) excludes the zero function, and Proposition 9.4 establishes sample-to-sample randomness. $\square$

Decay at infinity is not a closed condition in $C_{\text{loc}}((0, \infty))$. The almost-sure decay proved above can therefore coexist with nondecaying functions in the closed topological support. Appendix A gives a precise example: the constant-one function belongs to the closed topological support of Q_{crit} .

A The closed topological support

The decay assertion in Theorem 1.1 holds almost surely under Q_{crit} . The set of functions that decay at infinity is not closed in $C_{\text{loc}}((0, \infty))$, and in the present problem the distinction is substantive. We prove that the closed topological support contains a nondecaying function. For an edge configuration g , write $\mathcal{A}_g$ for the function in (2.7).

Proposition A.1. *The constant function $s \mapsto 1$ belongs to the closed topological support of Q_{crit} in $C_{\text{loc}}((0, \infty))$.*

The proof modifies finitely many edge points so that the conditioned law concentrates near $x_1 = \pm 1$. A smooth function distinguishing these two regions has vanishing form energy, while the resolvent contribution of the remaining coordinates vanishes. The finite-shift absolute continuity from Lemma 9.3 ensures that every function used in the approximation belongs to the closed support.

Proof. Let $K = \text{supp } Q_{\text{crit}}$ denote the closed topological support. Since $C_{\text{loc}}((0, \infty))$ is separable, $Q_{\text{crit}}(K) = 1$. Consider all shifts of finitely many edge points by rational amounts, restricted to the event that the shifted points remain strictly ordered. There are countably many such shifts. By Lemma 9.3, outside a single null set, every valid shift g' of g gives a configuration for which the conditioned measure, the form identities, and (2.7) are defined, and for which $\mathcal{A}_{g'}$ belongs to K . We fix such a configuration g , also satisfying the spectral bounds of Section 2. Choose and fix b with $g_1 + b > 0$, and put

$$l_a = (g_a + b)^{-1}, \quad D = D(b).$$

In particular, l_a is strictly decreasing, $\sum_a l_a = \infty$, and $l_a \leq C a^{-2/3}$ for all sufficiently large a . All constants below may depend on this fixed configuration and on b .

Choosing the modified variances. We leave g_1 fixed and modify $g_2, \dots, g_m$, along a sequence of integers $m \rightarrow \infty$. Let $J = 60$ and define

$$A_m = \sum_{i=2}^m l_i + D + l_1 - 1.$$

Then $A_m \rightarrow \infty$ and $A_m = O(m^{1/3})$. There are arbitrarily large m such that

$$l_{m+1} \leq \frac{A_m}{2m}. \quad (\text{A.1})$$

Indeed, if the reverse strict inequality held eventually, the identity $A_{m+1} = A_m + l_{m+1}$ would imply $A_{m+1} > A_m(1 + 1/(2m))$, and hence $A_m \geq cm^{1/2}$, contrary to the upper bound. Restrict henceforth to integers satisfying (A.1), and set $M = A_m/(m-1)$. For all sufficiently large such integers,

$$\frac{c}{m} \leq M \leq Cm^{-2/3}. \quad (\text{A.2})$$

Choose strictly decreasing positive numbers on indices $2, \dots, m$ whose sum is A_m , with the first $J-1$ numbers close to $2M$ and the others close to M . This is possible by adjusting the latter group by a relative amount $O(J/m)$ and then making small distinct perturbations of zero total sum. For large m these numbers are less than l_1 and greater than l_{m+1} , by (A.1). They therefore define an ordered modification of the edge points through $g'_i = 1/l'_i - b$. Approximate each of the finitely many shifts $g'_i - g_i$ by a rational number, sufficiently closely to retain strict order and all the strict comparisons just imposed. We continue to write $g^{(m)}$ and $l_a^{(m)}$ for the resulting configuration and variances. The approximation can also be chosen so that the following bounds hold, with constants independent of m :

$$l_1^{(m)} = l_1, \quad l_a^{(m)} = l_a \quad (a > m), \quad l_a^{(m)} \asymp M \quad (2 \leq a \leq m), \quad (\text{A.3})$$

$$\frac{l_a^{(m)}}{l_j^{(m)}} \leq q < 1 \quad (a > J, 1 \leq j \leq J), \quad (\text{A.4})$$

$$V_m := D^{(m)}(b) + l_1 \rightarrow 1, \quad \sum_{a>1} (l_a^{(m)})^2 \leq Cm^{-1/3}. \quad (\text{A.5})$$

Here $D^{(m)}$ is the value of (2.3) for $g^{(m)}$. To check the first conclusion in (A.5), the finite-change identity from that equation gives

$$D^{(m)}(b) - D(b) = - \sum_{i=2}^m (l_i^{(m)} - l_i).$$

Before rational approximation our choice of sum makes $V_m = 1$ exactly; afterwards we can require $|V_m - 1| \leq m^{-1}$. The square-sum bound follows from $mM^2 \leq Cm^{-1/3}$ and $\sum_{a>m} l_a^2 \leq Cm^{-1/3}$. By the choice of the original configuration, each $\mathcal{A}_{g^{(m)}}$ belongs to K .

Concentration after conditioning. Let γ_m be the product law of centered normals on indices $a > 1$ with variances $l_a^{(m)}$, and let

$$S_m = \sum_{a>1} (x_a^2 - l_a^{(m)}).$$

This series converges in $L^2(\gamma_m)$, and (A.5) implies $S_m \rightarrow 0$ in probability under γ_m . Write $\Pi_m = \Pi_{g^{(m)}}$. Eliminating x_1 in the conditioning prescription shows that the $a > 1$ marginal of Π_m has density

$$\begin{aligned} r_m &= Z_m^{-1} h_m(S_m), & Z_m &= \mathbb{E}_{\gamma_m} h_m(S_m), \\ h_m(v) &= \mathbf{1}_{\{v < V_m\}} (V_m - v)^{-1/2} \exp\left(-\frac{V_m - v}{2l_1}\right), \end{aligned} \quad (\text{A.6})$$

relative to γ_m . Under the full law, $x_1^2 = V_m - S_m$, and the sign of x_1 is independent and symmetric.

We give bounds near the singularity in (A.6), because concentration under the product law alone would not suffice. Put $t_m = m^{1/10}$. For either sign, the centered Gaussian square-sum formula and (A.5) give

$$\log \mathbb{E}_{\gamma_m} e^{\pm t_m S_m} \leq C t_m^2 \sum_{a>1} (l_a^{(m)})^2 \leq C, \quad (\text{A.7})$$

since $t_m \max_{a>1} l_a^{(m)} \rightarrow 0$. Under either exponential tilt the coordinates remain independent normals, with variances $l_a^{(m)}/(1 \mp 2t_m l_a^{(m)})$, comparable to the original variances. The sum of the squares of any fixed four coordinates among $2, \dots, J$ has density bounded by C/M : its characteristic function has absolute value at most $C(1 + M^2 u^2)^{-1}$, whose integral is $O(M^{-1})$. Convolution with the remaining coordinates does not increase this bound. The same assertion holds for the infinite sum by its L^2 convergence. Undoing the tilt and using (A.2)–(A.7), we obtain a density p_m of S_m satisfying

$$p_m(v) \leq C m e^{-t_m |v|}, \quad v \in \mathbb{R}. \quad (\text{A.8})$$

The factors $h_m(v)$ are uniformly bounded on the complement of a fixed neighborhood of $v = V_m$. Near that point, which stays bounded away from zero, (A.8) supplies an exponentially small factor, while the singularity $(V_m - v)^{-p/2}$ is integrable for every $p < 2$. Consequently, concentration of S_m at zero gives

$$Z_m \rightarrow e^{-1/(2l_1)}, \quad \sup_m \|r_m\|_{L^{3/2}(\gamma_m)} < \infty. \quad (\text{A.9})$$

The same estimate, integrated over $|S_m| > \varepsilon$, shows that for each fixed $\varepsilon > 0$ there are positive constants $C_\varepsilon, c_\varepsilon$ such that

$$\Pi_m\{|x_1^2 - 1| > \varepsilon\} \leq C_\varepsilon e^{-c_\varepsilon m^{1/10}} \quad (\text{A.10})$$

for all sufficiently large m . Integrating the negative tail of S_m also gives convergence of x_1^2 to 1 in every fixed moment. Finally, Hölder's inequality applied to (A.9) yields, for each fixed $p < \infty$,

$$\sup_m \sup_{a>1} \Pi_m \left[\left| x_a / \sqrt{l_a^{(m)}} \right|^p \right] < \infty. \quad (\text{A.11})$$

The first coordinate and the trace tail. Write $H_m = H_{\Pi_m}$ and $\mathcal{E}_m(F) = \int |\nabla F|^2 d\Pi_m$. Choose an odd smooth compactly supported function ϕ on $\mathbb{R}$ which equals 1 near 1 and -1 near -1 . The preceding concentration and moment estimates imply

$$\|x_1 - \phi(x_1)\|_{L^2(\Pi_m)} \rightarrow 0, \quad \|\phi(x_1)\|_{L^2(\Pi_m)}^2 \rightarrow 1, \quad \mathcal{E}_m(\phi(x_1)) \rightarrow 0.$$

For $0 \leq s \leq S$, the spectral theorem gives

$$0 \leq \|\phi(x_1)\|_2^2 - \langle \phi(x_1), e^{-sc_* H_m} \phi(x_1) \rangle \leq S c_* \mathcal{E}_m(\phi(x_1)).$$

Semigroup contraction therefore shows that the first coordinate's contribution to $\mathcal{A}_{g^{(m)}}(s)$ tends to 1, uniformly for s in bounded intervals. Coordinates $2, \dots, J$ contribute at most $\sum_{a=2}^J \Pi_m[x_a^2] = O(M)$, which tends to zero. It remains to control the infinitely many coordinates beyond J . We use the angular integration-by-parts estimate of Section 7. Here the angular identity (5.1) is exact, so there is no likelihood-score error. The remaining task is to make the coefficient and inverse-moment bounds uniform in m .

Let $R_0 = \sum_{j=1}^J x_j^2$. We first establish the uniform inverse moments needed for angular integration by parts. By (A.9), Hölder's inequality, and $R_0 \geq \sum_{j=2}^J x_j^2$,

$$\Pi_m[R_0^{-8}] \leq C \left(\mathbb{E}_{\gamma_m} \left[\left(\sum_{j=2}^J x_j^2 \right)^{-24} \right] \right)^{1/3} \leq CM^{-8} \leq Cm^8.$$

The Gaussian inverse moment is finite because $J - 1 = 59 > 48$. On $\{x_1^2 \geq 1/2\}$ the quantity R_0^{-4} is bounded by 16; on the complement, Cauchy-Schwarz and (A.10) absorb the preceding polynomial bound. Thus

$$\sup_m \Pi_m[R_0^{-4}] < \infty. \quad (\text{A.12})$$

For $a > J$ put

$$\delta_{aj} = (l_a^{(m)})^{-1} - (l_j^{(m)})^{-1}, \quad \mathcal{V}_a = \sum_{j=1}^J \frac{x_j}{R_0 \delta_{aj}} (x_j \partial_a - x_a \partial_j).$$

By (A.4), $\delta_{aj} \geq (1 - q)/l_a^{(m)}$. The calculation in (7.3), now in the limiting coordinates, gives

$$|\mathcal{V}_a| \leq Cl_a^{(m)}(1 + |x_a|/\sqrt{R_0}), \\ |\operatorname{div} \mathcal{V}_a| \leq Cl_a^{(m)}|x_a|/R_0.$$

Together with (A.11) and (A.12), these imply

$$\|\mathcal{V}_a\|_{L^2(\Pi_m)} + \|\operatorname{div} \mathcal{V}_a\|_{L^2(\Pi_m)} \leq Cl_a^{(m)}. \quad (\text{A.13})$$

Each rotation $x_j \partial_a - x_a \partial_j$ preserves the square-sum constraint. Its derivative of the Gaussian log density is $-\delta_{aj} x_a x_j$, so the corresponding derivative along $\mathcal{V}_a$ is exactly $-x_a$. Applying (5.1) with the coefficients defining $\mathcal{V}_a$ therefore gives

$$\Pi_m[x_a f] = \Pi_m[\mathcal{V}_a f] + \Pi_m[f \operatorname{div} \mathcal{V}_a] \quad (\text{A.14})$$

for smooth cylinder tests. To justify the singular coefficients, multiply them first by $\chi(R_0/\varepsilon)$, where χ is smooth, zero below 1, and one above 2. Direct calculation gives

$$\mathcal{V}_a R_0 = -2x_a \sum_{j=1}^J \frac{x_j^2}{R_0 \delta_{aj}}, \quad |\mathcal{V}_a \chi(R_0/\varepsilon)| \leq Cl_a^{(m)} \frac{|x_a|}{R_0} \mathbf{1}_{\{R_0 \leq 2\varepsilon\}}.$$

The last expression tends to zero in $L^2(\Pi_m)$ by (A.11) and (A.12). At infinity we use a radial cutoff in a finite coordinate set containing $1, \dots, J, a$ and all test coordinates. Every rotation in $\mathcal{V}_a$ preserves this radius, so differentiating that cutoff produces no error. This proves (A.14) on the cylinder core, which then extends the identity to the form domain.

Fix $z > 0$ and a finite set $I \subset \{J+1, J+2, \dots\}$. Applying (A.14) to a vector of tests $(f_a)_{a \in I}$ and using Cauchy–Schwarz and (A.13) yields

$$\left| \sum_{a \in I} \Pi_m[x_a f_a] \right| \leq C_z \left(\sum_{a > J} (l_a^{(m)})^2 \right)^{1/2} \left(\sum_{a \in I} \{z \|f_a\|_2^2 + c_* \mathcal{E}_m(f_a)\} \right)^{1/2}.$$

By the resolvent variational formula, followed by monotone convergence over finite I ,

$$\sum_{a > J} \langle x_a, (z + c_* H_m)^{-1} x_a \rangle_{\Pi_m} \leq C_z \sum_{a > J} (l_a^{(m)})^2 \longrightarrow 0. \quad (\text{A.15})$$

For every $s_0 > 0$ the scalar inequality $e^{-s\lambda} \leq C_{s_0, z}(z + \lambda)^{-1}$ holds for $s \geq s_0$ and $\lambda \geq 0$. Hence (A.15) also makes the semigroup trace beyond J tend to zero uniformly on compact subsets of $(0, \infty)$.

We have proved $\mathcal{A}_{g^{(m)}} \rightarrow 1$ in $C_{\text{loc}}((0, \infty))$. Each approximating function belongs to K , so closedness of K proves the proposition. $\square$

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